The undivided energy and its differentiation

$E = mc^2$ in the split-signature arena: the unitary energy, the symmetric conversion c^2, the differentiated mass, and the place of the Yang–Mills b-field

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Abstract

We test, inside the split-signature arena $\mathrm{Cl}(3,3)$ of [1], a reading of the rest-energy relation proposed by the author of that programme: in $E = mc^2$, c^2 is the symmetric part, m is what breaks the symmetry, the energy is one undivided six-dimensional entity whose internal differentiation produces the mass, and the mass is generated by the Yang–Mills b-field. The undivided energy exists: the only rotation- and duality-invariant quadratic energy of the six frame components is the positive form $E_6 = \frac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2)$, whose pointwise symmetry group is $U(3)$, and in the closure of the force law it is a Casimir of the momentum field. It differentiates without any interaction, $E_6^2 = E_0^2 + |\mathbf{S}|^2$, with $E_0 = \frac{1}{2}|q|$ the complete invariant of Lorentz transformations and duality and the minimum of E_6 over boosts; this is the field-point form of $E^2 = (pc)^2 + (mc^2)^2$. The constant c is a conversion forced by Schur's lemma, not a symmetry; the clause "c^2 is the symmetric part" holds exactly for the η-self-adjoint summand κG of the shift, $(\kappa G)^2 = -c^2/\ell^2$, which is the only summand that binds. A nonzero rest energy breaks duality to $\mathbb{Z}_2$ and, for matter, the chiral algebra to $\mathfrak{u}(1)_J$; it breaks neither Lorentz invariance nor $\mathrm{Spin}(3,3)$, since the Dirac mass form $e_1e_2e_3$ is invariant under the whole six-dimensional group. The b-field is the antisymmetric, connection summand of the same shift: classically it never generates the binding rate, a part commuting with G only splits the rest frequency into charge-labelled lines, and at three constituents the compatible algebra is $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$. What remains for it is the one scale the algebra leaves free, which dimensional transmutation could supply; that identification is a conjecture. We also give the arena's answers on particles and antiparticles (the two senses of one rotation, exchanged by charge conjugation), the vacuum and its zero-point energy, and the Goldstone bosons. Every finite-dimensional statement was checked twice, by independent scripts.

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1 The question

1.1 The hypothesis in its author's words

The starting point of this note is a reading of the rest-energy relation proposed by the author of the programme:

> *$E = mc^2$: consider c^2 as the symmetrical part of the equation and m as what breaks this symmetry; m is generated from a b particle, b standing for the b-particle of Yang–Mills.*

Three clarifications fix the intended reading. First, the energy is to be taken as *one undivided entity*, and the mass is to emerge through an *internal differentiation* of that unity, before any external interaction is applied. Second, c^2 corresponds to the variable X of the former momentum–force chapter, whose successor is the interior dynamics of Part IV of the paper [1] (Sections 15–17), where the shift operators are $\alpha_\pm = \partial_\tau \pm \kappa G$ with $\kappa = c/\ell$. Third, the energy is six-dimensional. The b-field is the non-abelian gauge potential of Yang and Mills [41], whose quanta carry isotopic spin; Yang and Mills left open whether those quanta are massive.

An earlier exploratory session settled several points that we take as given and do not repeat. The scalar c^2 is not by itself a symmetry, and if E and c^2 are invariant under a group then so is $m = E/c^2$, so a mass cannot *be* the breaking of a symmetry that E and c^2 share. A nonzero invariant mass does not break Lorentz invariance. The free six-dimensional scalar has an energy unbounded below ([1], Proposition 27.1), and positive four-dimensional mass cannot be obtained from momentum along the surplus times, since $m_4^2 = \mu_6^2 - (p_u^2 + p_v^2)/c^2$. Non-orthogonality of $\mathbf{E}$ and $\mathbf{B}$ generates no mass ([1], Section 14). The same session was rightly criticised for jumping to Yang–Mills interactions, condensates and gap equations before the unity of the energy and its differentiation had been built. This note does the work in the order the hypothesis asks for: the undivided energy first, its differentiation second, the dynamics of the differentiation third, and the b-field last, where it belongs.

1.2 The dictionary under test

All constructions are made inside the arena of [1]: $V = \mathbb{R}^{3,3}$ with $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the realisation of $\mathrm{Cl}(3,3)$ on $\mathbb{R}^8$, the binding root G ($Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$), which squares to $-\mathbf{1}$, is self-adjoint for η and acts as the Hodge star in the field-strength reading $X = \sum_i (B_i e_i + E_i e_{i+3})$, $z = \mathbf{B} + i\mathbf{E}$, $q(z) = z \cdot z$ ([1], Theorem 12.3). The working dictionary is

hypothesis	candidate object in the arena
E, undivided, six-dimensional	the positive, duality-invariant energy $E_6 = \frac{1}{2}\lvert z\rvert^2 = \frac{1}{2}(\lvert\mathbf{B}\rvert^2 + \lvert\mathbf{E}\rvert^2)$ of a frame vector; in the closure of the force law, the Casimir $\lvert p\rvert^2$
c^2, the symmetric part	the conversion $C = c\,\mathbf{1}$ between the temporal and the spatial triple forced by Schur's lemma ([1], Theorem 3.2), and the η-self-adjoint, factorising part κG of the shift, $(\kappa G)^2 = -c^2/\ell^2$
m, the differentiation	the Lorentz-class invariant $E_0 = \frac{1}{2}\lvert q\rvert$, the rest-frame energy density of the field; for Dirac matter the mass form $T = e_1 e_2 e_3$
b, the Yang–Mills field	the η-antisymmetric (transport) part of the shift, $\mathrm{End}(V) = \mathfrak{so}(3,3) \oplus \{X = X^\dagger\}$ ([1], Theorem 17.3); at three constituents the binding-compatible algebra $\mathfrak{so}(4)_\Delta \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ ([1], Theorem 17.21)

Every entry is a hypothesis to be tested, and several of them turn out to need correction.

1.3 How claims are classified

We follow the deflationary principle of [1] (Corollary 4.4): a statement about the arena carries information only if it could have come out otherwise. Every claim below belongs to one of six classes, named in the text. A claim is *fixed* when it is an identity or theorem of the algebra that could have failed; *constructed* when the algebra admits the object and nothing selects it; *obstructed* when it is impossible under stated hypotheses; *open* when it is neither established nor refuted. To these four classes of the paper we add two that the subject requires: *inherited*, for facts of ordinary relativistic physics that hold in any notation and are reproduced only because something else is stated in terms of them, and *interpretive*, for readings and dictionary entries, which are not theorems at all. The relation $E = mc^2$ itself is inherited. What can be new is only how the arena realises its three ingredients, and which of the realisations are forced.

Every finite-dimensional statement was checked by exact or high-precision computation. Each section has a verification script in `investigations/rest_energy/verify/`, written independently of a second, adversarial check script by a different author; the scripts print one `PASS` line per claim and exit with status zero only if all pass (Section 7).

2 The undivided energy and its differentiation

The hypothesis reads $E = mc^2$ [7] with c^2 as the symmetric part of the equation and m as what breaks that symmetry; the agreed clarification asks that energy first be one undivided, six-dimensional entity, and that mass appear through an internal differentiation of that unity before any interaction is applied. We follow that order: the undivided energy is built from the arena alone and differentiated with no other field and no coupling, c is restored, and the Yang–Mills field b [41] enters only in Section 2.9. The points settled in Section 1 (c^2 alone is no symmetry; a mass breaks no Lorentz invariance; Proposition 27.1, Example 27.3 and Section 14.4 of [1]) are used without repetition. Most of what follows is inherited electrodynamics or restates results of [1], and is labelled so; the verdicts are in Section 2.10.

Conventions. We use the field-strength reading of Theorem 12.3 of [1]: $X = \sum_i (B_i e_i + E_i e_{i+3})$, $z = \mathbf{B} + i\mathbf{E} \in \mathbb{C}^3$, the binding root G ($Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$) acts as i, $q(z) = z \cdot z = |\mathbf{B}|^2 - |\mathbf{E}|^2 + 2i\,\mathbf{E}\cdot\mathbf{B}$, $\eta(X,X) = \operatorname{Re} q$ and $\eta(X, GX) = -2\,\mathbf{E}\cdot\mathbf{B}$. The frame variable is minus the textbook field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$ (Section 1.4 of [1]). With $h(z,w) = \sum_j \bar{z}_j w_j$ we put

$$E_6 := \tfrac{1}{2}|z|^2 = \tfrac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2), \qquad E_0 := \tfrac{1}{2}|q(z)|, \qquad \mathbf{S} := \mathbf{E}\times\mathbf{B} = \tfrac{i}{2}\,\bar{z}\times z = -\mathbf{S}_{\mathrm{t}}, \quad \mathbf{S}_{\mathrm{t}} := \mathbf{E}\times\mathbf{B}_{\mathrm{t}}. \tag{1}$$

$\mathbf{S}$ is the frame Poynting vector of Section 12 of [1], $\mathbf{S}_{\mathrm{t}}$ the textbook one. Applied to the frame field X these are the frame field's energies; for the momentum field $\pi = \eta^{-1}p$ of the closure we write $E_6(\pi)$, $E_0(\pi)$, $\mathbf{S}(\pi)$. Units are Heaviside–Lorentz with $c = 1$ until Section 2.5. For cyclic (i,j,k) the diagonal rotations and boosts of $\mathfrak{l}_G$ are $L_i = \frac{1}{2}(e_{j+3}e_{k+3} - e_j e_k)$ and $K_i = \frac{1}{2}(e_k e_{j+3} - e_j e_{k+3})$, acting on vectors by the commutator: L_i by $\delta\mathbf{B} = e_i \times \mathbf{B}$, $\delta\mathbf{E} = e_i \times \mathbf{E}$, and K_i by $\delta\mathbf{E} = e_i \times \mathbf{B}$, $\delta\mathbf{B} = -e_i \times \mathbf{E}$; $\mathfrak{p}_G$ is the part of $\mathfrak{so}(3,3)$ anticommuting with G, so $\mathfrak{so}(3,3) = \mathfrak{l}_G \oplus \mathfrak{p}_G$. We write the Lorentz-class inertia of Theorem 17.11 of [1] as $N = (\nu_1\eta^{-1} + \nu_2 G\eta^{-1})/m$, the paper's (a,b) renamed, so that b denotes only the Yang–Mills field. These laws, Theorem 12.3, Example 12.4 and Lemma 3.4 of [1] ($\mathrm{Ad}_{e_1e_2e_3}|_V = \eta$, $G = -\mathcal{K}\circ\mathrm{Ad}_{e_1e_2e_3}$) were checked in the realisation of Section 2 of [1] (d1-0); identifiers d1-n refer to the verification script.

2.1 One undivided energy, with pointwise symmetry $U(3)$

Theorem 2.1 (Uniqueness of the undivided energy)**.** *1. The symmetric forms on V invariant under the diagonal rotations are spanned by $|\mathbf{B}|^2$, $|\mathbf{E}|^2$, $\mathbf{B}\cdot\mathbf{E}$, and split as $\mathbb{R}\,\mathbf{1} \oplus \operatorname{span}\{\eta, \eta G\}$, the first summand commuting and the second anticommuting with G.*

2. A quadratic energy invariant under rotations and under duality $e^{\vartheta G}$ is a multiple of E_6, which is positive definite.

3. The forms invariant under all of $\mathfrak{l}_G$ are $\operatorname{span}\{\eta, \eta G\}$, that is $|\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\mathbf{E}\cdot\mathbf{B}$; none is definite.

Proof. The diagonal rotations act on $V = V^+ \oplus V^-$ as $\mathbb{R}^2 \otimes \mathbb{R}^3$ with $SO(3)$ on the second factor, which is absolutely irreducible, so by Schur's lemma [31] the invariant forms are $N = \left(\begin{smallmatrix}\alpha & \gamma\\ \gamma & \beta\end{smallmatrix}\right) \otimes \mathbf{1}$. With $G = \left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right) \otimes \mathbf{1}$, $\frac{1}{2}(N - GNG) = \frac{1}{2}(\alpha+\beta)\mathbf{1}$ and $\frac{1}{2}(N + GNG) = \frac{1}{2}(\alpha-\beta)\eta - \gamma\,\eta G$. Since $G^{\top}G = \mathbf{1}$ and $G^{\top} = -G$, N is $e^{\vartheta G}$-invariant iff $G^{\top}N + NG = 0$ iff $[N, G] = 0$, which gives (2). For (3) the boost law gives $\delta(\alpha|\mathbf{B}|^2 + \beta|\mathbf{E}|^2 + 2\gamma\,\mathbf{B}\cdot\mathbf{E}) = -2(\alpha+\beta)\,\mathbf{v}\cdot(\mathbf{E}\times\mathbf{B})$, the γ term being unchanged, so $\beta = -\alpha$. □

This restates Theorem 17.11(iii) and Proposition 17.4 of [1] on V, and is inherited: for any binding root $G' = bGb^{-1}$, $b \in SO(3,3)$, the forms invariant under the compact part of $\mathfrak{l}_{G'}$ and under G' are the multiples of the positive $b^{-\top}b^{-1}$, so the statement could not have failed; it is the uniqueness of the energy density among rotation- and duality-invariant forms. The arena adds only that this form is the Euclidean one of its basis: G is η-self-adjoint and anticommutes with $\mathrm{Ad}_{e_1e_2e_3}|_V = \eta$, so $G^\top G = \eta G \eta G = \mathbf{1}$ (Lemma 3.4 of [1]; d1-1a, with an exact conjugated root). E_6 has duality weight 0, while $e^{\vartheta G}$ turns $(\mathrm{Re}\, q, \mathrm{Im}\, q)$ through 2ϑ: E_6 is undivided, distinguishing neither the magnetic triple from the electric one nor, as we now show, rest from motion.

Proposition 2.2 (The pointwise symmetry of the undivided energy is $U(3)$)**.** *(1) The maps of V preserving E_6 and commuting with G form $\mathfrak{u}(3)$, of dimension 9: $A = \left(\begin{smallmatrix} P & Q \\ -Q & P \end{smallmatrix}\right)$ acts on z by the anti-Hermitian matrix $P - iQ$. (2) The Lie algebra of $O(3,\mathbb{C}) \times U(1)_G$, the G-linear maps preserving q up to a phase, is $\mathfrak{l}_G \oplus \mathbb{R}G$, of dimension 7. (3) Their intersection is $\mathfrak{so}(3) \oplus \mathbb{R}G$, of dimension 4, the maximal compact subalgebra of $\mathfrak{l}_G \oplus \mathbb{R}G$; and $\mathfrak{u}(3) \cap \mathfrak{l}_G = \mathfrak{so}(3)$. (4) The stabiliser of the function E_0 in $U(3)$ is $U(1) \cdot O(3) = \{e^{i\varphi}O\}$; for instance the unitary map $(1,0,0) \mapsto \frac{1}{\sqrt{2}}(1,i,0)$ carries a field at rest ($E_0 = E_6 = \frac{1}{2}$) to a null field ($E_0 = 0$) at the same E_6.*

Proof. (1) Commuting with G forces the block form, which is multiplication by $P - iQ$ on z; $A^\top = -A$ means $P^\top = -P$, $Q^\top = Q$, so $P - iQ$ is anti-Hermitian; $3 + 6 = 9$. (2) $q(e^{tM}z) = e^{i\varphi t}q(z)$ gives $M^\top + M = i\varphi\,\mathbf{1}$, so $M = \frac{i\varphi}{2} + M_0$ with $M_0 \in \mathfrak{so}(3,\mathbb{C}) = \mathfrak{l}_G$ (Theorem 12.1 of [1]), and $i\,\mathbf{1}$ is G. (3) A complex antisymmetric anti-Hermitian matrix is real. (4) If $U \in U(3)$ preserves $|q|$, the form $z^\top U^\top U z$ vanishes on the cone $q = 0$; q is irreducible, so $U^\top U = e^{i\varphi}\mathbf{1}$, and $O = e^{-i\varphi/2}U$ is unitary and complex orthogonal, hence real. □

Parts (1)–(3) are inherited Hermitian linear algebra, which makes "unitary" literally true of the pointwise form. Part (4) is inherited as well, the stabiliser of $|z \cdot z|$ among unitary maps of the Riemann–Silberstein vector; read as a statement about the rest energy, it is the exact sense in which E_0 "breaks" something: $U(3)$ is reduced to $U(1) \cdot O(3)$, and the five-dimensional orbit $U(3)/U(1) \cdot O(3)$ parametrises the real structures $z \mapsto \bar{z}$ up to phase, of which the arena has chosen one (Proposition 2.11; d1-1b, d1-8b). But $U(3)$ is the symmetry of the pointwise form E_6 only. It is not a symmetry of Maxwell's equations on the frame, $\partial_t z = -i\nabla \times z$ (Proposition 12.10 of [1]): the unitary of (4) carries the wave $(1,i,0)e^{ik(x_3-t)}$ of Example 12.11 to $\frac{1+i}{\sqrt{2}}(1,1,0)e^{ik(x_3-t)}$, which does not solve them. Nor is it one of the split-form interior, since $G = i\,\mathbf{1} \in U(3)$ reverses η and $U(3) \cap O(\eta) = O(3)$; of the closures it is a symmetry only for $N = \mathbf{1}/m$, the $U(3)$-invariant forms being the multiples of $\mathbf{1}$.

2.2 The rest energy is the complete invariant of Lorentz and duality

Theorem 2.3 (Orbits)**.** *(1) Each level set $\{|q| = r^2\}$, $r > 0$, is one orbit of $O(3,\mathbb{C}) \times U(1)_G$, of real dimension 5, with two-dimensional stabilisers; every non-null z is carried to $(r,0,0)$, $r = \sqrt{2E_0}$. (2) The nonzero null fields form one orbit of $SO(3,\mathbb{C})$ alone, of dimension 4, and at a null point Gz is tangent to the Lorentz orbit. (3) Hence E_0 is a complete invariant of Lorentz and duality on non-null fields, and E_6 is the complete invariant of $U(3)$, whose orbits are the spheres.*

Proof. (1) Write $q(z) = r^2e^{i\varphi}$ and $w = e^{-i\varphi/2}z/r$ (a duality rotation), so $q(w) = 1$. Since q is nondegenerate on $w^\perp = \{f : f \cdot w = 0\}$, some $f \in w^\perp$ has $f \cdot f = 1$; with $f_3 = w \times f$, $(a \times b) \cdot (a \times b) = (a \cdot a)(b \cdot b) - (a \cdot b)^2$ gives $f_3 \cdot f_3 = 1$, so $A = (w\, f\, f_3)^\top \in SO(3,\mathbb{C})$ and $Ae^{-i\varphi/2}z = re_1$. An element $M_0 + \frac{i\varphi}{2}$ fixes e_1 iff $\varphi = 0$ and M_0 is a complex rotation about e_1: the stabiliser is $\mathfrak{so}(2,\mathbb{C})$ and the orbit has dimension $7 - 2 = 5$. (2) A rotation carries a null z to $\lambda(1,i,0)$, $\lambda > 0$; K_3 acts by $\delta z = i\,e_3 \times z$ and $i\,e_3 \times (1,i,0) = (1,i,0)$, so the Doppler boost $e^{\chi K_3}$, $\chi = -\log\lambda$, finishes. The Lorentz tangent space at a null z is its q-orthogonal complement,

which contains iz; at $(1, i, 0)$, $iz = -e_3 \times z$ is the rotation L_3, the helicity of Example 13.11 of [1]. (3) follows. □

Inherited: the classification of a field strength at a point [33, 17] (d1-2a, d1-2b). Without duality the complete invariant is q, whose phase (twice the complexion of Proposition 12.12 of [1]) duality removes.

2.3 The differentiation identity

Proposition 2.4 (Differentiation)**.** *For every $z \in \mathbb{C}^3$*

$$|z|^4 = |z \cdot z|^2 + |\bar{z} \times z|^2, \qquad \textit{equivalently} \qquad E_6^2 = E_0^2 + |\mathbf{S}|^2. \tag{2}$$

The rank-one Hermitian $H = zz^\dagger$ carries all three terms: $\operatorname{tr} H = 2E_6$, $(\operatorname{Im} H)_{jk} = \epsilon_{jkl} S_l$, $\operatorname{tr}(H\bar{H}) = |q|^2 = 4E_0^2$, $H^2 = (\operatorname{tr} H)H$. *Hence $E_0 \le E_6$, with equality iff $\mathbf{S} = 0$, iff $\bar{z} \parallel z$, iff $\mathbf{B} \parallel \mathbf{E}$ or one of them vanishes.*

Proof. $\bar{z} \times z = -2i\,\mathbf{E} \times \mathbf{B}$, and $(a \times b) \cdot (c \times d) = (a \cdot c)(b \cdot d) - (a \cdot d)(b \cdot c)$ at $(\bar{z}, z, z, \bar{z})$ gives $|\bar{z} \times z|^2 = |z|^4 - |q|^2$. Next $H_{jk} = B_j B_k + E_j E_k + i\epsilon_{jkl}(\mathbf{E} \times \mathbf{B})_l$ and $\operatorname{tr}(H\bar{H}) = |\sum_j z_j^2|^2$. The equality case is Cauchy–Schwarz for $|h(\bar{z}, z)| \le |z|^2$. □

Inherited, a complex Lagrange identity; the relation $u^2 - |\mathbf{S}|^2 = \frac{1}{4}|q|^2$ is stated in Section 14.3 of [1] (d1-3a). In the user's terms it is exact: the undivided E_6 is the hypotenuse of a right triangle whose legs are a Lorentz-invariant rest part E_0 and a motion part $|\mathbf{S}|$, and the split needs no interaction and no second field; in Minkowski terms E_0 is the Lorentz length of $(E_6, \mathbf{S})$, although $(E_6, \mathbf{S}_\mathrm{t})$ is not a four-vector (Proposition 2.5).

Proposition 2.5 (Rainich identity)**.** *For the stress–energy tensor of $(\mathbf{E}, \mathbf{B}_\mathrm{t})$, $T^{00} = \frac{1}{2}(|\mathbf{E}|^2 + |\mathbf{B}_\mathrm{t}|^2) = E_6$, $T^{0i} = (\mathbf{S}_\mathrm{t})_i$, $T^{ij} = T^{00}\delta_{ij} - E_i E_j - B_{\mathrm{t}i} B_{\mathrm{t}j}$, one has $T^\mu{}_\mu = 0$ and*

$$T^\mu{}_\lambda T^\lambda{}_\nu = E_0^2\, \delta^\mu{}_\nu, \qquad T_{\mu\nu} T^{\mu\nu} = 4E_0^2. \tag{3}$$

$T^{\mu\nu}$ is duality invariant, and for every unit timelike observer t the density $P(t)^\mu = T^\mu{}_\nu t^\nu$ is future causal with $\eta(P(t), P(t)) = E_0^2$.

Proof. Duality invariance is a polynomial identity; both sides of (3) are covariant, so by Theorem 2.3 it suffices to take $z = (r, 0, 0)$, where $T^\mu{}_\nu = \frac{r^2}{2}\operatorname{diag}(1, 1, -1, -1)$, and null fields follow by continuity. Causality of $P(t)$ is the dominant energy condition [14]. □

Inherited [29, 33] (d1-3b). The mass of a field point is the Lorentz scalar of the tensor whose 00 component is the undivided energy, but $(u, \mathbf{S}_\mathrm{t})$ is not a four-vector: each observer sees a different density $P(t)$, all of length E_0.

2.4 The rest energy as a minimum, and two boost factors

Theorem 2.6 (Variational form of the rest energy)**.**

1. *For non-null z, E_0 is the minimum of E_6 over the Lorentz orbit, and over the Lorentz–duality orbit, attained exactly where $\mathbf{S} = 0$; for null $z \neq 0$ the infimum 0 is not attained. Moreover $E_6 = E_0/\sqrt{1 - v_E^2}$ with $v_E := |\mathbf{S}|/E_6 < 1$.*

2. *If $\mathbf{B} \parallel \mathbf{E} \parallel \mathbf{n}$ and the boost $e^{\chi K}$ of rapidity χ has axis orthogonal to $\mathbf{n}$, then $E_6 = E_0 \cosh 2\chi$, $|\mathbf{S}| = E_0 \sinh 2\chi$ and $v_E = \tanh 2\chi = 2\beta/(1 + \beta^2)$, $\beta = \tanh \chi$; a boost along $\mathbf{n}$ changes neither E_6 nor $\mathbf{S} = 0$.*

3. *The observer moving with $\mathbf{v}_\mathrm{rest} = \mathbf{S}_\mathrm{t}/(E_6 + E_0)$ sees $\mathbf{S} = 0$ and $E_6 = E_0$; $|\mathbf{v}_\mathrm{rest}| = \beta$.*

Proof. (1) By (2) $E_6 \geq E_0$ with equality iff $\mathbf{S} = 0$, and Theorem 2.3 supplies points with $\mathbf{S} = 0$ in both orbits; for null z the Doppler boost drives E_6 to 0. Divide (2) by E_6^2. (2) Take $\mathbf{n} = e_1$, $X = \sigma_1 e_1 + \sigma_2 e_4$. By Example 12.4 of [1], $[K_3, e_1] = e_5$, $[K_3, e_5] = e_1$, $[K_3, e_4] = -e_2$, $[K_3, e_2] = -e_4$, so $\mathrm{ad}_{K_3}^3 = \mathrm{ad}_{K_3}$, $e^{\chi\,\mathrm{ad}_{K_3}} = \mathbf{1} + \sinh\chi\ \mathrm{ad}_{K_3} + (\cosh\chi - 1)\,\mathrm{ad}_{K_3}^2$, $\mathbf{B} = (\sigma_1 \cosh\chi, -\sigma_2 \sinh\chi, 0)$, $\mathbf{E} = (\sigma_2 \cosh\chi, \sigma_1 \sinh\chi, 0)$, and $\mathbf{S} = -E_0 \sinh 2\chi\, e_3$; likewise for K_2, while K_1 fixes e_1 and e_4. (3) $|\mathbf{S}|/(E_6 + E_0) = \sinh 2\chi/(\cosh 2\chi + 1) = \tanh\chi$. For the direction: $e^{\chi K_3}$ is the passage to a frame moving with $-\tanh\chi\, e_3$ (Section 12 of [1]), in which the rest frame moves along $+e_3$, and so does $\mathbf{S}_\mathrm{t} = -\mathbf{S}$. □

Inherited [17]; verified with the paper's own K_i, symbolically and numerically; the boost built from $\mathbf{E} \times \mathbf{B}$ instead of $\mathbf{S}_\mathrm{t}$ leaves a Poynting vector of at least $0.2\,E_0$ (d1-4a–d1-4c). This is the rest-frame velocity $\mathbf{S}/(u + \frac{1}{2}|q|)$, with $\frac{1}{2}|q| = u\sqrt{1 - v_E^2}$, of Section 14.4 of [1], where $\mathbf{S}$ is the textbook $\mathbf{S}_\mathrm{t}$, as Section 1.4 of [1] stipulates for formulas from the literature; with the frame vector $\mathbf{S} = \mathbf{E} \times \mathbf{B}$, which Section 12 also calls the Poynting vector, the direction would be reversed. A field point carries two boost factors, $E_6 = E_0 \cosh 2\chi$ against $E = mc^2 \cosh\chi$ for a particle, because the energy density is a component of a second-order tensor and is quadratic in the boost (the standard behaviour of T^{00}): the rest frame moves with β, the energy with $\beta \oplus \beta$, and an observer riding with the energy velocity sees the same E_6 and the reversed Poynting vector. E_0 is a rest-energy density of a field point, not the rest energy of a particle carried by the flux.

2.5 Restoring c: where c^2 enters

Adding the dimensionless G to a derivative along a clock coordinate needs a length ℓ (Section 3 of [1]):

$$\kappa = c/\ell, \qquad \alpha_\pm = \partial_\tau \pm \kappa G. \tag{4}$$

Proposition 2.7 (c^2 is an isotropic weight)**.** *Let the temporal triple carry* $\mathbf{B}$ *and the spatial triple* $\mathbf{E}$*,* $C : V^+ \to V^-$ *the conversion and* $G_C = \left(\begin{smallmatrix} 0 & -C^{-1} \\ C & 0 \end{smallmatrix}\right)$*,* $G_C^2 = -\mathbf{1}$*. (1)* G_C *commutes with the diagonal rotations iff* $C = c\,\mathbf{1}$*. (2) For* $C = c\,\mathbf{1}$ *the forms invariant under the diagonal rotations and under* $e^{\vartheta G_c}$ *are the multiples of* $c^2|\mathbf{B}|^2 + |\mathbf{E}|^2$*. (3)* $(\kappa G)^2 = -c^2/\ell^2$*, and* ℓ *is not fixed.*

Proof. (1) The lower-left block of $[G_C, R \oplus R]$ is $CR - RC$, and Schur's lemma gives $C = c\,\mathbf{1}$: Theorem 3.2 of [1]. (2) For $N = \left(\begin{smallmatrix} \alpha & \gamma \\ \gamma & \beta \end{smallmatrix}\right) \otimes \mathbf{1}$, $G_c^\top N + N G_c = \left(\begin{smallmatrix} 2c\gamma & c\beta - \alpha/c \\ c\beta - \alpha/c & -2\gamma/c \end{smallmatrix}\right) \otimes \mathbf{1}$, zero iff $\gamma = 0$ and $\alpha = c^2\beta$. (3) $G^2 = -\mathbf{1}$. □

Part (1) is Theorem 3.2 of [1], fixed there; part (2) is inherited; verified exactly, including the failure for $C = \mathrm{diag}(c_1, c_2, c_3)$ with distinct entries (d1-5b). Even the weight is a unit convention: $\Phi = \mathrm{diag}(c, c, c, 1, 1, 1)$ conjugates G_c to G and turns $c^2|\mathbf{B}|^2 + |\mathbf{E}|^2$ into $|\Phi X|^2$, so with $\mathbf{B}$ measured as $c\mathbf{B}$ (Heaviside–Lorentz units) c disappears from E_6 and appears only in $\rho_0 = u_0/c^2$ below.

Proposition 2.8 (SI form)**.** *With* $\varepsilon_0\mu_0 c^2 = 1$*,* $z_c := c\mathbf{B} + i\mathbf{E}$*,* $u = \frac{\varepsilon_0}{2}|z_c|^2$*,* $\mathbf{g} = \mathbf{S}_\mathrm{t}/c^2$ *with* $\mathbf{S}_\mathrm{t} = \mathbf{E} \times \mathbf{B}_\mathrm{t}/\mu_0$*,* $q_c := z_c \cdot z_c = c^2|\mathbf{B}_\mathrm{t}|^2 - |\mathbf{E}|^2 - 2ic\,\mathbf{B}_\mathrm{t} \cdot \mathbf{E}$ *and* $u_0 := \frac{\varepsilon_0}{2}|q_c|$*, Proposition 2.4 applied to* z_c *gives*

$$u^2 = c^2|\mathbf{g}|^2 + u_0^2, \qquad \rho_0 := u_0/c^2, \qquad [u] = [c\,\mathbf{g}] = [u_0] = \mathrm{J\,m^{-3}}, \quad [\rho_0] = \mathrm{kg\,m^{-3}}. \tag{5}$$

Inherited (d1-5a); calling ρ_0 a rest-mass density is a reading, since $(u, c\,\mathbf{g})$ is not a four-vector. This is the exact content of "c^2 is the symmetric part": c is the unique conversion between the triples that commutes with their common rotations, the residual symmetry of both E_6 and E_0; in SI units it enters E_6 as the isotropic weight c^2 of $|\mathbf{B}|^2$, which a choice of units removes. It is a unit conversion, not a symmetry, and nothing breaks it: u_0 uses the same c, and $\rho_0 = u_0/c^2$ only converts units.

2.6 Finite configurations: the mass of a whole

Proposition 2.9 (Mass of a whole)**.** *1. For a free field with compactly supported Cauchy data, $(U,\mathbf{P}) = \int (u,\mathbf{S}_{\mathrm{t}})\,d^3x$ is a four-vector and $M^2 := U^2 - |\mathbf{P}|^2 \geq U\int(u-|\mathbf{S}|)\,d^3x \geq 0$, so $M>0$ whenever $q\neq 0$ on a set of positive measure.*

2. In every frame $M \geq \int E_0\,d^3x$, with equality iff the densities $(u,\mathbf{S}_{\mathrm{t}})(x)$ are all proportional.

3. A plane wave has $M=0$. The waves $E_x = a\cos(kz\mp\omega t)$, $cB_{\mathrm{t}y} = \pm a\cos(kz\mp\omega t)$ superpose to $E_x = 2a\cos kz\cos\omega t$, $cB_{\mathrm{t}y} = 2a\sin kz\sin\omega t$; per cell (unit area, length $2\pi/k$) $U = 2\pi\varepsilon_0 a^2/k$ and $\mathbf{P}=0$, so $Mc^2 = U$, although each constituent has $q_c = 0$ and $U_\pm = |\mathbf{P}_\pm|c$. The total has $q_c = -2a^2(\cos 2kz + \cos 2\omega t)$, $\int_{\mathrm{cell}} T^{zz} = U$, $\int_{\mathrm{cell}} T^{xx} = -U\cos 2\omega t$, and the integrated rest energy $\int_{\mathrm{cell}} u_0$ oscillates between $\frac{2}{\pi}U$ and U.

4. With $c=1$ and $f(s) = (1-s^2)^2$ on $|s|\leq 1$, 0 elsewhere, $E_x = f(z-t)+f(z+t)$, $B_{\mathrm{t}y} = f(z-t)-f(z+t)$ has, per unit transverse area, $U = \frac{512}{315}$ and $\mathbf{P}=0$ at all times, so $M=U$, with (U,P_z) a two-vector under boosts along z only; for $|t|\geq 1$ the pulses are disjoint, $q\equiv 0$ and $\int E_0 = 0$.

Proof. (1) Conservation and finite propagation speed make $\int_\Sigma T^{\mu\nu}d\Sigma_\nu$ independent of the spacelike hyperplane Σ [19]; $u-|\mathbf{S}| = E_0^2/(u+|\mathbf{S}|) \geq 0$, and $M^2 = (U-|\mathbf{P}|)(U+|\mathbf{P}|)$. (2) Each $(u,\mathbf{S}_{\mathrm{t}})(x)$ is future causal of length $E_0(x)$ (Proposition 2.5), and future causal vectors obey $|P_1+P_2| \geq |P_1|+|P_2|$, with equality iff proportional. (3) Elementary cell integrals; for the last statement, with $s=\sin^2\omega t$ one has $q_c = 4a^2(s-\cos^2 kz)$, so $u_0 = 2\varepsilon_0 a^2|s-\cos^2 kz|$, whose mean is least at the median $s=\frac{1}{2}$ of the arcsine law, giving $\frac{2}{\pi}U$. (4) $u = f(z-t)^2+f(z+t)^2$ and $S_z = f(z-t)^2 - f(z+t)^2$, the cross terms cancelling, and $q = -4f(z-t)f(z+t)$. □

Items (1) and (3) are inherited, (1) being the statement of Section 14.4 of [1]. The cell in (3) is not isolated: $\int T^{zz} = U$ is the pressure on whatever confines it, so its $(U,\mathbf{P})$ is a four-vector only with the walls included, as for radiation in a box [19]. Items (2) and (4) refute the reading, proposed in the working dictionary of this investigation, that the mass of a whole "emerges from the interference that makes $q\neq 0$": interference suffices for $M>0$ but is not necessary. A whole gets its mass from the non-collinearity of its constituents' energy–momenta, as two photons do; the interference reading is obstructed. M is also blind to orthogonality, by the example of Section 14.4 of [1]: $(\mathbf{E},\mathbf{B}_{\mathrm{t}}) = (0,\mathbf{F})$ and $(\mathbf{F}\sin\vartheta, \mathbf{F}\cos\vartheta)$ have the same $T^{\mu\nu}$ (d1-6a–d1-6c).

2.7 The momentum reading of "E is six-dimensional"

Proposition 2.10 (Splitting the temporal energy lowers the mass)**.** *Let $P = \sum_i (\mathsf{E}_i/c)e_i + \sum_a p_a e_{a+3}$ with $\eta(P,P) = \mu^2c^2$, $\widehat{W}$ the symmetric clock of Proposition 3.5 of [1], $\mathsf{E}_W := (\mathsf{E}_1+\mathsf{E}_2+\mathsf{E}_3)/\sqrt{3}$ and $\mathsf{E}_\perp$ the part of $(\mathsf{E}_1,\mathsf{E}_2,\mathsf{E}_3)$ orthogonal to $(1,1,1)$, the relative-time energies. The mass read by the clock, $m_4^2c^4 := \mathsf{E}_W^2 - |\mathbf{p}|^2c^2$, is*

$$m_4^2c^4 = \mu^2c^4 - |\mathsf{E}_\perp|^2 \leq \mu^2c^4;$$

for the clock e_1, $m_4^2c^4 = \mu^2c^4 - \mathsf{E}_2^2 - \mathsf{E}_3^2$, and $\mathsf{E} = (3,4,0)$, $\mathbf{p} = (5,0,0)$, $c=1$ give $\mu = 0$, $m_4^2 = -16$.

Proof. $|(\mathsf{E}_1,\mathsf{E}_2,\mathsf{E}_3)|^2 = \mathsf{E}_W^2 + |\mathsf{E}_\perp|^2$ and $\mu^2c^4 = c^2\eta(P,P) = |(\mathsf{E}_1,\mathsf{E}_2,\mathsf{E}_3)|^2 - |\mathbf{p}|^2c^2$. □

Obstructed; this restates Proposition 27.1 and Example 27.3 of [1] in clock-covariant form (d1-7). In the momentum reading the differentiation of the temporal energy subtracts from the mass and can make it tachyonic, and under the surplus constraints of Theorem 30.1 of [1], $\mathsf{E}_\perp = 0$, nothing is left to differentiate; in the field-strength reading the motion part adds, $E_6 \geq E_0$. Only the latter carries the structure the user describes.

2.8 Three candidates for m in the arena's own structures

Proposition 2.11 (The mass form acts on the frame as parity)**.** *Let $T = e_1e_2e_3$ and $P := \mathrm{Ad}_T|_V$. (1) $P = \eta$ fixes $\mathbf{B}$ and reverses $\mathbf{E}$: it is parity on the frame; it commutes with the L_i and anticommutes with G and the K_i, so it normalises $\mathfrak{l}_G$ without being $\mathfrak{l}_G$-invariant. (2) PX corresponds to $\bar{z}$ and $q(z) = h(Pz, z)$, so $E_6 = \frac{1}{2}h(z, z)$ and $E_0 = \frac{1}{2}|h(Pz, z)| = \frac{1}{2}\max_\vartheta \eta(e^{\vartheta G}X, e^{\vartheta G}X)$. (3) As a Grassmann mass kernel on spinors T is invariant under all of $\mathfrak{so}(3,3)$, $\mathfrak{l}_G$ included, in the sense of Definition 18.2 of [1], and breaks nothing; only its sign changes, under conjugation by a spatial generator e_4, e_5, e_6.*

Proof. (1) Lemma 3.4 of [1]; $\eta G\eta = -G$ blockwise, the L_i preserve each triple and the K_i exchange them. (2) $z(\eta X) = \mathbf{B} - i\mathbf{E}$, $h(\bar{z}, z) = \sum_j z_j^2$, and $\eta(e^{\vartheta G}X, e^{\vartheta G}X) = \mathrm{Re}(e^{2i\vartheta}q)$. (3) By Lemma 2.1 of [1], e_ae_b commutes with $e_1e_2e_3$ when a, b have the same sign and anticommutes otherwise, and $T^\top = -T$; this is Corollary 18.4 of [1]. □

The identities are inherited or restate [1] (d1-8a); their consequence for the dictionary is negative, but as a type mismatch: "$m \leftrightarrow T$" fails on frame vectors, where the adjoint action of T is parity, while as a kernel on spinors T is $\mathfrak{l}_G$-invariant, (3). The Dirac mass lives on the interior reading, which Proposition 12.6 of [1] excludes on the same six directions, and even there it breaks nothing. What survives is a reading: the rest energy is the overlap of a field with its mirror image, and P, the arena's splitting into triples, is the real structure whose stabiliser in $U(3)$ is $U(1) \cdot O(3)$ (Proposition 2.2(4)).

Proposition 2.12 (The interior energy is not a rest energy)**.** *For the split-form interior with the binding root (Theorem 16.2 of [1]) every solution is $z(\tau) = c_-e^{-i\kappa\tau} + c_+e^{i\kappa\tau}$, and $\ddot{X} = -\kappa^2 X$ does not involve m. The Noether energy $\mathcal{E} = 2m\kappa^2\,\mathrm{Re}(c_- \cdot c_+)$ (Proposition 16.5(iii) of [1]) is indefinite and vanishes identically on the duality rotations of both senses, $z = e^{-i\kappa\tau}c_-$ (the orbits without momentum, $c_+ = 0$, of Proposition 16.5(i)) and $z = e^{i\kappa\tau}c_+$ (which carry the momentum $2i\kappa c_+e^{i\kappa\tau} \neq 0$); on both, the frame field's E_6 and E_0 are constant. In general*

$$|z|^2 = |c_-|^2 + |c_+|^2 + 2\,\mathrm{Re}\big(h(c_-, c_+)e^{2i\kappa\tau}\big), \qquad q(z) = e^{-2i\kappa\tau}c_- \cdot c_- + 2c_- \cdot c_+ + e^{2i\kappa\tau}c_+ \cdot c_+,$$

so the frame field's E_6 is constant iff $h(c_-, c_+) = 0$, and its E_0 oscillates: $c_- = e_1$, $c_+ = e_2$ give $\mathcal{E} = 0$, $E_6 = 1$ and $E_0 = |\cos 2\kappa\tau|$.

Proof. With $\eta(GX, GX) = -\eta(X, X) = -\,\mathrm{Re}\,q$, $\mathcal{E} = \frac{m}{2}\,\mathrm{Re}(\dot{z} \cdot \dot{z} + \kappa^2 z \cdot z) = 2m\kappa^2\,\mathrm{Re}(c_- \cdot c_+)$; the momentum is $\dot{z} + i\kappa z = 2i\kappa c_+e^{i\kappa\tau}$; expand $\bar{z} \cdot z$ and $z \cdot z$. □

Obstructed as an identification, verified symbolically (d1-9): in the interior m is an overall factor and an input independent of κ, the pure duality motion of either sense carries zero energy, and, with X a length and $\kappa = c/\ell$, $\mathcal{E} = mc^2 \cdot 2\,\mathrm{Re}(c_- \cdot c_+)/\ell^2$ has the form mc^2 times a dimensionless amplitude, but the amplitude is indefinite, vanishes on the pure duality orbits, and is selected by nothing, since ℓ is free. The entry "$c^2 \leftrightarrow \kappa G$" is right that κG lies in the factorising summand of Theorem 17.3 of [1], but nothing in the interior turns κ^2 into a rest energy.

Theorem 2.13 (In the closure the rest energy of the momentum field breaks the duality translation)**.** *In the Lie–Poisson closure of Proposition 17.8 of [1], $H = h^2/(2I_0) + \frac{1}{2}p^\top Np$, with momentum field $\pi = \eta^{-1}p$:*

1. *for every N, $E_6(\pi) = \frac{1}{2}|p|^2$, $E_0(\pi) = \frac{1}{2}|w(p)|$ and $\mathbf{S}(\pi)$ are Casimirs of the momentum field (constant already in the clamped interior, where $p = e^{-\kappa\tau G}p_0$);*
2. *for Lorentz-class $N = (\nu_1\eta^{-1} + \nu_2 G\eta^{-1})/m$ the reduced Hamiltonian is*

$$H = \frac{h^2}{2I_0} + \frac{|\nu_1 + i\nu_2|\,E_0(\pi_0)}{m}\cos(2\theta + \varphi_0), \qquad R_0e^{i\varphi_0} := (\nu_1 + i\nu_2)\,w(p_0),$$

so the rotor's translation $\theta \mapsto \theta + \text{const}$, the duality rotation of the momentum field, is broken with amplitude $|\nu_1 + i\nu_2|E_0(\pi_0)/m$; for $(\nu_1, \nu_2) \neq (0,0)$ the rate $\kappa = h/I_0$ is constant and nonzero iff $E_0(\pi_0) = 0$ and $h(0) \neq 0$ (Corollary 17.14 of [1]); for $E_0(\pi_0) > 0$ and $(\nu_1, \nu_2) \neq (0,0)$ it is constant only at the equilibria $h(0) = 0$, $2\theta(0) + \varphi_0 \in \pi\mathbb{Z}$, where it vanishes; for $(\nu_1, \nu_2) = (0,0)$ it is constant for all data;

3. *for the duality-class $N = \mathbf{1}/m$ the potential $\frac{1}{2}p^\top Np$ (the "frame energy" of [1]) is $E_6(\pi)/m$, the undivided energy of the momentum field, and the rate is conserved for all data;*

4. *the frame field is not conserved: with a free rotor in the rotating regime, Lorentz-class N, $(\nu_1, \nu_2) \neq (0,0)$ and $E_0(\pi_0) > 0$, the frame field's E_6 and E_0 grow without bound.*

Proof. (1) Along the flow $p = e^{-\theta G}p_0$, so $\pi = e^{\theta G}\pi_0$ and $z(\pi) = e^{i\theta}z(\pi_0)$, a duality rotation, which preserves E_6, E_0, $\mathbf{S}$; and $|p| = |\pi|$ as η is orthogonal. (2) is Theorem 17.13 of [1], $R_0 = |\nu_1 + i\nu_2|\,|w(p_0)| = 2|\nu_1 + i\nu_2|E_0(\pi_0)$, and $\dot{h} = (R_0/m)\sin(2\theta + \varphi_0)$: if $R_0 > 0$ and h is constant, $\sin(2\theta + \varphi_0) \equiv 0$, so θ is constant and $h = I_0\dot{\theta} = 0$; conversely these points are equilibria; the rest is Corollary 17.14. (3) $N = \mathbf{1}/m$ commutes with G, so $\dot{h} = -p^\top GNp = 0$. (4) By Theorem 17.16 of [1] the frame x (with $X = e^{-\theta G}x$, so $E_6(X) = E_6(x)$, $E_0(X) = E_0(x)$) is displaced by $\Delta x = \lambda e^{-\varphi_0 G}Np_0$, $\lambda > 0$, in each period of the rate, and $x = x_0 + n\Delta x$ after n periods, since $\dot{\theta}$ and $\dot{x} = e^{\theta G}Ne^{-\theta G}p_0$ depend on θ with period π. Since G acts as i, $z(Np_0) = (\nu_1 + i\nu_2)z(\pi_0)/m$ and $q(\Delta x) = \lambda^2 e^{-2i\varphi_0}(\nu_1 + i\nu_2)^2 q(\pi_0)/m^2 \neq 0$, so $|q(x)|$ and $|x|^2$ grow like n^2. □

Parts (1)–(3) restate Proposition 17.8, Theorem 17.13 and Corollary 17.14 of [1], fixed there, with $\frac{1}{2}|w|$ renamed $E_0(\pi)$; part (4) is a direct consequence of Theorem 17.16. Nothing in the theorem is new. Verified by integrating the full system (conservation to 10^{-9}), at the exact equilibria $\pi_0 = e_1$, $h(0) = 0$, $(\nu_1, \nu_2) = (\pm 1, 0)$, where the rate is constant with $E_0(\pi_0) = \frac{1}{2}$, and with the frame included, where the frame field's E_0 grows as n^2 (d1-10a, d1-10b). Read through a dictionary that is interpretive, not a theorem (the clamped rate $\kappa = c/\ell$, with ℓ free, for the user's c, and $E_0(\pi)$, which has the dimension of a momentum squared, for a rest energy), the momentum field's rate keeps its symmetry (constancy, invariance under the rotor's duality translation) when the momentum field is null, its rest part $E_0(\pi)$ is the amplitude that breaks it, and $E_6(\pi)$ is a Casimir and breaks nothing. But the breaking is explicit, from an inertia that Theorem 17.11 of [1] forbids from also conserving the rate; what is broken is the constancy of the rotor rate, a duality property, not Lorentz invariance and nothing carried by c; and with the undivided energy as Hamiltonian ($N = \mathbf{1}/m$) the rate is constant at the price of the frame drift of Theorem 17.12.

2.9 What transport by b does to the two energies

An admissible Yang–Mills generator is compact: Theorem 2.3 and Corollary 2.4 of [1] restrict gauge generators to $\mathfrak{so}(3)_T \oplus \mathfrak{so}(3)_S \subset \mathfrak{so}(3,3)$, the rotations of the two triples. Write such a b as $(A + D, A - D)$, $A + D$ acting on the temporal triple and $A - D$ on the spatial one; the diagonal part (A, A) lies in $\mathfrak{l}_G$ and the anti-diagonal part $(D, -D)$ in $\mathfrak{p}_G$. Boost-valued connections are legitimate metric connections (Proposition 16.7 of [1]) but not admissible Yang–Mills potentials.

Proposition 2.14 (What transport preserves)**.** *(1) $Y \in \mathfrak{so}(3,3)$ preserves $\mathbf{E} \cdot \mathbf{B}$ for all X, equivalently q, equivalently E_0, iff $[Y, G] = 0$, that is $Y \in \mathfrak{l}_G$. (2) A compact $b = (A + D, A - D)$ preserves E_6 and $\operatorname{Re} q = |\mathbf{B}|^2 - |\mathbf{E}|^2$; it preserves E_0 for all X iff $D = 0$; the anti-diagonal rotation through $\frac{\pi}{4}$ about e_3 carries the null field $(1, i, 0)$ to the field $\frac{1+i}{\sqrt{2}}(1, 1, 0)$ at rest, E_0 going from 0 to 1 at $E_6 = 1$. (3) At three constituents the part of $\mathfrak{ps}_W$ commuting with G_3, $\mathfrak{so}(4)_\Delta \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ of Theorem 17.21 of [1], consists of paired rotations, so it preserves the $\mathfrak{so}(9, \mathbb{C})$ invariants $\frac{1}{2}|Z|^2 = \frac{1}{2}\sum_k |z^{(k)}|^2$ and $q_3(Z) = \sum_k z^{(k)} \cdot z^{(k)}$, hence $\frac{1}{2}|q_3|$. (4) It preserves neither the constituents' rest energies nor that of their summed density: the rotation through*

$\pi/4$ generated by $\frac{1}{2}\mathrm{rot}(7,13)$ maps $(z^{(2)}, z^{(3)}) = ((1,i,0),(1,0,0))$ to $((0,i,0),(\sqrt{2},0,0))$, keeping $\frac{1}{2}|Z|^2 = \frac{3}{2}$ and $q_3 = 1$, while $\sum_k E_0^{(k)}$ goes from $\frac{1}{2}$ to $\frac{3}{2}$ and $(\sum_k E_6^{(k)}; \sum_k \mathbf{S}_{\mathrm{t}}^{(k)})$ from $(\frac{3}{2}; 0,0,1)$ to $(\frac{3}{2}; 0,0,0)$, its Lorentz length from $\frac{\sqrt{5}}{2}$ to $\frac{3}{2}$.

Proof. (1) For $Y \in \mathfrak{so}(3,3)$, $Y^\top \eta G + \eta G Y = \eta[G,Y]$. If Y preserves $\mathrm{Re}\, q$ and $|q|$, then $\mathrm{Im}\, q \cdot \delta(\mathrm{Im}\, q) = 0$, so $\delta(\mathrm{Im}\, q) = 0$ since polynomials form an integral domain. (2) b rotates each triple orthogonally, so it preserves $|\mathbf{B}|^2$ and $|\mathbf{E}|^2$; (A,A) commutes with G and $G(D,-D)G^{-1} = (-D,D)$, so by (1) E_0 is preserved for all X iff $D = 0$; the example is $\mathbf{B} = (1,0,0) \mapsto \frac{1}{\sqrt{2}}(1,1,0)$, $\mathbf{E} = (0,1,0) \mapsto \frac{1}{\sqrt{2}}(1,1,0)$. (3) By Theorem 17.19 of [1] the elements of $\mathfrak{so}(4)_\Delta$ are paired rotations, with $S = 0$ and R real antisymmetric, hence they preserve $|Z|^2$ and q_3. (4) Direct, with $[e_A e_B, e_C] = 2\eta_{BC} e_A - 2\eta_{AC} e_B$, checked against the realisation. □

Verified exactly (d1-11a, d1-11b). Parts (1) and (2) are inherited linear algebra, and they cut both ways. Pointwise, the anti-diagonal part of an admissible b does turn a null field into one at rest at fixed undivided energy, while the diagonal part, a rotation in $\mathfrak{l}_G$, changes no invariant. But the anti-diagonal part is the part that fails to commute with G: along a worldline it cannot be gauged away, unlike the diagonal part (Larmor's theorem [18]), and in the interior it produces Jordan blocks and secular growth unless screened by a diagonal part at equal rates, outside one exceptional ratio (Theorem 5.10); nothing here selects an amount of rest energy. Parts (3)–(4) are an obstruction for the $\mathfrak{so}(9,\mathbb{C})$ invariants only: no transport valued in the paired rotations changes $\frac{1}{2}|Z|^2$ or q_3. They are not one for the rest energy of the whole in the diagonal Lorentz reading, which the example changes, and $\mathrm{rot}(7,13)$ does not commute with the diagonal rotations; which is physical depends on the Lorentz action at three constituents, which is open (Theorem 17.20 of [1]); and $\mathfrak{so}(4)_\Delta$ is the part of $\mathfrak{ps}_W$ commuting with G_3, not the only part with bounded orbits (Section 5). Here b appears only as a constant transport element or a holonomy, with no field strength or b-field energy, so "where the b-field belongs" is, at this level, interpretive.

2.10 Assessment

Most of what was asked is inherited (Table 1), ordinary electrodynamics in the arena's variables, which by Corollary 4.4 of [1] predicts nothing. Nothing in this section is fixed beyond what [1] already fixes: one conversion constant (Theorem 3.2), the relation $G = -\mathcal{K} \circ \mathrm{Ad}_{e_1e_2e_3}$ that makes the binding root orthogonal for the arena's Euclidean form (Lemma 3.4), and the duality pendulum with $R_0 = 2|\nu_1 + i\nu_2|E_0(\pi_0)$ (Theorem 17.13). The uniqueness of E_6 and the stabiliser $U(1)\cdot O(3)$ are inherited. Five readings are obstructed. The c^2 clause gets less support than its wording suggests: in E_6 the weight c^2 is removed by the unit choice $\mathbf{B} \to c\mathbf{B}$; in the interior c^2 enters as $\kappa^2 = c^2/\ell^2$ and the Noether energy is mc^2 times an indefinite, unselected amplitude; and in the closure what $E_0(\pi)$ breaks is the constancy of the rotor rate, not anything carried by c.

Corrected so that it is true, the user's sentence becomes the following interpretive statement (d1-8c). The undivided energy is $E_6 = \frac{1}{2}|z|^2$, the one rotation- and duality-invariant quadratic energy of a six-component frame vector, with pointwise symmetry $U(3)$. It differentiates internally, with no interaction, as $E_6^2 = E_0^2 + |\mathbf{S}|^2$, into a Lorentz-invariant rest energy $E_0 = \frac{1}{2}|z \cdot z|$ and a motion part $|\mathbf{S}|$. What differentiates is the arena's splitting into a temporal and a spatial triple, the real structure $\mathrm{Ad}_{e_1e_2e_3}$, which reduces $U(3)$ to $U(1)\cdot O(3)$ (a symmetry of the pointwise form, not of any dynamics); nothing Lorentzian is broken. c is a conversion, not a symmetry, its weight in E_6 is a unit convention, and $m = E_0/c^2$ is a rest-mass density of a field point. The reading fails in three respects: a field point is not a particle (two boost factors; the mass of a composite is not interference); in the closure only the momentum field's energies are conserved, while the frame field's oscillate or grow; and b generates no selected amount of m: pointwise its anti-diagonal part moves a field between null and rest at fixed E_6, and at three

claim	statement	class
d1-1a	E_6 is the unique rotation- and duality-invariant energy	inherited (Thm. 17.11(iii) of [1])
d1-1b	pointwise symmetry $U(3)$; $\mathfrak{u}(3) \cap (\mathfrak{l}_G \oplus \mathbb{R}G) = \mathfrak{so}(3) \oplus \mathbb{R}G$	inherited
d1-2a,b	E_0 complete invariant of Lorentz $\times$ duality, E_6 of $U(3)$	inherited
d1-3a,b	$E_6^2 = E_0^2 + \lvert\mathbf{S}\rvert^2$; Rainich identity	inherited
d1-4a–c	$E_0 = \min E_6$; $E_6 = E_0 \cosh 2\chi$; rest-frame velocity with $\mathbf{S}_\mathrm{t}$	inherited
d1-5a	$u^2 = c^2\lvert\mathbf{g}\rvert^2 + u_0^2$	inherited
d1-5b	$C = c\,\mathbf{1}$; the weight c^2 of $\lvert\mathbf{B}\rvert^2$	fixed in [1] (Thm. 3.2); weight conventional
d1-6a,c	standing wave $Mc^2 = U$; M blind to orthogonality	inherited
d1-6b	the mass of a whole is not interference; $M \geq \int E_0$	obstructed
d1-7	the momentum reading lowers the mass	obstructed (restates Prop. 27.1)
d1-8a	T acts on the frame as parity; "$m \leftrightarrow T$" fails there	obstructed (type mismatch)
d1-8b	$\mathrm{Stab}_{U(3)}(E_0) = U(1) \cdot O(3)$	inherited; reduction reading interpretive
d1-9	the interior Noether energy is not a rest energy	obstructed
d1-10a	closure: $E_0(\pi)$ is the duality-breaking amplitude	restates [1] Thm. 17.13; reading interpretive
d1-10b	the frame field's E_6, E_0 grow under the free-rotor drift	restates [1] Thm. 17.16 (consequence)
d1-11a	compact b changes E_0 only through its anti-diagonal part	inherited
d1-11b	$\mathfrak{so}(4)_\Delta$ preserves $\lvert Z\rvert^2$, q_3; changes $\sum_k E_0^{(k)}$ and the summed density's length	obstructed (for q_3, $\lvert Z\rvert^2$ only)
d1-8c	"E undivided, m differentiated, c^2 symmetric"	interpretive

Table 1: The claims of Section 2 and their classes; identifiers refer to the verification script.

constituents $\mathfrak{so}(4)_\Delta$ moves rest energy between constituents at fixed $\frac{1}{2}|Z|^2$ and q_3 while changing the Lorentz length of the summed density. Whether some dynamics selects the rest frame or the value of E_0 is open; nothing here does.

3 What a mass breaks: duality, chirality, and the two senses of rotation

The hypothesis reads m as what breaks a symmetry of which c^2 is the symmetric part. The prior question is which symmetry a mass actually breaks in the arena, and which it leaves alone. The arena has two things called mass. In the field-strength reading (Theorem 12.3 of [1]) a vector $X = \sum_i (B_i e_i + E_i e_{i+3})$, $z(X) = \mathbf{B} + i\mathbf{E}$, has energy density $u(X) = \frac{1}{2}|z|^2$ (the E_6 of Section 2), complex invariant $q(X) = z \cdot z$ and rest-frame energy density $E_0(X) = \frac{1}{2}|q|$ (Section 14 of [1]). In the interior these pointwise functions are evaluated on two fields, the frame field $X(\tau)$ and the momentum field $\pi = \eta^{-1}p$; we always say which. In the Dirac reading the mass is the kernel $T = e_1e_2e_3 = \gamma^0 J$ of the Grassmann bilinear (Proposition 4.5 and Corollary 18.4 of [1]). In brief: a mass breaks no Lorentz generator and, in the Dirac reading, no generator of $\mathfrak{so}(3,3)$; it breaks the duality rotation generated by the binding root G, in the orbit-space sense of Proposition 3.1 and Theorem 3.4, and the chiral and Pauli–Gürsey symmetries of the massless kinetic term, down to $\mathfrak{u}(1)_J$; and the two senses of the Dirac rest rotation are exchanged

by charge conjugation, those of the frame's binding rotation by parity. Units are $\hbar = c = 1$ unless a symbol is shown. Every finite-dimensional statement proved here was checked, exactly or on dyadic-exact float64 data, by `verify/d2_symmetry_breaking.py`, and independently by `verify/x2_check.py`; the checks are listed in Table 4.

3.1 Duality and the rest energy of a field

Write $D_\vartheta = e^{\vartheta G} = \cos\vartheta\,\mathbf{1} + \sin\vartheta\,G$ (since $G^2 = -\mathbf{1}$). As G acts on z as multiplication by i (proof of Theorem 12.1 of [1]), D_ϑ is the duality rotation.

Proposition 3.1 (Duality on the invariants)**.** *For every X and ϑ:*

(i) $z(D_\vartheta X) = e^{i\vartheta} z(X)$, $u(D_\vartheta X) = u(X)$, $q(D_\vartheta X) = e^{2i\vartheta} q(X)$;

(ii) $D_\vartheta^\top D_\vartheta = \mathbf{1}$ *and* $D_\vartheta^\top \eta\, D_\vartheta = \cos 2\vartheta\,\eta + \sin 2\vartheta\,\eta G$*: D_ϑ preserves the positive form for every ϑ, and η only for $\vartheta \in \pi\mathbb{Z}$;*

(iii) if $q(X) = 0$ every ϑ fixes q; if $q(X) \neq 0$ exactly $\vartheta \in \pi\mathbb{Z}$ do;

(iv) if $q \neq 0$, there are Lorentz coordinates in which $z = (f_m + i f_e)\hat{e}_1$ with f_m, f_e real; then $q = (f_m^2 + f_e^2)e^{2i\varphi}$ with $\varphi = \arg(f_m + i f_e)$, so $\tan\varphi = f_e/f_m$: $\arg q$ is twice the Rainich complexion, and $D_{-\varphi}$ makes the field purely magnetic, $z = \sqrt{f_m^2 + f_e^2}\,\hat{e}_1$.

Proof. (i) is $G \leftrightarrow i$ and the quadratic character of q. (ii) G is antisymmetric, so D_ϑ is orthogonal; G is η-self-adjoint, so $G^\top = \eta G \eta$, and expanding $(\cos\vartheta + \sin\vartheta\,\eta G\eta)\eta(\cos\vartheta + \sin\vartheta\,G)$ with $G^2 = -\mathbf{1}$ gives the formula; η and ηG are linearly independent. (iii) is $e^{2i\vartheta} - 1 = 2i\sin\vartheta\,e^{i\vartheta}$. (iv) Proposition 12.12 of [1] supplies coordinates in which the field is an electric plane of strength f_e orthogonal to a magnetic one of strength f_m; then $e^{-i\varphi}(f_m + i f_e) = |f_m + i f_e|$. The principal branch of $\arctan(f_e/f_m)$ would give $-|f_m + i f_e|$ for $f_m < 0$. □

The proposition is inherited: (i) and (iii) are the classical duality rotation of the Maxwell invariants [29, 2, 5], restated from Section 14.3 of [1]; (iv) restates Proposition 12.12 of [1] with the branch of φ fixed; and (ii), recorded in Section 14.3 and after Theorem 17.23 of [1], is the textbook rotation of $\frac{1}{2}F_{\mu\nu}F^{\mu\nu}$ into $\frac{1}{2}F_{\mu\nu}{\star}F^{\mu\nu}$ under the isometry $X \mapsto F_X$ of Theorem 12.3 of [1], which carries G to the Hodge star. What the arena adds (Theorems 12.3, 16.2 and 17.3 of [1]) is that the generator of this rotation is the binding root, which lies in the η-self-adjoint summand of $\mathrm{End}(V) = \mathfrak{so}(3,3) \oplus \{A = A^\dagger\}$. For $\vartheta \notin \pi\mathbb{Z}$, $D_\vartheta \notin O(3,3)$ lifts to no element of $\mathrm{Pin}(3,3)$ (as noted after Theorem 17.23 of [1]). Duality is a symmetry of the positive form u; the split-form interior Lagrangian is rotated by it into its ηG partner.

Proposition 3.2 (The rest energy is the doubly invariant modulus)**.** *Let $\mathbf{S} = \mathbf{E} \times \mathbf{B}$ be the frame Poynting vector; as $\mathbf{B}_\mathrm{t} = -\mathbf{B}$ (Section 1.4 of [1]), $\mathbf{S} = -\mathbf{E} \times \mathbf{B}_\mathrm{t} = -\mathbf{S}_\mathrm{t}$ is minus the textbook one. Then $u^2 - |\mathbf{S}|^2 = \frac{1}{4}|q|^2$, so $u \geq E_0$, with equality exactly when $\mathbf{S} = 0$. Of u, $\mathrm{Re}\,q$, $\mathrm{Im}\,q$ and E_0, the first is invariant under $U(1)_G$ but not under $\mathfrak{l}_G$, the next two under $\mathfrak{l}_G$ but not under $U(1)_G$, and $E_0 = \frac{1}{2}|q|$ under both.*

Proof. $u^2 - |\mathbf{E}\times\mathbf{B}|^2 = \frac{1}{4}(|\mathbf{B}|^2 + |\mathbf{E}|^2)^2 - |\mathbf{E}|^2|\mathbf{B}|^2 + (\mathbf{E}\cdot\mathbf{B})^2 = \frac{1}{4}(|\mathbf{B}|^2 - |\mathbf{E}|^2)^2 + (\mathbf{E}\cdot\mathbf{B})^2$. The six generators of $\mathfrak{l}_G$ commute with G and lie in $\mathfrak{so}(3,3)$, so $\eta(X, AX) = \eta(X, GAX) = 0$; a boost K_i is symmetric for the positive form and $X^\top K_i X \neq 0$ for generic X. The duality statements are Proposition 3.1. □

This restates Section 14.3 of [1] and is inherited; with units restored it reads $u^2 - c^2|\mathbf{g}|^2 = E_0^2$, $\mathbf{g} = \mathbf{S}_\mathrm{t}/c^2$, the field form of $E^2 - p^2c^2 = m^2c^4$. Its reading for the hypothesis is interpretive. Lorentz orbits of nonzero vectors are labelled by $q \in \mathbb{C}$, since $\mathfrak{l}_G \cong \mathfrak{so}(3,\mathbb{C})$ is transitive on each level set of q minus the origin (proof of Theorem 14.5 of [1]); duality rotates the label through 2ϑ, fixes only the origin, and the stabiliser of a nonzero label is $\{D_0, D_\pi\} = \{\pm\mathbf{1}\}$. A nonzero

rest energy thus breaks $U(1)_G$ to $\mathbb{Z}_2$ on the space of Lorentz orbits, and $\arg q$ is the angle along the broken direction. But the breaking is a property of a configuration, not of a vacuum (u is minimised by the zero field and no potential selects $|q| \neq 0$, Section 14.3 of [1]), and the broken symmetry is not one of the interior Lagrangian. What the interior dynamics does with duality is sharper. Every solution of $\ddot{X} = -\kappa^2 X$ (Theorem 16.2 of [1]) is $z(\tau) = c_- e^{-i\kappa\tau} + c_+ e^{i\kappa\tau}$, $c_\pm \in \mathbb{C}^3$; the momentum is $p = m\eta(\dot{X} + \kappa G X)$, so the momentum field is $\pi = \eta^{-1}p = m\alpha_+ X$ with $\alpha_\pm = \partial_\tau \pm \kappa G$ (eq. (7) of [1]).

Proposition 3.3 (Frame field and momentum field in the interior)**.** *Along every solution:*

(i) $\mathcal{E}_\mathbb{C} := \frac{m}{2}(q(\dot{z}) + \kappa^2 q(z)) = 2m\kappa^2\, c_- \cdot c_+$ *is conserved, and the Noether energy* $\frac{m}{2}\eta(\dot{X},\dot{X}) - \frac{m\kappa^2}{2}\eta(GX,GX)$ *is* $\operatorname{Re}\mathcal{E}_\mathbb{C}$*; it vanishes when either sense is absent. A constant duality rotation maps solutions to solutions and multiplies* $\mathcal{E}_\mathbb{C}$ *by* $e^{2i\vartheta}$*.*

(ii) The momentum field is $z(\pi) = 2im\kappa\, c_+ e^{i\kappa\tau}$*, i.e.* $\pi(\tau) = e^{\kappa\tau G}\pi_0$*: it moves by pure duality, and* $u(\pi) = \frac{1}{2}|p|^2 = 2m^2\kappa^2|c_+|^2$*,* $E_0(\pi) = 2m^2\kappa^2|c_+ \cdot c_+|$ *and* $\mathbf{S}(\pi)$ *are constant.*

(iii) The frame field has $u(X) = \frac{1}{2}(|c_-|^2 + |c_+|^2) + \operatorname{Re}(\bar{c}_- \cdot c_+\, e^{2i\kappa\tau})$*, constant exactly when* $\bar{c}_- \cdot c_+ = 0$*, and* $q(X) = c_- \cdot c_-\, e^{-2i\kappa\tau} + 2c_- \cdot c_+ + c_+ \cdot c_+\, e^{2i\kappa\tau}$*, so* $E_0(X)$ *oscillates in general.*

(iv) Both pure senses, $z = e^{-i\kappa\tau}c_-$ *and* $z = e^{i\kappa\tau}c_+$*, are duality rotations of the frame field, with Noether energy* 0 *and constant* $u(X)$ *and* $E_0(X)$*. On the first, the zero-momentum orbits,* $p = 0$ *while* $u(X) = \frac{1}{2}|c_-|^2 > 0$*; on the second* $|p|^2 = 8m^2\kappa^2 u(X)$*.*

Proof. $\dot{z} + i\kappa z = 2i\kappa c_+ e^{i\kappa\tau}$ is $z(\alpha_+ X)$, since G acts as i; with $|\pi| = |p|$ (η is orthogonal) this gives (ii), and $\overline{z(\pi)} \times z(\pi)$, $\overline{z(\pi)} \cdot z(\pi)$ and $|z(\pi) \cdot z(\pi)|$ are invariant under $z(\pi) \mapsto e^{i\kappa\tau}z(\pi)$. In $q(\dot{z}) + \kappa^2 q(z)$ the terms in $e^{\mp 2i\kappa\tau}$ cancel, leaving $4\kappa^2 c_- \cdot c_+$; $\eta(GX,GX) = -\eta(X,X)$ identifies the Noether energy with $\frac{m}{2}\operatorname{Re}(q(\dot{z}) + \kappa^2 q(z))$; q has duality charge two. (iii) is the expansion of $|z|^2$ and $z \cdot z$, and (iv) is (i)–(iii) at $c_+ = 0$ or $c_- = 0$. Each identity was also checked exactly as a polynomial identity in the real and imaginary parts of $c_\pm$, modulo $\cos^2\kappa\tau + \sin^2\kappa\tau = 1$. □

Items (i) and (ii) restate Proposition 16.5(iii) and (ii) of [1] (fixed there), with the momentum field made explicit; the zero-momentum half of (iv) restates Proposition 16.5(i). Item (iii) and the other half of (iv) are added and fixed. The proposition corrects the working dictionary. The quantities that the interior conserves belong to the momentum field: $u(\pi)$, $E_0(\pi)$ and $\mathbf{S}(\pi)$ are constant already in the clamped interior, because π moves by a duality rotation, and they are the Casimirs of the closure (Theorem 4.1). The same functions of the frame field are not conserved: $u(X)$ moves unless $\bar{c}_- \cdot c_+ = 0$, and $E_0(X)$ oscillates. Nor is $u(X)$ a function of $|p|^2$: on the zero-momentum orbits $|p|^2 = 0$ while $u(X) > 0$, on the duality rotations of the other sense $|p|^2 = 8m^2\kappa^2 u(X)$. Duality acts along both pure senses, so duality orbits are not zero-momentum orbits; the zero-momentum sense is singled out by the momentum, not by duality. And the interior's conserved energy is not undivided: it is the real part of a constant of duality charge two, carried by neither sense alone.

3.2 Little groups: field points and momenta

On z, $\boldsymbol{\omega}\cdot L + \mathbf{v}\cdot K \in \mathfrak{l}_G$ acts as $z \mapsto (\boldsymbol{\omega} + i\mathbf{v}) \times z$, that is $\delta\mathbf{B} = \boldsymbol{\omega}\times\mathbf{B} - \mathbf{v}\times\mathbf{E}$, $\delta\mathbf{E} = \boldsymbol{\omega}\times\mathbf{E} + \mathbf{v}\times\mathbf{B}$ (Theorem 12.3 of [1]): $\mathfrak{l}_G = \mathfrak{so}(3,\mathbb{C})$ acting by complex cross products, and G acts as i.

Theorem 3.4 (Stabilisers of field points)**.** *Let* $z \neq 0$*.*

(i) $\operatorname{Stab}_{\mathfrak{l}_G}(z) = \{(\zeta z)\times\cdot : \zeta \in \mathbb{C}\}$ *is abelian of real dimension two. If* $q \neq 0$ *and* $\hat{z} = z/\sqrt{q}$*, then* $R = \hat{z}\times\cdot$ *has* $R^3 = -R$ *(a rotation) and* $(GR)^3 = GR$ *(a boost) about the common complex axis* $\hat{z}$*:* $\mathfrak{so}(2) \oplus \mathfrak{so}(1,1)$*. If* $q = 0$*, every element* N *of the stabiliser has* $N^3 = 0 \neq N^2$*: null rotations,* $\mathbb{R}^2$*.*

(ii) In $\mathfrak{l}_G \oplus \mathbb{R}G$ *a non-null* z *has the same stabiliser, with no* G *component. A null* $z = \mathbf{n}_1 + i\mathbf{n}_2$ *has a three-dimensional stabiliser* $\cong \mathfrak{e}(2)$*: the null rotations and* $L_{\mathbf{n}} + G$*, with* $\mathbf{n} = \mathbf{n}_1 \times \mathbf{n}_2/|\mathbf{n}_1|^2 = \mathbf{B} \times \mathbf{E}/|\mathbf{B} \times \mathbf{E}|$ *the propagation axis, acting on the null rotations with eigenvalues* $\pm i$*. For* $z = (1, i, 0)$*,* $(L_3 + G)z = 0$ *and* $e^{\vartheta G} z = e^{-\vartheta L_3} z$*.*

(iii) In $\mathfrak{l}_{\text{int}}$ *on the interior four-plane* $\langle e_1, e_4, e_5, e_6 \rangle$*,* $\text{Stab}(e_1) = \mathfrak{so}(3)$ *and* $\text{Stab}(e_1 + e_6) = \mathfrak{e}(2)$*, both three-dimensional.*

Proof. (i) $w \times z = 0$ iff $w \in \mathbb{C}z$. From $(z\times)^2 w = z(z \cdot w) - qw$, $(z\times)^3 = -q\,(z\times)$; with $\hat{z} \cdot \hat{z} = 1$ this is $R^3 = -R$, and $G \leftrightarrow i$ gives $(GR)^3 = GR$; for $q = 0$, $(z\times)^3 = 0 \neq (z\times)^2$. (ii) The condition is $w \times z + i\vartheta z = 0$; dotting with z gives $i\vartheta q = 0$. For null z, $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$ and $|\mathbf{n}_1| = |\mathbf{n}_2|$ give $\mathbf{n} \times \mathbf{n}_1 = \mathbf{n}_2$, $\mathbf{n} \times \mathbf{n}_2 = -\mathbf{n}_1$, so $\mathbf{n} \times z = -iz$, $(L_{\mathbf{n}} + G)z = 0$, and $[L_{\mathbf{n}}, z\times\cdot] = (\mathbf{n} \times z)\times\cdot = -i(z\times\cdot)$. (iii) is the classical computation. All items were also verified exactly at $z = (1 + i, 1, i)$, $q = 2i$, and at the null vectors $(1, i, 0)$ and $(1 + 2i, 2 + i, 2 - 2i)$. □

The theorem is inherited: (iii) is Wigner's classification [37, 36], (i)–(ii) the classical stabilisers of a bivector and the identity of helicity and duality on null fields [25, 2], recorded on the frame in Theorem 13.8 (the axis of a null field) and Example 13.11 (the G-charge is the helicity) of [1]. It separates two notions the hypothesis runs together: a particle's little group (of its momentum) is three-dimensional with or without mass, a field point's stabiliser two-dimensional with or without q. A mass changes the type of a Lorentz stabiliser, never its dimension, and breaks no Lorentz generator, since q and P^2 are invariants. What a nonzero q removes is the duality generator: a null field point is duality-invariant up to a rotation about its axis, so Wigner's $\mathfrak{e}(2)$ reappears once duality is admitted, with helicity as compact generator; for a non-null point duality is broken to $\{\pm\mathbf{1}\}$.

3.3 The Dirac mass: invariant in six dimensions, breaking the kinetic algebra

With $\gamma^0 = e_1$, $\gamma^a = e_{a+3}$ the kinetic kernels are $K^\mu = -\gamma^0\gamma^\mu$ and the mass kernel is $T = \gamma^0 J$ (Proposition 4.5 of [1]); a Grassmann bilinear $\psi^\top M\psi$ is invariant under $\delta\psi = Y\psi$ iff $Y^\top M + MY = 0$.

Proposition 3.5 (The mass form is a Spin(3, 3) invariant; the kinetic form is not)**.** *On the spinor module* $\mathbb{R}^8$*:*

(i) Each of the fifteen bivectors $Y = e_a e_b$ *satisfies* $Y^\top T + TY = 0$*, and* $(e^{sY})^\top T\, e^{sY} = T$ *for all real* s*. The antisymmetric* $M \in M_8(\mathbb{R})$ *satisfying the fifteen conditions form exactly* $\mathbb{R}\,T$*.*

(ii) An element $Y \in \mathfrak{so}(3, 3)$ *satisfies* $Y^\top K^\mu + K^\mu Y = \Lambda^\mu{}_\nu K^\nu$ *for some real* 4×4 *matrix* Λ *iff* $Y \in \mathfrak{l}_{\text{int}} \oplus \mathbb{R}J$*, of dimension* 7*.*

Proof. (i) By Lemma 2.1 of [1] the trivector T commutes with a bivector sharing two or no indices with $\{1, 2, 3\}$ and anticommutes with a mixed one; by Theorem 2.3 the former are the antisymmetric and the latter the symmetric bivectors, so $Y^\top T + TY$ is $[T, Y]$ or $\{T, Y\}$ and vanishes. For an antisymmetric Y, $e^{sY} = \cos s + \sin s\, Y$ commutes with T; for a symmetric one $e^{sY} = \cosh s + \sinh s\, Y$ and $(e^{sY})^\top T = Te^{-sY}$. Uniqueness is Corollary 18.4 of [1], rederived by solving the 960 linear conditions on the 28 antisymmetric entries. (ii) The 256 linear equations in the fifteen bivector coefficients and the sixteen entries of Λ have a seven-dimensional solution space, projecting injectively to the bivectors; $\mathfrak{l}_{\text{int}}$ and J lie in it by covariance and by $[J, \gamma^0\gamma^\mu] = 0$. □

Item (i) restates Corollary 18.4 of [1] (fixed there); item (ii) is added and fixed. Within Spin(3, 3) the Dirac mass breaks nothing, and what reduces the six-dimensional symmetry to the interior one is the kinetic term, where the clock is converted into the spatial triple and c

enters. In the six-dimensional sense the entry c^2 *symmetric, m breaking* is inverted; the breaking it describes lives in the internal symmetries of the kinetic term.

Proposition 3.6 (What the Dirac mass breaks)**.** *(i) The $Y \in \mathrm{End}_{\mathbb{R}}(\mathbb{R}^8)$ with $Y^\top K^\mu + K^\mu Y = 0$ for $\mu = 0, \dots, 3$ form the algebra $\mathfrak{g}_{\mathrm{massless}} = \mathrm{span}\{\Omega, J, e_2\Omega, e_3\Omega\} = \mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$, with trace form $-8\,\mathbf{1}$; adding $Y^\top T + TY = 0$ leaves $\mathbb{R}J$.*

(ii) On $\mathfrak{g}_{\mathrm{massless}}$ the orbit map $Y \mapsto Y^\top T + TY$ has rank 3 and image $\mathrm{span}\{T\Omega, Te_2\Omega, Te_3\Omega\}$, inside the six-dimensional space of $\mathfrak{l}_{\mathrm{int}}$-invariant kernels; $T\Omega$ commutes with J (pseudoscalar mass), $Te_2\Omega$ and $Te_3\Omega$ anticommute with it (Majorana kernels).

(iii) For $M = T(m_s + m_p\Omega)$ the field equation $K^\mu \partial_\mu \Psi = M\Psi$ of $\frac{1}{2}\Psi^\top (K^\mu \partial_\mu - M)\Psi$, with $\partial_a \Psi = J p_a \Psi$, is $\dot\Psi = -JH\Psi$ where

$$H = \gamma^0\gamma^a p_a + \gamma^0(m_s + m_p\Omega), \qquad H^\top = H, \qquad [H, J] = 0, \qquad H^2 = (|\mathbf{p}|^2 + m_s^2 + m_p^2)\mathbf{1}. \tag{6}$$

(iv) $e^{\xi\Omega}$ preserves each K^μ and sends $M \mapsto Me^{2\xi\Omega}$; the stabiliser of $M \neq 0$ in $U(1)_\Omega$ is $\{\pm\mathbf{1}\}$.

Proof. (i) The 256 conditions on the 64 entries of Y have exactly these solutions. At $\mu = 0$ the condition is $Y^\top = -Y$, so mass invariance is $[T, Y] = 0$, which only J satisfies. (ii) For antisymmetric Y the map is $[T, Y]$; by the blade rule T anticommutes with Ω, $e_2\Omega$, $e_3\Omega$ and commutes with J, giving $2T\Omega, 2Te_2\Omega, 2Te_3\Omega$; e_2, e_3 anticommute with J, T and Ω commute with it. (iii) Varying gives $K^\mu\partial_\mu\Psi = M\Psi$, since K^μ is symmetric and M antisymmetric. With $K^0 = -\mathbf{1}$ and J commuting with $\gamma^0\gamma^a$, γ^0, Ω: $\dot\Psi = (K^a J p_a - M)\Psi = -JH\Psi$. The five matrices $\gamma^0\gamma^a$, γ^0, $\gamma^0\Omega = e_4e_5e_6$ are symmetric, square to $\mathbf{1}$ and pairwise anticommute, which gives H^2. (iv) $\Omega \in \mathfrak{g}_{\mathrm{massless}}$ and $\{T, \Omega\} = 0$, so $e^{-\xi\Omega} T e^{\xi\Omega} = Te^{2\xi\Omega}$. □

Item (i) restates Proposition 5.5 of [1], and the count in (ii) restates Theorem 18.5(a) of [1] (both fixed there); the explicit broken directions of (ii) are added and fixed. Their physical content, that a Dirac mass breaks the axial $U(1)$ and the Pauli–Gürsey $SU(2)$ of a massless Dirac field to fermion number [24, 13], is inherited by Theorem 4.3 of [1]. Items (iii) and (iv) are inherited, the realification of $h = \boldsymbol{\alpha}\cdot\mathbf{p} + \beta(m_s + i\gamma_5 m_p)$ (Proposition 4.6 of [1]) and of the chiral rotation; (iii) was derived from the Grassmann kernels, and the commuting first-order form $\frac{1}{2}\Psi^\top J\dot\Psi - \frac{1}{2}\Psi^\top H\Psi$ of Section 4 of [1] with the same H has the same flow. In both readings the mass is the modulus of an order parameter of charge two under a $U(1)$ generated by a complex structure commuting with the Lorentz algebra (G for $\mathfrak{l}_G$, Ω for $\mathfrak{l}_{\mathrm{int}}$), and a nonzero modulus leaves $\{\pm\mathbf{1}\}$; but the Dirac mass is a coupling, or a value selected by a potential (Theorem 18.5 of [1]), while the rest energy of a frame vector is a property of a configuration that no potential selects.

3.4 Particle and antiparticle: the two senses of one rotation

At $\mathbf{p} = 0$ the real first-order form $\frac{1}{2}\Psi^\top J\dot\Psi - \frac{1}{2}\Psi^\top H\Psi$ of Section 4 of [1] has $H(0) = m\gamma^0 = me_1$.

Proposition 3.7 (The rest flow)**.** *$JH(0) = mT$, so $\Psi(\lambda) = e^{-m\lambda T}\Psi_0 = (\cos m\lambda - \sin m\lambda\, T)\Psi_0$. On the eigenspaces $e_1 = \pm 1$, each four-dimensional and J-invariant, $T = \pm J$: $\Psi(\lambda) = e^{\mp m\lambda J}\Psi_0$, with energy $\frac{1}{2}\Psi^\top H\Psi = \pm\frac{1}{2}m|\Psi|^2$. Moreover $\gamma^0 = -TJ$.*

Proof. $Je_1 = e_1e_2e_3$; $T = e_1J$ with $[e_1, J] = 0$ gives $T\Pi_\pm = \pm J\Pi_\pm$ for $\Pi_\pm = \frac{1}{2}(\mathbf{1} \pm e_1)$; $T^2 = -\mathbf{1}$ gives the exponential; $-TJ = -e_1J^2 = e_1$. □

This is inherited, the realification of $\psi = e^{\mp im\lambda}\psi_0$ on $\beta = \pm 1$, at angular frequency $mc^2/\hbar$ in restored units. The arena adds a name: the rest motion is the one-parameter group of the temporal volume element T, the unique $\mathrm{Spin}(3,3)$-invariant mass form, and the clock $\gamma^0 = -TJ$

is the quotient of that rotation by the charge rotation. Read interpretively, the single rotation $e^{-m\lambda T}$ splits into two senses of the charge rotation according to the eigenvalue of the clock. Below $\mathsf{C}, \mathsf{P}, \mathsf{T}$ name the operations; on the frame they are realised by $C_f = -\mathbf{1}$, $P_f = \eta$ ($\mathbf{E} \mapsto -\mathbf{E}$) and $T_f \colon X(\tau) \mapsto -\eta X(-\tau)$, a convention that bundles the reversal of τ with the field map $(\mathbf{B}, \mathbf{E}) \mapsto (-\mathbf{B}, \mathbf{E})$ of $C_f P_f$.

Proposition 3.8 (Which operation reverses the sense)**.** *(i) Every unit vector C of the surplus plane U has $C^2 = \mathbf{1}$, $[C, T] = 0$, $\{C, J\} = \{C, e_1\} = 0$ and $C^\top H(0) C = -H(0)$; it maps the solution $e^{-m\lambda J}\Psi_0$ on $e_1 = +1$ to the solution $e^{+m\lambda J} C\Psi_0$ on $e_1 = -1$ of the same flow, and $C^\top J C = -J$, $C^\top H(\mathbf{p}) C = -H(-\mathbf{p})$. The parities γ^0, $\gamma^0 J$ and the time reversal $T_{\mathrm{rev}} = e_2 e_4 e_5 e_6$ (Theorem 5.3 of [1]) keep the sign of the rest energy.*

(ii) On the frame, $P_f G P_f^{-1} = -G$, $\mathcal{L}_\kappa(P_f X) = \mathcal{L}_{-\kappa}(X)$ and $\mathcal{L}_\kappa - \mathcal{L}_{-\kappa} = -2m\kappa \frac{d}{d\tau}(\mathbf{E}\cdot\mathbf{B})$, while C_f and T_f leave $\mathcal{L}_\kappa = \frac{m}{2}\eta(\alpha_+ X, \alpha_+ X)$ invariant. On $z = c_- e^{-i\kappa\tau} + c_+ e^{i\kappa\tau}$: $P_f \colon (c_-, c_+) \mapsto (\bar{c}_+, \bar{c}_-)$, $C_f \colon (c_-, c_+) \mapsto (-c_-, -c_+)$, $T_f \colon (c_-, c_+) \mapsto (-\bar{c}_-, -\bar{c}_+)$.

(iii) The sense of the Dirac rest rotation is reversed by C, CP, CT, CPT; that of the frame's binding rotation by P, CP, PT, CPT.

Proof. (i) $C = \cos\sigma\, e_2 + \sin\sigma\, e_3$; by the blade rule it anticommutes with e_1, J and the γ^a and commutes with T and Ω, so $C^\top e_1 C = -e_1$, C commutes with $e^{-m\lambda T}$, and $CH(\mathbf{p})C^{-1} = -H(-\mathbf{p})$ (Theorem 5.1 of [1]). γ^0, $\gamma^0 J$ commute with e_1; T_{rev} commutes with e_1 and anticommutes with J and T, so $T_{\mathrm{rev}}\Psi(-\lambda)$ solves the same flow on the same eigenspace. (ii) $\eta G \eta = G^\top = -G$ and η is an η-isometry, so $\mathcal{L}_\kappa(P_f X) = \frac{m}{2}\eta(\dot{X} - \kappa G X, \dot{X} - \kappa G X)$; the total derivative is Theorem 16.6 of [1]; for T_f the reversal of τ compensates the sign. On z: $P_f \colon z \mapsto \bar{z}$, $C_f \colon z \mapsto -z$, $T_f \colon z(\tau) \mapsto -\overline{z(-\tau)}$. (iii) Signs multiply under composition (also checked on the matrices). □

Item (i) is inherited, realised by the arena's C (Theorems 5.1 and 5.3 of [1]); (ii) and (iii) are fixed, (ii) extending Theorem 16.6 of [1]. For Dirac matter the two senses are particle and antiparticle: solutions of one Lagrangian, of angular frequency m, each with energy of definite sign, exchanged by C, a symmetry of the Grassmann Lagrangian (it preserves the mass form and every K^μ, Theorem 5.1 of [1]; on the commuting first-order form it reverses the sign of the action). For the frame the two senses are the kernels of the two factors of $\alpha_-\alpha_+ = \partial_\tau^2 + \kappa^2$: one carries no momentum, the other all of it; each alone carries zero Noether energy (Proposition 3.3); they are exchanged by parity, a symmetry only up to a total derivative; and their frequency $\kappa = c/\ell$ is a conversion rate, not a mass. In the interior m and κ are independent inputs (m multiplies the Lagrangian, κ sets the period), so reading the frame's $\hbar\kappa$ as mc^2 would be an added identification, not a result. That the frame's binding is not a particle–antiparticle pair in the Dirac sense is an interpretive conclusion from these fixed facts. In both readings the reversal is CPT-odd.

3.5 Two vacua

Proposition 3.9 (The Fock vacuum of one momentum)**.** *Let $\theta_1, \dots, \theta_8$ be antisymmetric 16×16 integer matrices with $\theta_i^2 = -\mathbf{1}$, pairwise anticommuting, and $\rho(A) = -\frac{1}{4}\sum A_{ij}\theta_i\theta_j$, and let $\varepsilon = (|\mathbf{p}|^2 + m^2)^{1/2}$. Then $\chi_{2\rho(J)}(t) = t^6(t^2+4)^4(t^2+16)$, $\chi_{2\rho(JH)}(t) = t^6(t^2+4\varepsilon^2)^4(t^2+16\varepsilon^2)$; the joint spectrum of $Q = -i\rho(J)$ and $H = -i\rho(JH(\mathbf{p}))$ is the table of Theorem 5.7 of [1], with a unique charge-zero vacuum at -2ε and multiplicities $1, 4, 6, 4, 1$ for $H + 2\varepsilon$. Charge conjugation $C = e_2$ lifts to $\hat{C} = \theta(v_1)\cdots\theta(v_4)/4$, with v_k orthogonal integer vectors, $|v_k|^2 = 2$, spanning the -1 eigenspace of C; $\hat{C}$ is orthogonal, $\hat{C}^2 = \mathbf{1}$, $\hat{C}Q\hat{C}^{-1} = -Q$, $\hat{C}H(\mathbf{p})\hat{C}^{-1} = H(-\mathbf{p})$.*

Proof. The θ_i are tensor words in the 2×2 matrices of eq. (1) of [1], found by search; their 256 monomials have rank 256, so their commutant is $\mathbb{R}\mathbf{1}$ (Theorem 5.6 of [1]); ρ is a Lie homomorphism with $[\rho(A), \theta_k] = \sum_j A_{jk}\theta_j$. The characteristic polynomials were computed exactly

stage	order parameter	kept	broken directions	status here
$n=1$, Dirac	$T=\gamma^0 J$	$\mathfrak{u}(1)_J$	3: $T\Omega$, $Te_2\Omega$, $Te_3\Omega$ (global)	re-verified
$n=1$, Majorana	$M=\Gamma M=M\Gamma$	$\mathbb{R}(J-J\Gamma)$	3 (global)	re-verified
$n=3$, Pati–Salam	Φ_0 ($\nu_R\nu_R$)	14 of 23	9, eaten by Z', $W_R^{\pm}$, six X	re-verified
$n=3$, electroweak	Φ_0 + bidoublet	10 of 23	12 eaten, 1 axion-like $(1-\Gamma)$	re-derived (d2-24)

Table 2: Broken directions attached to the specified generators: at $n=1$ the four of $\mathfrak{g}_{\text{massless}}$, at $n=3$ the 21 Pati–Salam generators and the central J, $J\Gamma$ (Proposition 22.1 of [1]). The rows restate Theorems 18.5, 22.2 and 22.3 of [1].

at four momenta, the joint eigenspaces at two. $\hat{C}\rho(A)\hat{C}^{-1}=\rho(CAC^{\top})$, with $CJC^{\top}=-J$ and $CJH(\mathbf{p})C^{\top}=JH(-\mathbf{p})$ (Theorem 5.1 of [1]). □

This restates Theorem 5.7 and Proposition 5.8 of [1] and is inherited: the Fock space of four fermionic modes of one Dirac momentum. Its energy, $-2m$ per momentum at rest ($-2mc^2$ in restored units), is minus half the rest energies of the four modes; only in that sense does the vacuum carry the rest energy with the opposite sign, and only this vacuum does.

Proposition 3.10 (The frame vacuum)**.** *The interior energy is* $\mathcal{E}=\frac{m}{2}(\eta(\dot{X},\dot{X})+\kappa^2\eta(X,X))=\sum_a \eta_{aa}\,\frac{m}{2}(\dot{x}_a^2+\kappa^2 x_a^2)$*; a spatial mode* x *has* $p_x=-m\dot{x}$ *and Hamiltonian* $-(p_x^2/2m+m\kappa^2x^2/2)$*. With canonical, positive-norm quantisation a temporal mode contributes* $\hbar\kappa(j+\frac{1}{2})$ *and a spatial one* $-\hbar\kappa(j+\frac{1}{2})$*,* $j=0,1,\dots$*, so on a coordinate subspace with* k *temporal and* l *spatial directions the zero-point energy is* $\frac{\hbar\kappa}{2}(k-l)$*:* 0 *on* $\mathbb{R}^{3,3}$*,* $\pm\frac{3}{2}\hbar\kappa$ *on the temporal and spatial triples,* $\pm\frac{1}{2}\hbar\kappa$ *on the* $(2,1)$ *and* $(1,2)$ *planes.*

Proof. The first identity uses $\eta(GX,GX)=-\eta(X,X)$; the spatial Hamiltonian is the Legendre transform of $-\frac{m}{2}(\dot{x}^2-\kappa^2x^2)$. With $x=\sqrt{\hbar/2m\kappa}\,(\hat{a}+\hat{a}^{\dagger})$, $p_x=i\sqrt{\hbar m\kappa/2}\,(\hat{a}^{\dagger}-\hat{a})$, $p_x^2/2m+m\kappa^2x^2/2=\frac{\hbar\kappa}{2}(\hat{a}\hat{a}^{\dagger}+\hat{a}^{\dagger}\hat{a})$, checked exactly on truncated ladder matrices. The cross term is a total derivative; it shifts the momenta by a canonical transformation and leaves the spectrum unchanged. □

Corollary 3.11 (No ground state)**.** *With positive norm the spectrum of the full frame is* $\hbar\kappa\mathbb{Z}$*, every level is infinitely degenerate, and every state has one of lower energy (add a spatial quantum). Making the spatial energies positive instead requires negative norm (Section 18 of [1]).*

The proposition restates Theorem 18.6 of [1] (fixed there; the per-mode quantisation is the inherited indefinite oscillator): the zero-point energy vanishes because the signature is neutral, the condition under which factorisation and binding coexist (Theorem 16.2 of [1]). The corollary, which makes explicit for the full space the remark after Theorem 18.6 of [1], is an obstruction under the hypothesis of positive norm, the familiar fate of indefinite oscillators [23]. The two vacua differ in kind: the Dirac vacuum has energy -2ε per momentum and is bounded below; the frame's zero-point energy cancels between positive temporal and negative spatial oscillators, and there is no ground state for it to be the energy of.

3.6 Goldstone bookkeeping across the tower

Proposition 3.12 (Goldstone directions)**.** *(i) The* $\mathfrak{l}_{\text{int}}$*-invariant kernels supported on* $\Gamma=+1$ *form a complex line; each* M *in it anticommutes with* J*, has rank four,* $M^{\top}M\propto\frac{1}{2}(\mathbf{1}+\Gamma)$ *and* $\operatorname{Tr}(M^{\top}M)^2/(\operatorname{Tr}M^{\top}M)^2=\frac{1}{4}$*; its stabiliser in* $\mathfrak{g}_{\text{massless}}$ *is* $\mathbb{R}(J-J\Gamma)=\mathbb{R}(J+\Omega)$ *and its orbit is three-dimensional.*

(ii) At three constituents, with the generators E_A *of* $\mathrm{Cl}(9,9)$ *and* P_**,* Φ_0*,* t_3^R *of Appendix A of [1] (its* $\mathfrak{su}(2)$ *generators normalised as in Section 8, as the corrected Appendix has them):* $[J,\Phi_0]=[J\Gamma,\Phi_0]=2[t_3^R,\Phi_0]\neq 0$*, and the orbits of the* 21 *Pati–Salam generators and of all* 23 *have tangent spaces of the same dimension* 9 *at* Φ_0*.*

symmetry	broken by a mass?	class	source in [1] / here
Lorentz $\mathfrak{l}_{\text{int}}$	no: P^2, T invariant; little groups stay 3-dim	inherited	Theorem 3.4(iii)
Lorentz $\mathfrak{l}_G$	no: q invariant; stabilisers stay 2-dim	inherited	Sect. 14.3; Theorem 3.4(i)
$\mathrm{Spin}(3,3)$	no: T invariant (the kinetic term keeps 7)	fixed	Cor. 18.4; Proposition 3.5(ii) new
duality $U(1)_G$	yes, to $\{\pm\mathbf{1}\}$ on Lorentz orbits, iff $q \neq 0$	inherited	Sect. 14.3
axial $U(1)_\Omega$	yes, to $\{\pm\mathbf{1}\}$	inherited	Props. 4.6, 5.5
Pauli–Gürsey $\mathfrak{su}(2)$	yes, to $\mathfrak{u}(1)_J$	inherited	Prop. 5.5, Thm. 18.5(a)
$U(1)_{1-\Gamma}$	by Dirac type; not by Majorana on $\Gamma = +1$	inherited	Thms. 18.5(b), 22.3

Table 3: What a mass breaks in the arena.

(iii) If a kernel M commutes with J and anticommutes with Γ (Dirac type), $[J - J\Gamma, M] = -2J\Gamma M \neq 0$; if it lives on $\Gamma = +1$ and anticommutes with J, $[J - J\Gamma, M] = 0$. At one constituent $J - J\Gamma = J + \Omega$.

Proof. (i) and (ii) are exact computations, at three constituents on $\mathbb{R}^{512}$ with dyadic matrices and ranks over $\mathbb{Q}$; the reconstruction reproduces Theorem 19.2(v) of [1]: Φ_0 antisymmetric, $\{J, \Phi_0\} = 0$, $\Phi_0^\top \Phi_0 = P_\nu$, the rank-four ν_R projector of the block (on $\mathbb{R}^{512}$, P_* times a rank-eight joint eigenprojector), $[t_3^R, \Phi_0] = J\Phi_0$. The identity in (ii) has a one-line reason: Φ_0 lives on $\Gamma = +1$, so $J\Gamma\Phi_0 = J\Phi_0$ and $\Phi_0 J\Gamma = -J\Phi_0$. (iii) $[J, \Gamma] = 0$; if $\{M, \Gamma\} = 0 = [M, J]$ then $MJ\Gamma = -J\Gamma M$; if M lives on $\Gamma = +1$ and $\{M, J\} = 0$ then $[J, M] = 2JM = [J\Gamma, M]$; and $J\Gamma = J^2\Omega = -\Omega$. The bidoublet kernel $P_* E_1 E_7 E_8 P_*$ served as an explicit Dirac-type instance. □

Items (i) and (ii) restate Theorems 18.5(b) and 22.2 of [1] (fixed there), re-verified: the would-be Majoron is eaten. Item (iii) is inherited, the two-line fact that a charge vanishing on one chirality is kept by a Majorana mass of that chirality and broken by every mass pairing opposite chiralities; what the arena adds is the cross-scale identification: one element, $J - J\Gamma$, of charge $1 - \Gamma$, is kept by a Majorana mass on $\Gamma = +1$ and broken by every Dirac-type mass, at one constituent, where it is $J + \Omega$ (Theorem 18.5(b) of [1]), and at three. That at the generic electroweak vacuum the broken $1 - \Gamma$ direction escapes the gauge orbit restates Theorem 22.3 of [1]. We re-derived its whole table in integer arithmetic (d2-24: the 2176 kernel blades, the up- and down-type $(\mathbf{1}, \mathbf{2}, \mathbf{2})$ kernels K_{1u}, K_{1d}, and orbit dimensions $(12, 13)$, $(12, 12)$, $(5, 6)$, the uneaten tangent being that of $J - J\Gamma$); the adversarial check x2-24 re-derived it numerically.

3.7 What the hypothesis gets right, and where the b-field belongs

Table 3 collects the answer. *m breaks the symmetry* is true for duality in the frame, in the orbit-space sense and for a configuration rather than a vacuum, and for the chiral and Pauli–Gürsey symmetries of Dirac matter; it is false for every spacetime symmetry, the six-dimensional one included: a mass changes which little group a state has and removes the duality generator from the stabiliser of a field.

c^2 is the symmetric part is an interpretive entry that needs correction. The constant c is the Schur-forced conversion $c\,\mathbf{1}$ between the triples (Theorem 3.2 of [1]), and a constant generates nothing. It enters the interior in two places: in η itself, through the conversion that makes η dimensionally consistent (Theorem 3.2 and the opening of Section 3 of [1]), and in the shift κG, $\kappa = c/\ell$ (eq. (7) of [1]), where $(\kappa G)^2 = -c^2/\ell^2$ is definitional ($G^2 = -\mathbf{1}$ is adopted in Theorem 16.2 of [1]). Of the two, the conversion inside η (Lorentz class) and the shift κG, the

statement	class	source in [1]	checks
Proposition 3.1	inherited	Sect. 14.3, Prop. 12.12	d2-1, d2-2
Proposition 3.2	inherited	Sect. 14.3	d2-3
Proposition 3.3 (i), (ii); (iv) at $c_+ = 0$	fixed there	Prop. 16.5	d2-4, d2-4b
Proposition 3.3 (iii); (iv) at $c_- = 0$	fixed (new)	—	d2-4b
Theorem 3.4	inherited	Thm. 13.8, Ex. 13.11	d2-5a, d2-5, d2-6, d2-7
Proposition 3.5 (i) / (ii)	fixed there / fixed (new)	Cor. 18.4 / —	d2-8 / d2-8b
Proposition 3.6 (i), (ii)	fixed there; directions new	Prop. 5.5, Thm. 18.5(a)	d2-9
Proposition 3.6 (iii), (iv)	inherited	Prop. 4.6	d2-10
Proposition 3.7	inherited	Sect. 4	d2-11
Proposition 3.8 (i) / (ii), (iii)	inherited / fixed (new)	Thms. 5.1, 5.3 / 16.6	d2-12 / d2-13, d2-14
Proposition 3.9	inherited	Thm. 5.7, Prop. 5.8	d2-15a, d2-15b, d2-16, d2-17
Proposition 3.10 / Corollary 3.11	fixed there / obstructed	Thm. 18.6	d2-18 / d2-19
Proposition 3.12 (i), (ii) / (iii)	fixed there / inherited	Thms. 18.5(b), 22.2	d2-20, d2-21a, d2-21 / d2-22
electroweak count	fixed there	Thm. 22.3	d2-24 (exact), x2-24
$(\kappa G)^2 = -c^2/\ell^2$	constructed (definitional)	Thm. 16.2, eq. (7)	d2-23
readings of Section 3.7	interpretive	—	—

Table 4: Claims of this section, their class ("fixed there": a restated result of [1], fixed in the paper), and the labels of the checks printed by `d2_symmetry_breaking.py` (and, for x2-24, by `x2_check.py`). Check d2-0 is a consistency check of the constructed realisation (eq. (1) of [1]), not a claim.

second generates duality. But G is not an infinitesimal isometry of η (Proposition 3.1): duality is a symmetry of the positive form u, while the interior Lagrangian is built on η. The split form and the positive form lie in the Lorentz and the duality class of Theorem 17.11 of [1], which exclude each other (remark after that theorem); c sits in both, and the symmetric part and the Lorentz-invariant dynamics cannot be one object. In the Dirac reading the symmetric part is the massless kinetic algebra $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$, which the mass cuts to $\mathfrak{u}(1)_J$; for Spin(3, 3) the roles are reversed (Proposition 3.5). Which term is symmetric depends on which symmetry is meant, and the hypothesis has to say which.

Nor is the undivided energy the interior's conserved energy. Of the frame field, $u(X)$ depends on the Lorentz frame and is not conserved by the binding flow, and $E_0(X)$ oscillates; the Noether energy has duality charge two and lives on the overlap of the two senses; the conserved $u(\pi)$, $E_0(\pi)$ and $\mathbf{S}(\pi)$ belong to the momentum field (Proposition 3.3). Pointwise, the only listed quantity invariant under both Lorentz transformations and duality is $E_0 = \frac{1}{2}|q|$, the infimum of u over Lorentz transformations (Proposition 3.2). If an undivided energy whose differentiation is the mass is wanted, the arena's candidate is u of the momentum field, where it is conserved, with its split $u^2 = E_0^2 + |\mathbf{S}|^2$, the inherited field form of $E^2 = p^2c^2 + m^2c^4$; the differentiation it describes is the choice of a rest frame, not a breaking.

Finally the b-field. An admissible Yang–Mills potential takes compact values (Theorem 2.3 and Corollary 2.4 of [1]; Section 5). The broken generators at one constituent are compact, but none of them can carry a Yang–Mills field: duality is not a symmetry of the split-form Lagrangian (Proposition 3.1) and has no spin lift, $U(1)_\Omega$ is anomalous, and the Pauli–Gürsey $\mathfrak{su}(2)$ acts on a single doublet and mixes a field with its conjugate (Proposition 5.5 of [1], [39]). The one gaugeable generator there, J, is abelian and unbroken by the Dirac mass ($[T, J] = 0$); in the projector model of Section 26 of [1], where the charge is $J_N/2$, the normal-plane form of

J, it acquires a Stückelberg mass exactly when the vacuum projector fails to commute with it (Theorem 26.1 and Corollary 26.2 of [1]). In the paper's vacuum analysis broken non-abelian generators are first gauged at three constituents, where the nine Goldstone directions of Φ_0 are eaten by Z', $W_R^{\pm}$ and six leptoquarks (Proposition 3.12; Theorems 21.1 and 22.2 of [1]). There the gauge quanta of Yang and Mills [41] acquire their masses from the breaking, by the inherited mechanism of [8, 15]. The same condensate gives the right-handed neutrino its Majorana mass $|y|r$ (Theorem 21.1(d) of [1], with the single Yukawa coupling y that Assumption 19.4 of [1] adopts), while the charged-fermion and Dirac-neutrino masses require the bidoublet values of the electroweak stage, which are inputs (Proposition 21.2 of [1]); there the $SU(2)_L$ gauge fields, the closest analogue of the isotopic b-field of [41], eat three further directions ($W^{\pm}$, Z; Theorem 22.3 of [1]). In the arena the b-particle receives mass from the differentiation; it does not generate it. That is where the b-field belongs, and it reverses the order the hypothesis proposes.

4 The differentiation in motion: rotor, rate and duality angle

This section follows the undivided energy and its differentiation in motion inside the one dynamics that the arena closes by itself, the Lie–Poisson closure of the force law (Section 17 of [1]), in which the rate $\kappa = c/\ell$ of $\alpha_{\pm} = \partial_\tau \pm \kappa G$, where the hypothesis puts c^2, is the angular velocity of a rotor. No Yang–Mills field is used. As in Proposition 3.1, two fields are kept apart: the *momentum field* $\pi = \eta^{-1}p$ ($\pi = m\alpha_+ X$ in the clamped interior) and the *frame field* X. In brief: the closure freezes the undivided energy, rest energy and Poynting vector of the momentum field, so no Hamiltonian generates or changes the rest energy of that field, while the rest energy of the frame field oscillates in the clamped interior and grows without bound under the free-rotor drift (Theorem 4.1 and Corollary 4.2). The conserved total splits into a rate part $\frac{1}{2}(I_0/\ell^2)c^2$ and a Lorentz-invariant frame part with an amplitude set by the rest energy of the momentum field, and the duality angle swings with Ω^2 proportional to it (Propositions 4.3, 4.4 and 4.7 and Corollary 4.5, restating Theorem 17.13 and Corollaries 17.14–17.15 of [1]). A constant rate is the heavy-rotor limit (Proposition 4.8); nothing in the algebra fixes ℓ (Proposition 4.12); and $\hbar\kappa = mc^2$, equivalently $\ell = \hbar/(mc)$, is an added identification, not a result, under which a quantised rotor has the first level $\frac{1}{2}mc^2$ (Propositions 4.10 and 4.16).

Conventions. As in Section 17 of [1]: $V = \mathbb{R}^{3,3}$, $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$, so $G^2 = -\mathbf{1}$, $G^{\top} = -G$, $G^{\dagger} := \eta^{-1}G^{\top}\eta = G$, $\eta(GX, GY) = -\eta(X,Y)$ (Proposition 3.3, Theorem 16.2 of [1]), and $G = -\mathcal{K}\circ \mathrm{Ad}_{e_1e_2e_3}$, $\mathcal{K} = \frac{1}{2}\,\mathrm{ad}_{\mathrm{P}}$ (Lemma 3.4 of [1]). For $Y \in V$ with components $(\mathbf{B};\mathbf{E})$ along $(e_1,e_2,e_3;e_4,e_5,e_6)$, $z(Y) = \mathbf{B} + i\mathbf{E}$, so G acts as i (Theorem 12.3 of [1]), $q(z) = z\cdot z = |\mathbf{B}|^2 - |\mathbf{E}|^2 + 2i\,\mathbf{B}\cdot\mathbf{E}$, and

$$E_6(Y) := \tfrac{1}{2}|z(Y)|^2, \qquad E_0(Y) := \tfrac{1}{2}|q(z(Y))|, \qquad \mathbf{S}(Y) := \tfrac{i}{2}\,\overline{z(Y)} \times z(Y) = \mathbf{E}\times\mathbf{B}$$

are the undivided energy, the rest energy (the rest-frame energy density of Section 14.4 of [1]) and the frame Poynting vector of Y; as $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$ (Section 12 of [1]), $\mathbf{S} = -\mathbf{E}\times\mathbf{B}_{\mathrm{t}}$ is minus the textbook one, which the rest-frame velocity of Section 14.4 uses; only $|\mathbf{S}|$ enters below. A covector p is a column, $C(p) = p^{\top}\eta^{-1}p$, $D(p) = p^{\top}G\eta^{-1}p$, $w(p) := C - iD = q(z(\pi))$, so $E_0(\pi) = \frac{1}{2}|w(p)|$, $E_6(\pi) = \frac{1}{2}|p|^2$. Lorentz-class inertia (Theorem 17.11(ii) of [1]) is written $N = (\nu_1\eta^{-1} + \nu_2 G\eta^{-1})/m$, $\nu := \nu_1 + i\nu_2$ (the (a,b) of [1]), as in Theorem 2.13, keeping b for the Yang–Mills field. Units are $\hbar = c = 1$ unless shown; τ is a time; $\mathrm{K}(\mu)$, $\mathrm{E}(\mu)$ are the complete elliptic integrals in the parameter μ [6]. d3-n refers to `verify/d3_closure_dynamics.py` (integrations: eighth-order Runge–Kutta, relative tolerance 10^{-12}; d3-0 checks the conventions), x3-n to the independent `verify/x3_check.py`.

4.1 What the closure conserves

The only $\mathfrak{l}_G$-equivariant closure in which the rotor rotates the frame covector without dilating it is $\mathfrak{s} = \mathbb{R}\hat{g} \ltimes_G V$, $[\hat{g}, X] = GX$, $[X, Y] = 0$ (Theorem 17.7 of [1]). On $\mathfrak{s}^*$, with coordinates (h, p), $F_h = \partial F/\partial h$ and $\nabla F = \partial F/\partial p$, the Lie–Poisson equations of an arbitrary Hamiltonian H, and the bracket for which they read $\dot{F} = \{F, H\}$, are

$$\dot{p} = H_h\, G^\top p, \qquad \dot{h} = -\,p^\top G\,\nabla H, \qquad \{F_1, F_2\} = p^\top\,(F_{2,h}\,G\nabla F_1 - F_{1,h}\,G\nabla F_2). \tag{7}$$

For $H = h^2/(2I_0) + \frac{1}{2}p^\top N p$ this is Proposition 17.8 of [1], $\dot{p} = (h/I_0)G^\top p$, $\dot{h} = -p^\top GNp$; re-derived from the structure constants, with H conserved for every symmetric N and $|p|^2$ for every H (d3-1).

Theorem 4.1 (The momentum field is frozen up to its duality phase)**.** *(i) A function $f(p)$ is a Casimir of $\mathfrak{s}^*$ iff it is invariant under $p \mapsto e^{tG}p$, i.e. under the duality rotations $z(\pi) \mapsto e^{-it}z(\pi)$ of the momentum field. So $E_6(\pi) = \frac{1}{2}|p|^2$, $E_0(\pi)$ and the vector $\mathbf{S}(\pi)$ are Casimirs; C, D are not.*

(ii) At $p \neq 0$ the symplectic leaves are two-dimensional, so there are five independent local Casimirs; the real and imaginary parts of the $\bar{z}_j z_k$, $z = z(\pi)$, provide them (Jacobian rank five), and $E_6(\pi)$, $E_0(\pi)$, $\mathbf{S}(\pi)$ account for four, with $E_6(\pi)^2 = E_0(\pi)^2 + |\mathbf{S}(\pi)|^2$.

(iii) Along the flow of any Hamiltonian, $p(\tau) = e^{-\theta(\tau)G}p_0$ with $\dot{\theta} = H_h$, $\theta(0) = 0$, so $z(\pi(\tau)) = e^{i\theta(\tau)}z(\pi_0)$ and $w(p(\tau)) = e^{2i\theta(\tau)}w(p_0)$.

Proof. (i) For $f = f(p)$, (7) gives $\{f, F\} = F_h\, p^\top G\nabla f = -F_h\,(Gp)\cdot\nabla f$, zero for every F iff f is constant along $t \mapsto e^{tG}p$. As $\eta G = -G\eta$, $\eta^{-1}e^{tG}p = e^{-tG}\pi$, on which G acts as i. $|z|^2$, $|z\cdot z|$ and $\bar{z}\times z$ are phase invariant, while $C = \operatorname{Re} w$ and $D = -\operatorname{Im} w$ rotate into each other; $|\pi| = |p|$ as η is orthogonal. (ii) $\{p_i, p_j\} = 0$ and $\{h, p_j\} = (Gp)_j$, so the Poisson tensor has rank two at $p \neq 0$ and $\dim\mathfrak{s}^* - 2 = 5$. The $\bar{z}_j z_k$ are phase invariant with Jacobian rank five at a generic point; the relation is the Lagrange identity $u^2 - |\mathbf{S}|^2 = \frac{1}{4}|q|^2$ of Section 14.3 of [1]. (iii) $\dot{p} = -H_h GP$ integrates to $p = e^{-\theta G}p_0$, so $\pi = e^{\theta G}\pi_0$, and w is quadratic in z. □

The theorem is fixed, and in substance a corollary of Proposition 17.8 of [1], which identifies the coadjoint orbit through (h, p_0) with the cylinder $\mathbb{R}\times\{e^{\theta G}p_0\}$, so that every function of p constant on these circles is a Casimir; (iii) is the phase law of Theorem 17.13 of [1] and the relation in (ii) is inherited. What is added is the naming and the count of the Casimirs. Already in the clamped interior $p = e^{-\kappa\tau G}p_0$, so these functions of the momentum field were constant before the closure (Theorem 2.13). All items were checked exactly (d3-2); along five random Lorentz-class flows H, $|p|^2$, $E_6(\pi)$, $E_0(\pi)$, $\mathbf{S}(\pi)$ and (iii) hold to 7×10^{-12} (d3-3).

Corollary 4.2 (What the closure can and cannot change)**.** *(i) For every Hamiltonian on $\mathfrak{s}^*$, quadratic or not, of Lorentz class or not, $E_0(\pi)$, $E_6(\pi)$ and $\mathbf{S}(\pi)$ of the momentum field are constant along every solution: a null momentum field stays null, and one of rest energy E_0 keeps it.*

(ii) The same functions of the frame field are not constants of motion. In the clamped split-form interior, $z(X) = c_-e^{-i\kappa\tau} + c_+e^{i\kappa\tau}$ (Proposition 16.5(ii) of [1]) gives

$$E_6(X) = \tfrac{1}{2}(|c_-|^2+|c_+|^2)+\operatorname{Re}(\bar{c}_-\cdot c_+\,e^{2i\kappa\tau}), \quad q(z(X)) = c_-\cdot c_-\,e^{-2i\kappa\tau}+2\,c_-\cdot c_+ + c_+\cdot c_+\,e^{2i\kappa\tau},$$

so $E_6(X)$ is constant iff $\bar{c}_-\cdot c_+ = 0$, and $c_- = (1, i, 0)$, $c_+ = (1, 0, 0)$ give $|q(z(X))|^2 = 9 - 8\sin^2\kappa\tau$. Under the free-rotor drift of Example 4.13 with $x_0 = 0$, the frame field $X = e^{-\theta G}x$ has $E_0(X) = 0$ at $\tau = 0$ and $\frac{1}{2}(3\,\Delta x)^2 \approx 1.807$ after three full turns; on every rotating solution with $R_0 > 0$, $E_0(X)$ and $E_6(X)$ grow without bound.

Proof. (i) Casimirs Poisson-commute with every Hamiltonian (Theorem 4.1(i)). (ii) Expand $|z|^2$ and $z \cdot z$; in the example $c_- \cdot c_- = 0$, $c_- \cdot c_+ = c_+ \cdot c_+ = 1$, so $|q|^2 = |2 + e^{2i\kappa\tau}|^2$. In Example 4.13 the motion is 2π-periodic in θ, so at $\theta = 2\pi k$, $e^{-\theta G} = \mathbf{1}$ and $x = k\,\Delta x$, a real multiple of e_1, whence $q(z(X)) = k^2 \Delta x^2$. The growth is Theorem 2.13, item 4, from Theorem 17.16 of [1]. □

Item (i) is an obstruction for the momentum field, checked with the generic Hamiltonian $h^2/(2I_0) + \frac{1}{10}h^4 + \sin(p^\top M_1 p) + \frac{1}{5} h\, p^\top M_2 p$, $M_{1,2}$ random symmetric: the Casimirs and H stay constant to 5×10^{-12} while the phase of w moves through 6.3 radians (d3-4; x3-3). Item (ii) is fixed, from Proposition 16.5(ii) and Theorem 17.16 of [1] (d3-22; x3-4, x3-23). This first verdict cuts both ways. For the momentum field, $E_6(\pi)$ and its split $E_6(\pi)^2 = E_0(\pi)^2 + |\mathbf{S}(\pi)|^2$ are labels of the leaf: its rest energy is carried by the dynamics, not produced by it. For the frame field, to which the dictionary's $u = \frac{1}{2}|z|^2$ and $m \leftrightarrow \frac{1}{2}|q|$ refer, the closure does change the rest energy, by an oscillation or an unbounded drift, not by a mechanism that selects a value; and $E_6(X)$ is not a function of $\frac{1}{2}|p|^2 = E_6(\pi)$ (on zero-momentum orbits with $X \neq 0$, $|p|^2 = 0 < E_6(X)$, Proposition 3.1). What moves is the duality angle θ and the rate h/I_0.

4.2 The duality pendulum

From now on $N = (\nu_1 \eta^{-1} + \nu_2 G \eta^{-1})/m$ is of Lorentz class and $\nu\, w(p_0) = R_0 e^{i\varphi_0}$, $R_0 \geq 0$. On a leaf (θ, h) are canonical and $H = h^2/(2I_0) + U(\theta)$ (Proposition 17.8 of [1]).

Proposition 4.3 (The potential of the duality angle)**.** $\frac{1}{2} p^\top N p = \mathrm{Re}(\nu\, w(p))/(2m)$*, and with* $\psi := 2\theta + \varphi_0$*,*

$$U(\theta) = \frac{R_0}{2m} \cos\psi, \qquad R_0 = |\nu|\,|w(p_0)| = 2|\nu|\, E_0(\pi_0), \qquad I_0 \ddot{\theta} = \dot{h} = \frac{R_0}{m} \sin\psi.$$

Proof. $\frac{1}{2} p^\top N p = (\nu_1 C + \nu_2 D)/(2m) = \mathrm{Re}(\nu(C - iD))/(2m)$, and $w(p(\theta)) = e^{2i\theta} w(p_0)$ by Theorem 4.1(iii). For $p(\theta) = e^{-\theta G} p_0$, $dU/d\theta = -p^\top N G p = p^\top G N p$ (N symmetric, G antisymmetric), so $\dot{h} = -dU/d\theta$; and $\dot{\theta} = h/I_0$. □

This restates Theorem 17.13 of [1] (fixed there), re-verified exactly and along three random flows (d3-5); $R_0 = 2|\nu| E_0(\pi_0)$ only renames $\frac{1}{2}|w|$, and $E_6(\pi)$, $\mathbf{S}(\pi)$ do not enter.

Proposition 4.4 (Small oscillations)**.** *The stable equilibrium is* $\psi = \pi$, $U = -R_0/(2m)$*. With* $\psi = \pi + \xi$, $\ddot{\xi} = -\Omega^2 \sin\xi$ *and*

$$\Omega^2 = \frac{2R_0}{mI_0} = \frac{4|\nu|\, E_0(\pi_0)}{mI_0};$$

θ oscillates at the same frequency with half the amplitude, and at amplitude $0 < \xi_0 < \pi$ *the period is* $4\mathrm{K}(\sin^2(\xi_0/2))/\Omega$*.*

Proof. $\ddot{\psi} = 2\ddot{\theta} = (2R_0/mI_0)\sin\psi$ and $\sin(\pi + \xi) = -\sin\xi$. The period follows from $\frac{1}{2}\dot{\xi}^2 = \Omega^2(\cos\xi - \cos\xi_0)$ with $\sin(\xi/2) = \sin(\xi_0/2)\sin t$. □

Fixed, as a corollary of Theorem 17.13 of [1], whose pendulum it linearises; the period is the textbook one. At amplitude 10^{-3} the measured period exceeds $2\pi/\Omega$ by 6.25×10^{-8}, the shift $\xi_0^2/16$; at amplitude 2 it matches the elliptic formula to 5×10^{-14} (d3-6).

Corollary 4.5 (When the rate is constant)**.** *(i) If* $\nu \neq 0$*, the rate* h/I_0 *is constant and nonzero iff* $R_0 = 0$*, that is* $E_0(\pi_0) = 0$ *(a null momentum field,* $|\mathbf{B}| = |\mathbf{E}|$, $\mathbf{B} \cdot \mathbf{E} = 0$ *[33]), and* $h(0) \neq 0$*; then* $U \equiv 0$ *and* $\theta = h\tau/I_0$*, an exact flat direction. For* $E_0(\pi_0) > 0$ *the rate is constant only at the equilibria* $h(0) = 0$, $\psi(0) \in \pi\mathbb{Z}$*, where it vanishes and the frame moves at the constant velocity* Np_0, $|Np_0| = |\nu||p_0|/m \neq 0$*.*

(ii) If $\nu = 0$*, then* $U \equiv 0$ *and the rate is constant for all data.*

(iii) For duality-class inertia, $[N, G] = 0$ (nine dimensions), the rate is constant for all data and $U = \frac{1}{2} p_0^\top N p_0$ is constant, in general nonzero.

Proof. (i) $\dot{h} = (R_0/m) \sin\psi$. If $R_0 > 0$ and h is constant, $\sin\psi \equiv 0$, so θ is constant and $h = I_0 \dot{\theta} = 0$; conversely these points are equilibria, where Theorem 17.12 of [1] ($N_c = 0$, $\theta(0) = 0$) gives $\dot{x} = N p_0$, and $N^2 = |\nu|^2/m^2$. (ii) $N = 0$. (iii) GN is antisymmetric, so $\dot{h} = -p^\top GNp = 0$, and $U(\theta) = \frac{1}{2} p_0^\top e^{\theta G} N e^{-\theta G} p_0 = \frac{1}{2} p_0^\top N p_0$. □

Items (i)–(ii) restate Corollary 17.14 of [1] (fixed there), with the equilibria explicit as in Theorem 2.13; (iii) restates Theorems 17.11(i),(iii) and 17.12 of [1]. Numerically $|h - h_0| < 10^{-14}$ over $\tau \in [0, 30]$ on null data and for a random duality-class N, and at the equilibria h stays 0 while $x = \tau N p_0$ (d3-7).

Remark 4.6 (A pseudo-Goldstone reading, and its limits)**.** On a leaf θ is the angle of the duality group $U(1)_G$ of the momentum field, broken by a nonzero rest energy (Proposition 3.1): in the duality class it is a symmetry of H, in the Lorentz class it is broken on each leaf in proportion to $E_0(\pi)$. Exactly (d3-8),

$$I_0\, \Omega^2 = U''(\theta_{\min}) = \frac{2R_0}{m} = \frac{4|\nu|\, E_0(\pi_0)}{m}.$$

The first equality defines Ω for every one-degree-of-freedom oscillator; the closure-specific content is the second, $U'' = 2R_0/m$, which is Theorem 17.13 of [1]. With I_0 for f_π^2 and $|\nu| E_0(\pi_0)/m$ for the explicit breaking it has the shape of the Gell-Mann–Oakes–Renner relation [10, 35], as every pendulum does; the reading is interpretive. The breaking is explicit, by the inertia, not spontaneous: there is no vacuum manifold, $E_0(\pi)$ is a label fixed by the data (Corollary 4.2), not an order parameter (Section 14.3 of [1]), and with one degree of freedom "mass" means a frequency. What survives: on a null momentum field the angle is an exact zero mode, and a rest energy of the momentum field gives it the frequency $\Omega \propto \sqrt{E_0(\pi_0)}$.

4.3 The energy budget: a rate part and a frame part

Proposition 4.7 (Rate part and frame part)**.** *With Lorentz-class inertia, along every solution:*

(i) $H = K_{\rm rot} + U$, $K_{\rm rot} = \frac{1}{2} I_0 \kappa^2 \geq 0$, $\kappa = h/I_0$, $U = \frac{R_0}{2m} \cos\psi = \mathrm{Re}(\nu\, w(p))/(2m)$, and $\dot{K}_{\rm rot} = -\dot{U} = \frac{R_0}{m}\, \kappa \sin\psi$;

(ii) in the rotating regime $K_{\rm rot}$ runs through exactly $[H - \frac{R_0}{2m}, H + \frac{R_0}{2m}]$ in every period of the rate, so the energy exchanged is $R_0/m = 2|\nu| E_0(\pi_0)/m$; in the librating regime it runs through $[0, H + \frac{R_0}{2m}]$;

(iii) $K_{\rm rot}$ and U are separately invariant under $\mathfrak{l}_G$ acting by $(h, p) \mapsto (h, \Lambda^{-\top} p)$, $\Lambda = e^r$, $r \in \mathfrak{l}_G$, and the flow commutes with it;

(iv) $(\theta, h)(\tau)$ depends on p_0 only through $\nu\, w(p_0)$: $E_6(\pi_0)$ and $\mathbf{S}(\pi_0)$ do not enter the motion.

Proof. (i) By Proposition 4.3, $\dot{K}_{\rm rot} = h\dot{h}/I_0$ and $\dot{U} = -(R_0/2m)\, 2\kappa \sin\psi$. (ii) Rotating, $\dot{\psi} = 2\kappa$ never vanishes and $\cos\psi$ covers $[-1, 1]$ in each period; librating, $K_{\rm rot} = 0$ at the turning points and is largest at $\psi = \pi$. (iii) The Lorentz requirement $rN + Nr^\top = 0$ (Definition 17.10 of [1]) integrates to $\Lambda^{-1} N \Lambda^{-\top} = N$, so $U(\Lambda^{-\top} p) = U(p)$; $\Lambda^{-\top}$ commutes with $G^\top$ and maps solutions to solutions. (iv) $\pi \mapsto \Lambda\pi$ as $\Lambda \in O(3, 3)$, and $\mathfrak{l}_G \cong \mathfrak{so}(3, \mathbb{C})$ preserves q (Theorem 12.1 of [1]), so $w(\Lambda^{-\top} p) = w(p)$; the equations for (θ, h) contain only R_0 and φ_0. □

Fixed; (i) restates the energy $E_\theta = H$ of Theorem 17.13 of [1], (ii)–(iv) are added. Numerically $H = K_{\rm rot} + U$ to 3×10^{-11}, the range of $K_{\rm rot}$ equals R_0/m to 3×10^{-10}, and a Lorentz transformation of p_0 changed $E_6(\pi_0)$ by 5.0 and $\theta(\tau)$ by less than 10^{-11} (d3-9). So the conserved total differentiates into two Lorentz-invariant parts whose exchange is fixed by the rest energy of the momentum field alone; neither is the undivided $E_6(\pi)$, which is inert.

Units, and the factor one half. With $[\tau] = \mathsf{T}$, $[\kappa] = \mathsf{T}^{-1}$ (Section 3 of [1]) and $[X] = \mathsf{L}$, a convention by Proposition 4.11(ii) that makes m a mass (d3-10): $[I_0] = \mathsf{ML}^2$, $[h] = \mathsf{ML}^2\mathsf{T}^{-1}$ (an action), $[p] = \mathsf{MLT}^{-1}$, $[N] = \mathsf{M}^{-1}$, $[H] = [K_{\mathrm{rot}}] = [U] = \mathsf{ML}^2\mathsf{T}^{-2}$, $[M_{\mathrm{rot}}] = \mathsf{M}$ for $M_{\mathrm{rot}} := I_0/\ell^2$; and $E_0(\pi)$ is a momentum squared, $|\nu|E_0(\pi)/m$ an energy. With the clamped value $\kappa = c/\ell$, $K_{\mathrm{rot}} = \frac{1}{2}I_0\kappa^2 = \frac{1}{2}M_{\mathrm{rot}}c^2$, the "$c^2$ part" of the conserved total (interpretive): literally the Newtonian kinetic energy of a mass M_{rot} on a circle of radius ℓ at rim speed c, outside Newtonian kinetics. It is not $E = mc^2$: the $\frac{1}{2}$ comes from an energy quadratic in the rate, where Planck's relation is linear; M_{rot} is not tied to m (Proposition 4.11(iv)); and $\kappa = c/\ell$ holds only for the clamped rotor.

4.4 When the rate is constant

Proposition 4.8 (The multiscale criterion)**.** *Let $R_0 > 0$, $\kappa_0 = \kappa(0) \neq 0$, $\psi_0 = \psi(0)$ and $\varepsilon := R_0/(mI_0\kappa_0^2) = U_{\max}/K_{\mathrm{rot}}(0)$, $U_{\max} = R_0/(2m)$. Then $\kappa^2 = \kappa_0^2[1 + \varepsilon(\cos\psi_0 - \cos\psi)]$, so $|\kappa^2/\kappa_0^2 - 1| \leq 2\varepsilon$; the rotor turns for all time iff $\varepsilon(1 - \cos\psi_0) < 1$, which holds for every phase when $\varepsilon < \frac{1}{2}$; for $\varepsilon(1 - \cos\psi_0) > 1$ it librates, and equality is the separatrix.*

Proof. Divide the conservation of $H = \frac{1}{2}I_0\kappa^2 + \frac{R_0}{2m}\cos\psi$ by $\frac{1}{2}I_0\kappa_0^2$. The least value of κ^2, at $\cos\psi = 1$, is $\kappa_0^2[1 - \varepsilon(1 - \cos\psi_0)]$; if it is negative, κ vanishes where $\sin\psi \neq 0$, hence $\ddot{\theta} \neq 0$, and changes sign. □

This restates Corollary 17.15 of [1] (fixed there), checked along four flows to 8×10^{-12} and in five cases on both sides of the criterion (d3-11); the form $\varepsilon = U_{\max}/K_{\mathrm{rot}}(0)$ is ours. The reading is interpretive (d3-12): if c is the fixed conversion of Theorem 3.2 of [1], a varying $\kappa = c/\ell$ is a varying length $\ell(\tau) = c/\kappa(\tau)$, with relative spread $\varepsilon + O(\varepsilon^2)$ along a rotating solution (measured at $\varepsilon = 10^{-1}, 10^{-2}, 10^{-3}$). A constant "$c/\ell$" requires $\frac{1}{2}M_{\mathrm{rot}}c^2 \gg |\nu|E_0(\pi_0)/m$: the rotor energy must dominate the rest-energy amplitude of the momentum field. The clamped split-form interior of Section 16 of [1] is the limit $\varepsilon \to 0$ (Proposition 17.9 and Corollary 17.15 of [1]); the positive-form interior of Section 15 has duality-class inertia and an exactly constant rate (Theorem 17.11(iii) of [1]). The "symmetric" c^2/ℓ^2 is constant only insofar as the differentiated part is negligible beside the rotor, and ε is an input.

4.5 The rest energy as a rate of rotation

Proposition 4.9 (Two rotations at rest)**.** *(i) For the real first-order Dirac form of Section 4.4 of [1] at $\mathbf{p} = 0$, $H(0) = me_1$, $Je_1 = T = e_1e_2e_3$, $T^2 = -\mathbf{1}$, and $\Psi(\lambda) = \cos(m\lambda)\Psi_0 - \sin(m\lambda)Je_1\Psi_0 = e^{-m\lambda T}\Psi_0$: angular frequency m, that is $mc^2/\hbar$. (ii) The zero-momentum orbits $X(\tau) = e^{-\kappa G\tau}X_0$ satisfy $\alpha_+X = 0$, $\ddot{X} = -\kappa^2X$: angular frequency $\kappa = c/\ell$, $z \mapsto e^{-i\kappa\tau}z$, $q \mapsto e^{-2i\kappa\tau}q$. The orbits $X(\tau) = e^{\kappa G\tau}X_0$, $z = e^{i\kappa\tau}c_+$, are duality rotations of the other sense at the same rate, with momentum $\alpha_+X = 2\kappa GX \neq 0$.*

Proof. (i) $\dot{\Psi} = -JH\Psi$ with $JH(0) = mT$, since e_1 commutes with $J = e_2e_3$. (ii) $\frac{d}{d\tau}e^{\mp\kappa G\tau} = \mp\kappa Ge^{\mp\kappa G\tau}$, $G^2 = -\mathbf{1}$. □

Item (i) is inherited, the realification of $\psi = e^{-imc^2t/\hbar}\psi_0$, de Broglie's internal clock [4] (compare Proposition 3.7); (ii) restates Proposition 16.5(i)–(ii) of [1] (fixed there); both checked exactly (d3-13). Duality acts along both senses at the same rate: the zero-momentum sense is singled out by the momentum, not by duality.

Proposition 4.10 (No operator relates the two rates)**.** *(i) $\mathrm{Ad}_{e^{\vartheta T}}$ fixes e_1, e_2, e_3 and maps a spatial e_a to $\cos 2\vartheta\, e_a + \sin 2\vartheta\, Te_a$, a vector only for $\vartheta \in \frac{\pi}{2}\mathbb{Z}$: the Dirac rest flow is not a motion of V, and T, of grade three, is not in $\mathfrak{so}(3,3)$. (ii) $(e^{\theta G})^{\top}\eta\, e^{\theta G} = \cos 2\theta\,\eta + \sin 2\theta\,\eta G$, so $e^{\theta G} \in O(3,3)$, and lifts to $\mathrm{Pin}(3,3)$, only for $\theta \in \pi\mathbb{Z}$. Hence neither one-parameter group is the image of the other under the spin homomorphism, so this route supplies no relation between m and κ, which remain independent inputs.*

Proof. (i) T commutes with the temporal and anticommutes with the spatial generators (Lemma 2.1 of [1]), so $e^{\vartheta T} e_a e^{-\vartheta T} = e^{2\vartheta T} e_a$, and Te_a has grade four. (ii) Expand with $e^{\theta G} = \cos\theta + \sin\theta\, G$, $G^\top \eta = \eta G$ symmetric, $G^\top \eta G = -\eta$. □

Item (ii) is the remark of Sections 14.3 and 17.6 of [1]. The proposition obstructs one route only, a map of the arena from one rate to the other (d3-15); m and κ are independent because both are free inputs. What remains is an added identification, interpretive (d3-14): with $E = \hbar\times$rate, $\hbar\kappa = mc^2$ holds iff $\ell = \hbar/(mc)$, the reduced Compton wavelength. The dynamics does not supply it. In the interior $\ddot{X} = -\kappa^2 X$ does not contain m; and the rescaling $(\tau, \kappa, I_0, m) \mapsto (s\tau, \kappa/s, sI_0, sm)$ of Proposition 4.11, which maps solutions to solutions and preserves the action (hence any quantisation of h), keeps $m\kappa$ and multiplies m/κ by s^2, so $m = \hbar\kappa/c^2$ picks one point of that orbit by fiat. The rate is also ambiguous by a factor two (q turns at 2κ, z at κ, the Dirac bilinears at 0 or $2m$): $2\hbar\kappa = mc^2$ gives $\ell = 2\hbar/(mc)$, and the algebra decides neither.

4.6 Scale freedom

Proposition 4.11 (Scalings)**.** *(i) η, G, $G\eta^{-1}$, ηG, $\mathcal{K}$ and the structure constants of the complex cross product β_0 are integer matrices.*

(ii) $\ddot{X} = -\kappa^2 X$ is invariant under $X \mapsto \sigma X$, which scales $\frac{m}{2}\eta(\alpha_+ X, \alpha_+ X)$ by σ^2, absorbed by $m \mapsto m/\sigma^2$, and under $(\tau, \kappa) \mapsto (s\tau, \kappa/s)$, which with $m \mapsto sm$ preserves the action.

(iii) The closure is invariant under $(\tau, I_0, m) \mapsto (s\tau, sI_0, sm)$ with (θ, h, p) unchanged as functions of the rescaled time; $\kappa \mapsto \kappa/s$, and $m\kappa$, ε and the action are invariant.

(iv) For $s, t > 0$, $(I_0, m) \mapsto (sI_0, tm)$ at fixed initial rate κ_0 ($h_0 \mapsto sh_0$) keeps $I_0 > 0$ and N of Lorentz class and multiplies ε by $1/(st)$ (by s/t at fixed h_0).

Proof. (i) $\beta_0(e_i, e_j) = \epsilon_{ijk} e_k$, $\beta_0(e_i, e_{j+3}) = \epsilon_{ijk} e_{k+3}$, $\beta_0(e_{i+3}, e_{j+3}) = -\epsilon_{ijk} e_k$; the rest is Lemma 3.4 of [1]. (ii) Linearity; $Y(\tau') = X(\tau'/s)$ solves $Y'' = -(\kappa/s)^2 Y$, the Lagrangian scales by s^{-1} and $d\tau' = s\, d\tau$. (iii) With $d/d\tau' = s^{-1} d/d\tau$, $I_0' = sI_0$, $N' = N/s$, both equations of Proposition 17.8 of [1] are reproduced, $H' = H/s$, and $\kappa_0' = \kappa_0/s$ gives $\varepsilon' = \varepsilon$. (iv) R_0 does not depend on (I_0, m), and $\varepsilon = R_0/(mI_0\kappa_0^2)$ with κ_0 held fixed; at fixed h_0, $\kappa_0 \mapsto \kappa_0/s$. □

Items (i) and (iv) restate the remark closing Section 17.5 of [1] (fixed there; the fixed-κ_0 qualifier is item 12 of `investigations/errata_2026-10-17`); (ii) and (iii) are added; all are fixed. The closure rescaled by $s = 2.5$ reproduces (θ, h, p) to 7×10^{-12}, and ε measured from the range of κ^2 goes from 0.27209155 to 0.04534858 $= \varepsilon/6$ under $(I_0, m) \mapsto (2I_0, 3m)$ at fixed κ_0, and to $\frac{2}{3}\varepsilon$ at fixed h_0 (d3-16, d3-17).

Proposition 4.12 (Nothing in the algebra fixes ℓ)**.** *(i) Schur's lemma [31] fixes only the form $\mathcal{C} = c\,\mathbf{1}$ of the conversion operator (Theorem 3.2 of [1]), and c converts units of time into units of length. (ii) No monomial $c^{n_1}\hbar^{n_2}$ is a length, a mass or a time; with a mass m adjoined, the only length is $\hbar/(mc)$ up to a number. (iii) The interior systems with different $\kappa > 0$ are carried into each other by a rescaling of τ that fixes every Clifford datum (Proposition 4.11(ii)); so no condition formulated in the Clifford data distinguishes one value of κ, or of $\ell = c/\kappa$, and in the closure none fixes ε (Proposition 4.11(iv)).*

Proof. (i) The commutant of the vector representation of $\mathfrak{so}(3)$ in $M_3(\mathbb{R})$ is $\mathbb{R}\,\mathbf{1}$. (ii) In the basis $(\mathsf{M}, \mathsf{L}, \mathsf{T})$, $[c] = (0, 1, -1)$ and $[\hbar] = (1, 2, -1)$ span a plane containing none of $(0, 1, 0)$, $(1, 0, 0)$, $(0, 0, 1)$; adjoining $[m] = (1, 0, 0)$ gives an invertible matrix, with $(0, 1, 0) = -[c] + [\hbar] - [m]$. (iii) The data are integer matrices, and the rescaling acts on τ and κ only. □

This is inherited dimensional analysis, already implicit in equation (kappa) of Section 3 of [1] ("their sum requires a scale") and in the remark closing Section 17.5 (d3-18). It is the one exact form of "m breaks the symmetry of c^2" available here: c carries no scale, the shift κG needs a length ℓ that the algebra cannot supply, and through $\ell = \hbar/(mc)$ that length is a mass. What it "breaks" is the change of the unit of τ (Proposition 4.11(iii)), not a symmetry of a vacuum; the reading is interpretive.

4.7 The frame drifts

Example 4.13 (Example 17.17 of [1], re-verified)**.** Take $p_0 = e^1$, $m = I_0 = \kappa_0 = 1$, $\nu = 1$: $w = 1$, $\varphi_0 = 0$, $\varepsilon = 1$, $\dot\theta^2 = 1 + 2\sin^2\theta$, and by Theorem 17.12 of [1] ($N_c = 0$) $\dot x = e^{2\theta G} N p_0 = \cos 2\theta\, e_1 + \sin 2\theta\, e_4$. The e_4 part integrates to zero ($\theta \mapsto \pi - \theta$), and with $(1 - 2\sin^2\theta)/\sqrt{1 + 2\sin^2\theta} = 2/\sqrt{1 + 2\sin^2\theta} - \sqrt{1 + 2\sin^2\theta}$,

$$\Delta x\big|_{\theta:0\to\pi} = \int_0^\pi \frac{\cos 2\theta\, d\theta}{\sqrt{1 + 2\sin^2\theta}}\, e_1 = (4\mathrm{K}(-2) - 2\mathrm{E}(-2))e_1 \approx 0.3168040511\, e_1.$$

Over a full turn, $\theta : 0 \to 2\pi$, $\Delta x = (8\mathrm{K}(-2) - 4\mathrm{E}(-2))e_1 \approx 0.6336081022\, e_1$, the value of Example 17.17 of [1], whose corrected wording (erratum item 13) makes the turn $\theta \mapsto \theta + 2\pi$, two periods of the rate. After half a turn h has returned and $p = -p_0$ ($e^{\pi G} = -\mathbf{1}$); after a full turn both have returned.

Fixed, as a re-verification (d3-19): quadratures agree with the elliptic expressions to 10^{-25}, and integrating (θ, h, p, x) to the events $\theta = \pi, 2\pi$ reproduces both values and the returns of h, p to 3×10^{-14}. By Theorem 17.16 of [1] every rotating solution with a nonzero rest energy of the momentum field has an unbounded frame; with a free rotor, a constant nonzero rate and a bounded frame coexist exactly on null data (end of Section 17.5 of [1]; at the equilibria of Corollary 4.5 the rate is zero and the frame moves uniformly), or, for several constituents, on the balance surface (Proposition 17.18 of [1]). In motion the differentiation unbinds the frame, and the frame field's rest energy grows with it (Corollary 4.2(ii)).

4.8 A quantum rotor, conditionally

Proposition 4.14 (The half turn)**.** *(i) $e^{\pi G} = -\mathbf{1}$, $e^{2\pi G} = \mathbf{1}$, and for $p_0 \neq 0$ the leaf circle $\{e^{\theta G} p_0\}$ has period exactly 2π. (ii) For every symmetric N, $U(\theta+\pi) = U(\theta)$. (iii) $\Gamma = e_1 e_2 \cdots e_6$ anticommutes with every e_a and $\Gamma^2 = \mathbf{1}$, so $\mathrm{Ad}_\Gamma = -\mathbf{1}$ on V: Γ implements $e^{\pi G} = -\mathbf{1}$ on spinors; by Proposition 4.10(ii) the multiples of the half turn are the only rotor elements that lift to* $\mathrm{Pin}(3,3)$.

Proof. (i) $(e^{\theta G} - \mathbf{1})p_0 = (\cos\theta - 1)p_0 + \sin\theta\, G p_0$ with $p_0 \perp G p_0$, $|G p_0| = |p_0|$. (ii) $p(\theta+\pi) = -p(\theta)$ and U is quadratic. (iii) A product of all six anticommuting generators anticommutes with each, and $(e_1 \cdots e_6)^2 = (-1)^{15} \prod_a \eta_{aa} = \mathbf{1}$. □

This restates the remark of Section 17.6 of [1] (fixed there; in the realisation $\Gamma = J\, e_1 e_4 e_5 e_6 = +e_1 \cdots e_6$), re-verified exactly (d3-20). The quantum statement it permits is conditional on an assumption.

Assumption 4.15 (A quantised leaf)**.** The leaf T^*S^1 of Theorem 4.1(ii) is quantised by wave functions with $\Phi(\theta + 2\pi) = e^{2\pi i\delta}\Phi(\theta)$, $h = -i\hbar\,\partial_\theta$, and the Casimirs of the leaf are treated as numbers.

Proposition 4.16 (A conditional rate ladder)**.** *Under Assumption 4.15: (i) h has spectrum $\hbar(\mathbb{Z} + \delta)$; on flat leaves (Corollary 4.5) the levels are $E_n^{\mathrm{rot}} = \hbar^2(n+\delta)^2/(2I_0)$ (plus the constant $\frac{1}{2}p_0^\top N p_0$ on duality-class leaves), $\kappa_n = (n+\delta)\hbar/I_0$ and $E_{n+1}^{\mathrm{rot}} - E_n^{\mathrm{rot}} = \frac{\hbar}{2}(\kappa_n + \kappa_{n+1})$. (ii) If $R_0 > 0$ the Hamiltonian is a Mathieu operator [20, 6] whose potential couples $e^{i(n+\delta)\theta}$ only to*

$e^{i(n\pm2+\delta)\theta}$, *so for every* δ *the parity of* n *is conserved; it labels the eigenvalues* $\pm e^{i\pi\delta}$ *of the half-turn translation* $\theta \mapsto \theta+\pi$ *(*±1 *for* $\delta=0$*). For* $q_{\mathrm{M}} := I_0R_0/(2m\hbar^2) \gg 1$ *the lowest gap in either parity sector is* $\hbar\Omega - \hbar^2/(2I_0) + O(q_{\mathrm{M}}^{-1/2}\hbar^2/I_0)$*, and the two sectors are degenerate up to tunnelling. (iii) If moreover* $\delta=0$*, and, as a separate assumption, the first level* $h=\hbar$ *is matched by* $\hbar\kappa_1 = mc^2$ *with* $\ell = \hbar/(mc)$*, then* $I_0 = \hbar^2/(mc^2)$*,* $M_{\mathrm{rot}} = m$*, and* $E_1^{\mathrm{rot}} = \frac{1}{2}mc^2$*.*

Proof. (i) Spectrum of $-i\hbar\partial_\theta$ on twisted periodic functions; for $[N,G]=0$, U is the constant of Corollary 4.5(iii). (ii) $\cos(2\theta+\varphi_0)$ shifts n by ±2, and $e^{i(n+\delta)(\theta+\pi)} = (-1)^n e^{i\pi\delta}e^{i(n+\delta)\theta}$. Dividing the Schrödinger equation by $-\hbar^2/(2I_0)$ gives $\Phi'' + (a_{\mathrm{M}} - 2q_{\mathrm{M}}\cos(2\theta+\varphi_0))\Phi = 0$, $a_{\mathrm{M}} = 2I_0E/\hbar^2$, with $a_{\mathrm{M}} \sim -2q_{\mathrm{M}} + 2(2r_{\mathrm{M}}+1)\sqrt{q_{\mathrm{M}}} - \frac{1}{8}((2r_{\mathrm{M}}+1)^2+1)$ as $q_{\mathrm{M}}\to\infty$ (DLMF 28.8.1); the levels $r_{\mathrm{M}} = 0, 1$ lie in the same sector and differ by $4\sqrt{q_{\mathrm{M}}} - 1$, and $\hbar\Omega = (\hbar^2/2I_0)\,4\sqrt{q_{\mathrm{M}}}$. (iii) $\hbar/I_0 = c/\ell = mc^2/\hbar$ and $E_1^{\mathrm{rot}} = \hbar^2/(2I_0)$. □

Checked (d3-21; independently, x3-21): at $R_0 = 0$ the ladder is exact for $\delta = 0$ and 0.3; at $R_0/m = 2000$, $\hbar = I_0 = 1$, the lowest gap in each sector is 62.7415 for both δ, against $\hbar\Omega - \hbar^2/(2I_0) = 62.7456$ (62.7416 with the next term of DLMF 28.8.1), and the sectors' ground states agree to 3×10^{-12}. The proposition is constructed: the quantisation of the leaf exists (the orbit method, δ a character of the stabiliser $2\pi\mathbb{Z}$ in the universal cover of the rotor group), and nothing selects δ, I_0 or the matchings; whether it is physical is open. Missing are a reason for the assumption (a quantum theory of the momentum field, which the closure does not give and the split-form frame obstructs, having no ground state, Corollary 3.11), a value of I_0, a coupling to matter (only the $\mathbb{Z}_2$ of the half turn reaches spinors, as Γ, and nothing ties the parity of n to chirality), and any derivation of the two matchings of (iii). $E_1^{\mathrm{rot}} = \frac{1}{2}mc^2$ is $K_{\mathrm{rot}} = \frac{1}{2}h\kappa$ at $h=\hbar$: the $\frac{1}{2}$ is built in, and mc^2 appears only as $\hbar$ times the rate put in, neither as a level nor as a transition energy.

4.9 Assessment

Against the working dictionary (Table 5) the closure gives four answers. *E as the Casimir* $|p|^2$: $E_6(\pi) = \frac{1}{2}|p|^2$ is a definition; the identification of $u = \frac{1}{2}|z|^2$ of the frame with $\frac{1}{2}|p|^2$ fails, since the frame drifts and its u is not conserved (Corollary 4.2(ii)); and the Casimir does not enter the rotor's motion. What is conserved and differentiates in motion is $H = K_{\mathrm{rot}} + U$. c^2 *as the factorising shift* κG: in the closure c^2 appears as the rotor term $\frac{1}{2}(I_0/\ell^2)c^2$ and, clamped, as $(\kappa G)^2 = -c^2/\ell^2$; c is fixed up to units by Schur's lemma, ℓ is free, κ is dynamical and constant only when $\varepsilon \ll 1$, and the factor $\frac{1}{2}$ is that of a rotor. m *as the rest energy* $\frac{1}{2}|q|$: confirmed in a precise dynamical sense for the momentum field, whose $E_0(\pi)$ is the amplitude of the duality potential, sets Ω^2 and the energy exchanged, and vanishes on the flat direction; but there it is a Casimir, carried and not generated. For the frame field $\frac{1}{2}|q(X)|$ is not conserved even in the clamped interior (Proposition 16.5(ii) of [1]) and grows without bound under the free-rotor drift, which changes it without selecting a value; and $mc^2 = \hbar\kappa$, which would tie m to the rotor, is an added identification (Proposition 4.10). *b, the Yang–Mills field*: the closure contains none and needs none (its frame is abelian, Theorems 17.6–17.7 of [1]); a Yang–Mills field can only be added as a transport. An admissible one is compact (Theorem 2.3 and Corollary 2.4 of [1]): $b = (A_{\mathrm{d}} + A_{\mathrm{a}}, A_{\mathrm{d}} - A_{\mathrm{a}})$ on the temporal and spatial triples, rotation vectors $\mathbf{a}_{\mathrm{d}}, \mathbf{a}_{\mathrm{a}}$, diagonal part in $\mathfrak{l}_G$, anti-diagonal part in $\mathfrak{p}_G$. In the interior with the shift $b + \kappa G$ (Section 5) the diagonal part alone gives the bounded lines κ, $\kappa \pm |\mathbf{a}_{\mathrm{d}}|$, removable along a worldline [18] and so not a gauge-invariant spectrum; the anti-diagonal part alone gives Jordan blocks at the rate $\sqrt{\kappa^2 + |\mathbf{a}_{\mathrm{a}}|^2}$ and secular growth; $\mathbf{a}_{\mathrm{d}}\cdot\mathbf{a}_{\mathrm{a}} = 0$, $\mathbf{a}_{\mathrm{d}} \neq 0$ screens it, every orbit bounded although $[b,G] \neq 0$, except on the ratio $|\mathbf{a}_{\mathrm{d}}|^2 = \kappa^2 + |\mathbf{a}_{\mathrm{a}}|^2$, where a static mode produces a Jordan block (Theorem 5.10); a generic compact b is unbounded (d3-23). At three constituents $\mathfrak{so}(4)_\Delta$ is the part of $\mathfrak{ps}_W$ commuting with G_3 (Theorem 17.21 of [1]), not the only bounded part ($b = R_{7,13} \pm R_{10,11}$ is bounded with $[b, G_3] \neq 0$, Section 5); it preserves the total $\frac{1}{2}|Z|^2$ and q_3, hence $\frac{1}{2}|q_3|$, but changes the constituents' rest energies, their sum included ($\frac{1}{2} \to \frac{3}{2}$),

claim	statement	class	check
Theorem 4.1	$E_6(\pi)$, $E_0(\pi)$, $\mathbf{S}(\pi)$ are Casimirs; only the duality phase moves	fixed (cf. Prop. 17.8)	d3-1–3
Corollary 4.2	(i) no Hamiltonian changes $E_0(\pi)$; (ii) $E_0(X)$ oscillates or grows	(i) obstructed, (ii) fixed	d3-4, 22
Proposition 4.3	$U = \frac{R_0}{2m}\cos(2\theta+\varphi_0)$, $R_0 = 2\lvert\nu\rvert E_0(\pi_0)$	fixed (restated)	d3-5
Proposition 4.4	$\Omega^2 = 4\lvert\nu\rvert E_0(\pi_0)/(mI_0)$; elliptic period	fixed (corollary)	d3-6
Corollary 4.5	constant rate: null data, $\nu = 0$, equilibria, duality class	fixed (restated)	d3-7
Remark 4.6	$U'' = 2R_0/m$ / θ as a pseudo-Goldstone angle	restated / interpretive	d3-8
Proposition 4.7	$H = K_{\mathrm{rot}} + U$, exchange R_0/m, Lorentz invariance, $E_6(\pi)$ inert	fixed	d3-9
units	$K_{\mathrm{rot}} = \frac{1}{2}M_{\mathrm{rot}}c^2$ is the "c^2 part"; not $E = mc^2$	interpretive	d3-10
Proposition 4.8	κ^2 law, bound 2ε, rotation criterion	fixed (restated)	d3-11
heavy rotor	constant c/ℓ needs $\frac{1}{2}M_{\mathrm{rot}}c^2 \gg \lvert\nu\rvert E_0(\pi_0)/m$	interpretive	d3-12
Proposition 4.9	rest energy as a rotation rate (Dirac; frame, both senses)	inherited / restated	d3-13
matching	$\hbar\kappa = mc^2 \iff \ell = \hbar/(mc)$, an added identification	interpretive	d3-14
Proposition 4.10	no Spin or $O(3,3)$ map relates the two rates	obstructed (one route)	d3-15
Proposition 4.11	integer data; scalings of X, τ, I_0, m	fixed	d3-16, 17
Proposition 4.12	nothing in the algebra fixes ℓ (a mass scale)	inherited	d3-18
Example 4.13	$\Delta x = (8\mathrm{K}(-2) - 4\mathrm{E}(-2))e_1$ per full turn	fixed (restated)	d3-19
Proposition 4.14	$e^{\pi G} = -\mathbf{1}$, implemented by Γ	fixed (restated)	d3-20
Proposition 4.16	conditional ladder $\kappa_n = (n+\delta)\hbar/I_0$; $E_1^{\mathrm{rot}} = \frac{1}{2}mc^2$	constructed	d3-21
b-field	compact transports (Sections 2 and 5): Zeeman, Jordan, screening; $\mathfrak{so}(4)_\Delta$	fixed (restated)	d3-23

Table 5: The claims of Section 4, their classes and the checks of `d3_closure_dynamics.py` (d3-0 verifies the conventions). "Fixed (restated)" and "restated": fixed in [1] (or in another section of this note), restated and re-verified here.

and the Lorentz length of the summed density (Proposition 2.14; d3-23, x3-22). The conclusion, interpretive and drawn from fixed statements: the undivided energy and its differentiation in motion are complete before any b-field appears, and where the hypothesis needs the closure to produce a rest energy the reading fails: for the momentum field it cannot, and the frame field's rest energy changes only by an oscillation or an unbounded drift, with no value selected.

5 The b-field: the antisymmetric part of the shift

The hypothesis places the b-field of Yang and Mills [41] at the origin of the mass. This section asks where a connection enters the interior dynamics of Sections 15–17 of [1], and what it can do there. A constant shift of the interior Lagrangian splits into the η-self-adjoint binding term κG and an η-antisymmetric part $b \in \mathfrak{so}(3,3)$, and only b is a connection (Theorem 5.2). A Yang–Mills b-field is moreover compact (Theorem 2.3 and Corollary 2.4 of [1]), $b = (A+C, A-C)$ on the two triples. Its diagonal part splits the binding frequency into Zeeman lines that can be gauged away along any worldline (Theorem 5.6 and Proposition 5.7); its anti-diagonal part alone tears the binding (Theorem 5.8); together they keep every orbit bounded exactly when the triples rotate at equal rates about axes that are not opposite, outside one exceptional ratio (Theorem 5.10). On its own b does not bind, and nothing in the interior dynamics ties the mass m to the rate κ (Proposition 5.12). Units are $\hbar = c = 1$ unless a symbol is shown; τ has dimension

T, $\kappa = c/\ell$ (equation (7) of [1]) and every eigenvalue of b are rates, and $\hbar\kappa$ is a quantum of interior energy. Every finite-dimensional statement was checked by `verify/d4_bfield.py` and independently by `verify/x4_check.py` (Table 6).

Notation. For $a < b$, $R_{ab} \in \mathfrak{so}(3,3)$ is defined by $R_{ab}e_b = \eta_{bb}e_a$ and $R_{ab}e_a = -\eta_{aa}e_b$; on vectors it equals $\frac{1}{2}\operatorname{ad}(e_ae_b)$. The generators of $\mathfrak{l}_G$ (Theorems 12.1 and 12.3 of [1]) are $L_i = \frac{1}{2}\operatorname{ad}(e_{j+3}e_{k+3} - e_je_k)$ and $K_i = \operatorname{ad}(\frac{1}{2}(e_ke_{j+3} - e_je_{k+3}))$ for cyclic (i,j,k). On $z = \mathbf{B} + i\mathbf{E}$ they act as $z \mapsto \mathbf{e}_i \times z$ and $z \mapsto i\,\mathbf{e}_i \times z$, so $K_i = GL_i$; the L_i are rotations and the K_i boosts, and

$$b = \textstyle\sum_i (w_i^{\mathrm{r}} L_i + w_i^{\mathrm{b}} K_i) \in \mathfrak{l}_G \quad \text{acts as} \quad z \mapsto w \times z, \qquad w = w^{\mathrm{r}} + i\,w^{\mathrm{b}} \in \mathbb{C}^3. \tag{8}$$

In particular $R_{12} - R_{45} = -L_3$. The grading root is $\mathcal{K} = \frac{1}{2}\operatorname{ad}(\sum_i e_ie_{i+3})$, with $G = -\mathcal{K}\eta$ (Lemma 3.4 of [1]). We write σ_j for the boost exchanging e_j and e_{j+3}, $F^\dagger = \eta^{-1}F^\top\eta$, $[\mathbf{v}]_\times$ for the matrix of $\mathbf{x} \mapsto \mathbf{v} \times \mathbf{x}$ on $\mathbb{R}^3$, $\mathcal{E}_\mu(T) = \ker(T - \mu)$, and $h(y,y') = \bar{y}^\top\eta y'$. The interior Lagrangian with a constant shift is

$$\mathcal{L} = \frac{m}{2}\,\eta(u,u), \qquad u = \dot{X} + SX, \qquad S = b + \kappa G, \qquad b \in \mathfrak{so}(3,3), \tag{9}$$

with $p = m\eta u$ and $\pi = \eta^{-1}p = mu$. For $b = 0$ this is the split-form Lagrangian of Theorem 16.2 of [1].

5.1 Two summands, and which one is a connection

Proposition 5.1 (Two summands)**.** $\operatorname{End}(V) = \mathfrak{so}(3,3) \oplus \{F : F = F^\dagger\}$*, of dimensions* $15 + 21$*. The part of* $S = b + \kappa G$ *in the first summand is* b*, and the part in the second is* κG*. The binding root lies in the second summand (*$G^\dagger = G$*) and the grading root in the first (*$\mathcal{K}^\dagger = -\mathcal{K}$*).*

Proof. $F \mapsto F^\dagger$ is an involution, and F lies in its ∓ 1 eigenspace exactly when ηF is antisymmetric (respectively symmetric). □

This restates Theorem 17.3 of [1] at $n = 1$, and the placement of the two roots is Proposition 3.3 and Theorem 17.3 of [1] (fixed there); the decomposition itself is linear algebra valid for any nondegenerate form. The next theorem gives it dynamical meaning.

Theorem 5.2 (Covariant factorisation)**.** *Let* $b \in \mathfrak{so}(3,3)$ *be constant and* $D := \frac{d}{d\tau} + b$*.*

(i) The Euler–Lagrange equations of (9) *are* $(D - \kappa G)\pi = 0$ *with* $\pi = m(D + \kappa G)X$*, that is*

$$(D - \kappa G)(D + \kappa G)X = 0 \qquad \Longleftrightarrow \qquad D^2X + \kappa^2 X + \kappa\,(D_\tau G)\,X = 0, \qquad D_\tau G := [b, G]. \tag{10}$$

(ii) Let $X = g(\tau)Y$ *with* $g(\tau) \in O(3,3)$*. Then* $(\frac{d}{d\tau} + S)(gY) = g(\frac{d}{d\tau} + S^g)Y$*, where* $S^g = g^{-1}Sg + g^{-1}\dot{g}$*. The* $\mathfrak{so}(3,3)$ *part* $b^g = g^{-1}bg + g^{-1}\dot{g}$ *transforms as a connection, the self-adjoint part* $\kappa g^{-1}Gg$ *as a tensor, and* $(D_\tau G)^g = g^{-1}(D_\tau G)g$*.*

Proof. (i) $\partial\mathcal{L}/\partial\dot{X} = p$ and $\partial\mathcal{L}/\partial X = S^\top p$, so $\dot{p} = S^\top p$, that is $\dot{\pi} = S^\dagger\pi = (-b + \kappa G)\pi$, which is $(D - \kappa G)\pi = 0$. Moreover $\pi = m(D + \kappa G)X$, $D(GX) - GDX = [b, G]X$ and $G^2 = -\mathbf{1}$. (ii) Differentiating $g^\top\eta g = \eta$ gives $(g^{-1}\dot{g})^\dagger = -g^{-1}\dot{g}$, and $(g^{-1}Fg)^\dagger = g^{-1}F^\dagger g$ preserves both summands; finally $\frac{d}{d\tau}(g^{-1}Gg) + [b^g, g^{-1}Gg] = g^{-1}[b, G]g$. Checked with fifteen symbolic coordinates for b and a Cayley curve in $SO(3,3)$. □

Part (ii) is inherited, the transformation law of a connection and of an endomorphism field in any gauge theory. Part (i) is fixed: the split-form, internal-connection version of the curved extension at the end of Section 15 of [1], with the single new term $\kappa(D_\tau G)X$. That the binding part is a tensor restates Theorem 17.3 of [1] ($G = G^\dagger$): no gauge transformation creates, removes or rescales it. The dictionary entry "$b \leftrightarrow$ transport part of the shift" is correct, and forced.

Proposition 5.3 (Commuting and anticommuting parts)**.** $\mathrm{Ad}_G : b \mapsto GbG^{-1}$ *is an involution of* $\mathfrak{so}(3,3)$*, so* $\mathfrak{so}(3,3) = \mathfrak{l}_G \oplus \mathfrak{p}_G$*, where* $\mathfrak{l}_G = \{[b, G] = 0\}$ *has dimension* 6 *and* $\mathfrak{p}_G = \{bG = -Gb\}$ *has dimension* 9*. In the variable* z*,*

$$\mathfrak{l}_G = \{z \mapsto (P + iQ)z\} \cong \mathfrak{so}(3, \mathbb{C}), \qquad \mathfrak{p}_G = \{z \mapsto (C + i\Sigma)\bar{z}\},$$

with P, Q, C *real antisymmetric and* Σ *real symmetric. The Killing form* $4\,\mathrm{Tr}$ *has signature* $(3,3)$ *on* $\mathfrak{l}_G$*. On* $\mathfrak{p}_G$ *it is negative on the three directions* $\Sigma = 0$*, compact counter-rotations of the two triples such as* $R_{12} + R_{45}$*, and positive on the six boosts* $C = 0$*. Both* $\mathcal{K}$ *(*$z \mapsto -i\bar{z}$*) and the pairing boosts* σ_j *lie in* $\mathfrak{p}_G$*.*

Proof. $G^{-1} = -G$ and $G^\dagger = G$ give $(GbG^{-1})^\dagger = -GbG^{-1}$ and $\mathrm{Ad}_G^2 = \mathbf{1}$. In $(\mathbf{B}, \mathbf{E})$ blocks, commuting with G means the form $\left(\begin{smallmatrix} P & -Q \\ Q & P \end{smallmatrix}\right)$ and anticommuting means $\left(\begin{smallmatrix} C & \Sigma \\ \Sigma & -C \end{smallmatrix}\right)$; $b^\dagger = -b$ gives the symmetry types. The Killing form was computed in the adjoint representation. □

This is fixed; the description of $\mathfrak{l}_G$ restates Theorem 12.1 of [1]. The pairing boosts that tear the binding at a threshold (Proposition 16.7 of [1]) are anticommuting, and so is the grading root.

Gauge admissibility. By Theorem 2.3 and Corollary 2.4 of [1] ("no gauge boson is a boost"), a Yang–Mills b-field with a positive kinetic term takes values in the compact part $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$ of $\mathfrak{so}(3,3)$, spanned by the R_{ab} with a, b in the same triple. On the two triples,

$$b = \mathrm{diag}(A + C,\, A - C), \qquad A = [\mathbf{a}]_\times, \quad C = [\mathbf{c}]_\times, \tag{11}$$

so the triples rotate about $\mathbf{a} + \mathbf{c}$ and $\mathbf{a} - \mathbf{c}$. The diagonal part (A, A) lies in $\mathfrak{l}_G$, with $w = \mathbf{a}$ real in (8); the anti-diagonal part $(C, -C)$ lies in $\mathfrak{p}_G$, with $\Sigma = 0$ ($z \mapsto \mathbf{c} \times \bar{z}$). Boost-valued b below (K_i, σ_j, $\mathcal{K}$, the Σ-part of $\mathfrak{p}_G$) are metric connections in the sense of Proposition 16.7 of [1], legitimate in (9), but not gauge fields.

5.2 The phase-space generator and a criterion for boundedness

Lemma 5.4 (Bookkeeping)**.** *On* (X, π) *the flow is generated by* $M = \left(\begin{smallmatrix} -S & \mathbf{1}/m \\ 0 & S^\dagger \end{smallmatrix}\right)$*, with* $S^\dagger = -b + \kappa G$*. It is the Hamiltonian flow of the Noether energy* $H(X, p) = \frac{1}{2m} p^\top \eta^{-1} p - p^\top S X$*. H vanishes on the invariant Lagrangian subspace* $p = 0$*, and* $\det(\lambda - M) = \chi_S(\lambda)\chi_S(-\lambda)$*, where* $\chi_S(\lambda) = \det(\lambda - S)$ *(since* $\dot{X} = \pi/m - SX$*,* $\dot{\pi} = S^\dagger \pi$*, and* $S^\dagger$ *is similar to* $S^\top$*).*

This is Hamiltonian bookkeeping, inherited. The classification rests on the following criterion.

Theorem 5.5 (When the interior motion stays bounded)**.** *Let* S *be any constant real* 6×6 *matrix.*

(i) M *is semisimple if and only if* S *is semisimple and, for every* $\mu \in \mathbb{C}$*,* $\mathcal{E}_\mu(S^\dagger)$ *and* $\mathcal{E}_{-\mu}(S^\dagger)$ *are orthogonal for the bilinear extension of* η*.*

(ii) Every solution is bounded on $\mathbb{R}$ *if and only if, in addition,* $\mathrm{spec}\, S \subset i\mathbb{R}$*. Then (i) says that every eigenspace of* $S^\dagger$ *is totally isotropic for* h*. Equivalently, every momentum normal mode* $\pi = \mathrm{Re}(y e^{i\omega\tau})$ *with* $\omega \neq 0$*, superpositions within a degenerate frequency included, has period mean* $\langle \eta(\pi, \pi) \rangle = \frac{1}{2} h(y, y) = 0$*, and the static momenta (*$\omega = 0$*) span a totally* η*-null space.*

Proof. M is semisimple if and only if $-S$ (its restriction to $\pi = 0$) and $S^\dagger$ (the quotient map) are, and M is similar to $\mathrm{diag}(-S, S^\dagger)$, which by Roth's theorem [30] means that $-S\Xi - \Xi S^\dagger = -\mathbf{1}/m$ is solvable. With $-S = F_1 \Lambda_1 F_1^{-1}$ and $S^\dagger = F_2 \Lambda_2 F_2^{-1}$ diagonal, this holds if and only if

$(F_1^{-1}F_2)_{ij} = 0$ whenever $(\Lambda_1)_i = (\Lambda_2)_j =: \mu$. A row $\ell^\top$ of F_1^{-1} has $S^\top \ell = -\mu\ell$, and $S^\top = \eta S^\dagger \eta^{-1}$ puts $\eta^{-1}\ell$ in $\mathcal{E}_{-\mu}(S^\dagger)$; the condition is $\ell^\top y = \eta(\eta^{-1}\ell, y) = 0$ for $y \in \mathcal{E}_\mu(S^\dagger)$. (ii): a linear flow is bounded on $\mathbb{R}$ exactly when it is semisimple with imaginary spectrum, and then $\mathcal{E}_{-\mu}(S^\dagger) = \overline{\mathcal{E}_\mu(S^\dagger)}$. □

The theorem is fixed, though valid for any nondegenerate η and any S. Its necessity half is the split virial identity (Theorem 16.1 of [1]: a bound orbit has null momentum on average); the new content is that the averaged virial condition, imposed mode by mode, is also sufficient. For $S = \kappa G$ it is the isotropy of Proposition 16.3 of [1]. On ten test connections, exact semisimplicity, exact Sylvester solvability over $\mathbb{Q}$ and numerical isotropy (to 10^{-8}) agree.

5.3 Connections commuting with G: a Zeeman pattern without a scale

Theorem 5.6 (Connections commuting with G)**.** *Let $b \in \mathfrak{l}_G$ act as in (8).*

(i) $\Xi = -G/(2\kappa m)$ *solves* $-S\Xi - \Xi S^\dagger = -\mathbf{1}/m$*, so M is semisimple if and only if S is.*

(ii) $\chi_S(\lambda) = (\lambda^2 + \kappa^2)\big((\lambda - i\kappa)^2 + w \cdot w\big)\big((\lambda + i\kappa)^2 + \overline{w \cdot w}\big)$.

(iii) In the frame in which b is constant, every solution is bounded if and only if $e^{-b\tau}$ is bounded, that is, if and only if $w = 0$ or $w \cdot w > 0$. The frequencies are then κ and $|\kappa \pm s|$, where $s = \sqrt{w \cdot w}$. For $w \cdot w = 0 \neq w$ the zero-momentum flow grows like τ^2; for $w \cdot w \notin [0, \infty)$ it grows exponentially. For a gauge-admissible b, $w = \mathbf{a}$ is real and every solution is bounded.

(iv) For the rotation $b = \beta(R_{12} - R_{45}) = -\beta L_3$, the spectrum of M is $\pm i\kappa$, $\pm i(\kappa+\beta)$, $\pm i(\kappa-\beta)$, each double. For the boost $b = \beta K_3$, the exponents are $\pm\beta \pm i\kappa$: complex rather than real, so the orbits spiral outward at the binding frequency. The growth is the boost factor $e^{-\beta K_3 \tau}$ of Proposition 5.7, and K_3 is not a gauge generator (Corollary 2.4 of [1]).

Proof. (i) With $[b, G] = 0$: $-(b + \kappa G)\Xi - \Xi(-b + \kappa G) = \frac{1}{2\kappa m}(bG - Gb + 2\kappa G^2) = -\frac{1}{m}$. (ii) On $(V, G) \cong \mathbb{C}^3$, S is $z \mapsto i\kappa z + w \times z$, and $(w\times)^3 = -(w \cdot w)(w\times)$ gives $w\times$ the eigenvalues $0, \pm is$; the real characteristic polynomial of a complex-linear map is its complex one times the conjugate. (iii) By Proposition 5.7, $X = e^{-b\tau}Y$ with Y a solution at $b = 0$, bounded by (i)–(ii); conversely the zero-momentum solutions are $e^{-b\tau}e^{-\kappa G\tau}X_0$ with $e^{\kappa G\tau}$ bounded. And $e^{-b\tau}$ is bounded exactly when $w\times$ is semisimple with imaginary eigenvalues: s real, and not $w \cdot w = 0 \neq w$, where $N = w\times$ has $N^2\bar{w} = |w|^2 w \neq 0 = N^3$. For real w, $w\times$ is real antisymmetric. (iv) $s = \pm\beta$, respectively $\pm i\beta$; semisimplicity was checked exactly at $(\kappa, \beta) = (1, \frac{3}{7})$ and $(1, \frac{3}{2})$. □

The theorem is fixed, but its content deflates. Criterion (iii) is a property of the frame in which b is constant, not of the binding: in the co-moving frame of Proposition 5.7 the binding survives every commuting connection, as the paper warns after Proposition 16.7 of [1] ("growth on one worldline proves nothing"), and for admissible b it holds automatically. In that frame a rotation lifts the degeneracy noted there ("the single frequency is degenerate across the three pairs") into the triplet $\kappa - \beta$, κ, $\kappa + \beta$. Read as a frame vector through the Lie isomorphism of the frame with $\mathfrak{l}_G$ (Theorem 17.5 of [1]), the connection has invariant $q = w \cdot w$; the interpretive reading is that the lab-frame motion stays bounded exactly for a connection of magnetic type, $\mathbf{E}_b \cdot \mathbf{B}_b = 0$ and $|\mathbf{B}_b| > |\mathbf{E}_b|$, while the binding survives every commuting connection, and an admissible one has $\mathbf{E}_b = 0$.

Proposition 5.7 (The commuting part is a moving frame)**.** *Let $b \in \mathfrak{l}_G$. Then $(\frac{d}{d\tau} + b + \kappa G)(e^{-b\tau}Y) = e^{-b\tau}(\dot{Y} + \kappa G Y)$. The exponents are therefore $\pm i\kappa$ shifted by the eigenvalues of $-b$, and with $p_X = e^{b^\top \tau}p_Y$,*

$$H_{\text{lab}} = H_{\text{co}} - p_Y^\top b Y.$$

Proof. $e^{-b\tau}$ is an isometry that commutes with b and G; moreover $e^{b\tau}\eta^{-1}e^{b^\top\tau} = \eta^{-1}$ and $e^{b\tau}(b + \kappa G)e^{-b\tau} = b + \kappa G$. □

For compact b this is Larmor's theorem [18], inherited (a rotating frame trades the field for an angular velocity), and the first sentence of Proposition 16.7 of [1]; for an $\mathfrak{l}_G$ boost, $e^{-b\tau}$ is a uniformly boosted frame instead. $Q_b = p_Y^\top bY$ is the Noether charge of the $\mathfrak{l}_G$ generator, so H_{lab} carries a chemical potential for it. The reading is interpretive but sharp: the Zeeman pattern labels the levels of the binding frequency by their $\mathfrak{l}_G$ charge, has no gauge-invariant content on one worldline, and is observable only against a reference that b does not rotate, as the Zeeman effect is against the laboratory. A commuting b supplies a spectrum but no invariant scale; β/κ is free.

5.4 Connections anticommuting with G, and screening

Theorem 5.8 (Connections anticommuting with G)**.** *Let $b \in \mathfrak{p}_G$ with $b \neq 0$.*

(i) $(\pm b + \kappa G)^2 = b^2 - \kappa^2$, *and* $M^2 = \operatorname{diag}(b^2 - \kappa^2, b^2 - \kappa^2) + N$ *with* $N = \left(\begin{smallmatrix} 0 & -2b/m \\ 0 & 0 \end{smallmatrix}\right) \neq 0$. *Hence M is never semisimple, and some solutions grow at least linearly.*

(ii) For the admissible counter-rotation $b = \beta(R_{12} + R_{45})$, which is $(C, -C)$ with $\mathbf{c} = -\beta\mathbf{e}_3$, $\det(\lambda - M) = (\lambda^2 + \kappa^2)^2(\lambda^2 + \kappa^2 + \beta^2)^4$. *The exponents $\pm i\sqrt{\kappa^2 + \beta^2}$ have algebraic multiplicity four and geometric multiplicity two, and $\pm i\kappa$ are semisimple. The bounded initial data are exactly those with $\pi_1 = \pi_2 = \pi_4 = \pi_5 = 0$, that is, with no momentum on the support of b. The zero-momentum orbits are bounded and turn at $\sqrt{\kappa^2 + \beta^2}$ on the support.*

(iii) For a pairing-plane boost $b = \nu\,\sigma_j$, the exponents in that plane are $\pm\sqrt{\nu^2 - \kappa^2}$, and there $M^2 - (\nu^2 - \kappa^2) = \left(\begin{smallmatrix} 0 & -2\nu\sigma_j/m \\ 0 & 0 \end{smallmatrix}\right)$*: Jordan blocks for every $\nu \neq 0$, below, at and above the threshold (Proposition 16.7 of [1] and the computation after it, where ν is called c).*

(iv) The grading root as a connection satisfies $(\nu\mathcal{K} + \kappa G)^2 = (\nu^2 - \kappa^2)\mathbf{1}$ on all of V.

Proof. (i) $\{b, G\} = 0$ gives both squares, and M^2 has off-diagonal block $-2b/m$. With $b^2 - \kappa^2 = Q_s + Q_n$ (Jordan–Chevalley), $M^2 = \operatorname{diag}(Q_s, Q_s) + (\operatorname{diag}(Q_n, Q_n) + N)$ with commuting summands (Q_n is a polynomial in b), so by uniqueness M^2, and hence M, is semisimple only if $\operatorname{diag}(Q_n, Q_n) + N = 0$, impossible for off-diagonal $N \neq 0$. (ii)–(iv): exact ranks and null spaces over $\mathbb{Q}(i)$ at $(\kappa, \beta) = (4, 3)$ and at $\nu = \frac{1}{2}, 1, 2$, $\kappa = m = 1$ ($\operatorname{rank}(M^2 + \frac{3}{4}) = 2$ on the plane at $\nu = \frac{1}{2}$), with symbolic characteristic polynomials; (i) with nine symbolic coordinates on $\mathfrak{p}_G$. □

Part (i) is an obstruction: no purely anticommuting connection, admissible or not, is compatible with bounded normal modes. Parts (ii) and (iv) are fixed; (iii) re-verifies Proposition 16.7 of [1] and the computation following it. The rate changes to $\sqrt{\kappa^2 + |\mathbf{c}|^2}$ for an admissible element and $\sqrt{\kappa^2 - \nu^2}$ for a pairing boost (a metric connection, not a gauge field), and the momentum sector behaves as under the positive form of Theorem 15.5 of [1]: exponents carried by Jordan blocks, imaginary for a compact element and for a boost below the threshold.

Proposition 5.9 (What a gauge transformation removes)**.** *Let $b = b_l + b_p \in \mathfrak{l}_G \oplus \mathfrak{p}_G$ be constant. Some $g(\tau) \in O(3,3)$ carries (b, G) to $(0, G')$ with G' constant if and only if $[b, G] = 0$. For $b \in \mathfrak{p}_G$, $e^{b\tau}Ge^{-b\tau} = e^{2b\tau}G$. The gauge-covariant remainder is $D_\tau G = [b, G] = 2b_pG$: along a worldline, b_l is pure gauge and b_p is not.*

Proof. $b^g = 0$ forces $g = e^{-b\tau}g_0$, and then $G' = g_0^{-1}e^{b\tau}Ge^{-b\tau}g_0$ has derivative $g_0^{-1}e^{b\tau}[b, G]e^{-b\tau}g_0$. If b anticommutes with G, then $Gf(b) = f(-b)G$ for every power series f. □

This is fixed. A connection on a line has no curvature (the non-abelian $[b_\mu, b_\nu]$ needs two directions; inherited); once the binding root is present, the invariant is $D_\tau G$, exactly the anticommuting part. One might expect boundedness to force $b \in \mathfrak{l}_G$. It does not, and the admissible case shows it.

Theorem 5.10 (Compact screening)**.** *Let b be admissible, as in (11), and $\kappa > 0$.*

(i) If every solution is bounded, then $\mathbf{a}\cdot\mathbf{c} = 0$: the two triples rotate at equal rates, $|\mathbf{a}+\mathbf{c}| = |\mathbf{a}-\mathbf{c}|$.

(ii) Let $\mathbf{a}\cdot\mathbf{c} = 0$ and $\mathbf{a} \neq 0$. Then $\operatorname{spec} S = \{\pm i\omega_1, \pm i\omega_2, \pm i\omega_3\}$, where the ω_j are the roots of

$$f(\omega) = \omega^3 + \kappa\omega^2 - (\kappa^2 + |\mathbf{a}|^2 + |\mathbf{c}|^2)\,\omega - \kappa(\kappa^2 + |\mathbf{c}|^2 - |\mathbf{a}|^2),$$

and if $|\mathbf{a}|^2 \neq \kappa^2 + |\mathbf{c}|^2$, every solution is bounded, although $[b, G] = 2b_p G \neq 0$ when $\mathbf{c} \neq 0$.

(iii) If $\mathbf{a}\cdot\mathbf{c} = 0$, $\mathbf{c} \neq 0$ and $|\mathbf{a}|^2 = \kappa^2 + |\mathbf{c}|^2$, the vector $X_0 = (\mathbf{n}, (\mathbf{a}+\mathbf{c})\times\mathbf{n}/\kappa)$, $\mathbf{n} = \mathbf{a}\times\mathbf{c}$, is static ($SX_0 = 0$) with $\eta(X_0, X_0) = -2|\mathbf{c}|^2|\mathbf{n}|^2/\kappa^2 \neq 0$, and some solutions drift linearly. For $\mathbf{a} = 0$ see Theorem 5.8.

Proof. $S = b + \kappa G$ is real antisymmetric, since b and G are; so S and $S^\dagger = -\eta S\eta = -b + \kappa G$ are normal with imaginary spectrum, and by Theorem 5.5 the motion is bounded if and only if $P_\mu \eta P_\mu = 0$ for the orthogonal spectral projectors P_μ of $S^\dagger$. (i) Then $\operatorname{Tr}(\eta(S^\dagger)^2) = \sum_\mu \mu^2 \operatorname{Tr}(P_\mu \eta P_\mu) = 0$. But $\{b, G\}$ is block-off-diagonal, $\operatorname{Tr}\eta = 0$ and $\operatorname{Tr}[\mathbf{v}]_\times^2 = -2|\mathbf{v}|^2$, so $\operatorname{Tr}(\eta(S^\dagger)^2) = \operatorname{Tr}(A+C)^2 - \operatorname{Tr}(A-C)^2 = -8\,\mathbf{a}\cdot\mathbf{c}$. (ii) Let ρ be the half-turn about $\mathbf{a}$ and $\hat\rho = \operatorname{diag}(\rho, \rho)$. Since $\rho\mathbf{a} = \mathbf{a}$ and $\rho\mathbf{c} = -\mathbf{c}$, $\hat\rho b \hat\rho = \operatorname{diag}(A - C, A + C) = G^{-1} b G$, so $G' := G\hat\rho$ commutes with b and G, hence with S and $S^\dagger$. Moreover $G'^2 = -\mathbf{1}$ and $G'^\dagger = G'$, so, as for G in Proposition 16.3 of [1], each eigenspace $\mathcal{V}_\pm$ of G' (eigenvalue $\pm i$) is totally h-isotropic: $-i\,h(y, y') = h(G'y, y') = h(y, G'y') = i\,h(y, y')$ on $\mathcal{V}_+$. In the $(\mathbf{B}, \mathbf{E})$ blocks, $\mathcal{V}_+ = \{(\mathbf{x}, -i\rho\mathbf{x})\}$, and S acts there as $\mathbf{x} \mapsto (\mathbf{a}+\mathbf{c})\times\mathbf{x} + i\kappa\rho\mathbf{x}$; after a common rotation of both triples (which commutes with G and η) to $\mathbf{a} = |\mathbf{a}|\mathbf{e}_3$, $\mathbf{c} = |\mathbf{c}|\mathbf{e}_1$, one finds $\det(i\omega - [\mathbf{a}+\mathbf{c}]_\times - i\kappa\rho) = -if(\omega)$. So S has eigenvalues $i\omega_j$ on $\mathcal{V}_+$ and $-i\omega_j$ on $\mathcal{V}_- = \overline{\mathcal{V}_+}$; since η exchanges $\mathcal{V}_\pm$, $S^\dagger$ has the same eigenvalues there. Every eigenspace of $S^\dagger$ splits along $\mathcal{V}_\pm$, and lies in one of them unless $f(\omega) = f(-\omega) = 0$ for some ω. But $\frac{1}{2}[f(\omega) + f(-\omega)] = \kappa(\omega^2 - \kappa^2 - |\mathbf{c}|^2 + |\mathbf{a}|^2)$ and $\frac{1}{2}[f(\omega) - f(-\omega)] = \omega(\omega^2 - \kappa^2 - |\mathbf{a}|^2 - |\mathbf{c}|^2)$, so a common root forces $\mathbf{a} = 0$, or $\omega = 0$ and $|\mathbf{a}|^2 = \kappa^2 + |\mathbf{c}|^2$. (iii) With $\mathbf{u} = \mathbf{a}+\mathbf{c}$, $\mathbf{v} = \mathbf{a}-\mathbf{c}$: $SX_0 = (\mathbf{u}\times\mathbf{n} - \mathbf{u}\times\mathbf{n},\ \mathbf{v}\times(\mathbf{u}\times\mathbf{n})/\kappa + \kappa\mathbf{n}) = 0$, because $\mathbf{v}\cdot\mathbf{n} = 0$ and $\mathbf{u}\cdot\mathbf{v} = |\mathbf{a}|^2 - |\mathbf{c}|^2 = \kappa^2$; and $\eta(X_0, X_0) = |\mathbf{n}|^2 - |\mathbf{u}|^2|\mathbf{n}|^2/\kappa^2$. So $\mathcal{E}_0(S^\dagger) = \eta\,\mathcal{E}_0(S)$ is not η-null, and Theorem 5.5 fails at $\mu = 0$. Exact checks: the identities symbolically, boundedness at four rational points (one with $\mathbf{a}, \mathbf{c}$ in general position), and the exceptional case at $(\kappa, |\mathbf{c}|, |\mathbf{a}|) = (3, 4, 5)$ and $(\frac{3}{5}, \frac{4}{5}, 1)$. □

The theorem is fixed, and it is the case that matters for a gauge field. Its mechanism is a second binding root $G' = G\hat\rho$ commuting with S. The family was found by the adversarial check (x4-N1: 500 of 500 random $\mathbf{a} \perp \mathbf{c}$ bounded, none of 500 generic compact b); part (i) explains the second count, and the exceptional ratio, of measure zero, was found here. At $\mathbf{c} = 0$ it gives the Zeeman lines of Theorem 5.6. For $\mathbf{c} \neq 0$ the screened frequencies cannot be removed along a worldline (Proposition 5.9), but they are κ times functions of $|\mathbf{a}|/\kappa$ and $|\mathbf{c}|/\kappa$: screening supplies no scale.

Proposition 5.11 (Gyroscopic screening by a boost)**.** *Let $b = \alpha L_3 + b_\Sigma$, where $b_\Sigma \in \mathfrak{p}_G$ is the boost $z \mapsto i\Sigma\bar z$ with $\Sigma = \left(\begin{smallmatrix} y_1 & y_2 & 0 \\ y_2 & -y_1 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, and let $r^2 = y_1^2 + y_2^2 > 0$. Then $[b, G] \neq 0$ and*

$$\chi_S(\lambda) = (\lambda^2 + \kappa^2)\big(\lambda^2 + (\kappa - \alpha)^2 - r^2\big)\big(\lambda^2 + (\kappa + \alpha)^2 - r^2\big).$$

If $\alpha \neq 0$ and $r < \min(|\kappa - \alpha|, |\kappa + \alpha|)$, every solution is bounded; for $\alpha = 0$ (Theorem 5.8) and for $|\kappa - \alpha| < r < |\kappa + \alpha|$ some are not. Numerically, near bounded points at which the five conditions of the proof are independent, the set of $b \in \mathfrak{so}(3, 3)$ with all motion bounded is a 10-dimensional submanifold; at the tested points of $\mathfrak{l}_G$ and of this family the conditions are degenerate (Jacobian rank four), so these lie on its singular locus.

Proof. On $\langle e_1, e_2, e_4, e_5\rangle$, $K_3 = GL_3$ is an η-antisymmetric involution commuting with G, L_3 and b_Σ, so its eigenplanes $\mathcal{U}_\pm$ are totally η-null; there $L_3 = \mp G$, $S = (\kappa \mp \alpha)G + b_\Sigma$ and $S^\dagger = (\kappa \pm \alpha)G - b_\Sigma$, each squaring to a negative multiple of $\mathbf{1}$ ($b_\Sigma^2 = r^2$, $\{b_\Sigma, G\} = 0$). For $\alpha \neq 0$ the planes carry different frequencies, so every eigenvector of $S^\dagger$ lies in $\mathcal{U}_+$, $\mathcal{U}_-$ or the b-free plane $\langle e_3, e_6\rangle$ and is h-isotropic; Theorem 5.5 applies (exact checks at $(\alpha, y_1, y_2) = (\frac{1}{2}, \frac{1}{4}, 0)$, $(\frac{1}{3}, \frac{1}{5}, \frac{1}{7})$, $\kappa = 1$). For simple frequencies Theorem 5.5(ii) imposes five real conditions (two for imaginary spectrum after the trace relation, three for isotropy); their finite-difference Jacobian has rank five at four least-squares solutions and at an exact compact screened point, and rank four (residual $\sim t^2$ along kernel directions) at three points of this family and two elliptic points of $\mathfrak{l}_G$. □

The family is fixed; the dimension statements are numerical. This is gyroscopic stabilisation of a saddle [34, 16]: the rotation detunes the two light-like planes of K_3 in which the boost acts, so $\eta(\pi, \pi) = 2\eta(\pi_+, \pi_-)$ averages to zero. Since b_Σ is a boost, the family is not a gauge field. With Theorem 5.10 it corrects the working dictionary: an anticommuting part need not cause secular growth, and boundedness does not single out $\mathfrak{l}_G$; gauge removability does (Proposition 5.9).

Proposition 5.12 (A connection alone does not bind)**.** *(i) Let $\kappa = 0$ and let $b \in \mathfrak{so}(3,3)$ be any constant connection, $b = \nu\mathcal{K}$ included. Then $X = e^{-b\tau}Y$ gives $\mathcal{L} = \frac{m}{2}\eta(\dot{Y}, \dot{Y})$ and $\ddot{Y} = 0$. The generator $M = \left(\begin{smallmatrix} -b & \mathbf{1}/m \\ 0 & -b \end{smallmatrix}\right)$ has $\det(\lambda - M) = \chi_{-b}(\lambda)^2$ and is never semisimple: the motion is free drift, and there is no binding frequency.*

(ii) The rescaling $(\tau, \kappa, b, m) \mapsto (\vartheta\tau, \kappa/\vartheta, b/\vartheta, \vartheta m)$ maps M to M/ϑ, solutions to solutions and the action $\int \mathcal{L}\, d\tau$ to itself, and leaves η, G, $\mathfrak{l}_G$, $\mathfrak{p}_G$ and $\mathfrak{so}(4)_\Delta$ unchanged. No value of κ, hence of $\ell = c/\kappa$, is selected, with or without a constant b.

(iii) $M(m) = TM(1)T^{-1}$ with $T = \mathrm{diag}(\mathbf{1}, m\mathbf{1})$: the spectrum of M does not involve m, and the rescaling of (ii) keeps $m\kappa$, not m/κ, fixed.

Proof. (i) $\dot{X} + bX = e^{-b\tau}\dot{Y}$. Write $-\mathrm{diag}(b, b) = S' + N'$ (Jordan–Chevalley). Then $M = S' + (N' + N)$ with $N = \left(\begin{smallmatrix} 0 & \mathbf{1}/m \\ 0 & 0 \end{smallmatrix}\right)$, and $N' + N \neq 0$ because N is off-diagonal and N' is not. (ii) and (iii) are direct. □

Part (i) is inherited (every connection on a line is pure gauge); its consequence is an obstruction: a Yang–Mills potential on a worldline, with no self-adjoint shift, produces free motion. Part (ii) is dimensional analysis, inherited; it restates [1] ("κ is not fixed", end of Section 16; no ratio among the inertia constants and m is fixed by the Clifford structure, Section 17.5). Part (iii) is fixed and elementary: nothing in the interior dynamics ties m to κ, and $mc^2 = \hbar\kappa$, that is $\ell = \hbar/(mc)$, is a matching added from outside, never a result.

5.5 The split virial identity and the vacuum

Proposition 5.13 (Split virial with a connection)**.** *For every constant S, and in particular for every $b \in \mathfrak{so}(3,3)$, $\frac{d}{d\tau}\eta(X, \pi) = \eta(\pi, \pi)/m$ along every solution.*

Proof. $\eta(\pi/m - SX, \pi) + \eta(X, S^\dagger\pi) = \eta(\pi, \pi)/m$; also checked in integral form along numerical solutions (DOP853, error below 10^{-8}). □

This restates and re-verifies Theorem 16.1 of [1] (fixed there), whose metric-compatible connection is our b.

Theorem 5.14 (Krein signatures and the zero-point energy)**.** *Suppose every solution is bounded, and let $\omega > 0$ have multiplicity k_ω in $\operatorname{spec} S$. On the eigenspace $\mathcal{E}_{i\omega}(M)$, the energy form $v \mapsto \bar{v}^\top \mathcal{H} v$, where $\mathcal{H}$ is the Hessian of H, has signature (k_ω, k_ω). Hence, with positive-norm quantisation of the normal modes, the zero-point energy $\sum_j (\pm)\frac{1}{2}\hbar\omega_j$ cancels frequency by frequency, and every level has a lower one.*

Proof. On $\mathcal{E}_{i\omega}(M)$, $\mathcal{H}v = -i\omega\mathbb{J}v$, so the energy form equals ω times the Krein form $-i\bar{v}^{\top}\mathbb{J}v$, which is nondegenerate there because eigenspaces of distinct imaginary eigenvalues are orthogonal for it [40]. The zero-momentum part of $\mathcal{E}_{i\omega}(M)$ has dimension k_ω (the multiplicity of $i\omega$ in $-S$), and H vanishes on it (Lemma 5.4). Since M is semisimple, $\dim \mathcal{E}_{i\omega}(M) = 2k_\omega$. A nondegenerate Hermitian form with a half-dimensional isotropic subspace has signature (k, k), and a mode of energy sign $\pm$ quantises to $\pm\hbar\omega(n + \frac{1}{2})$. □

The theorem is fixed, though structural (any nondegenerate η); it was checked numerically (to 10^{-9}) on the Zeeman triplet, on $w = (2, i, 0)$, on the gyroscopic family, on a compact screened connection and at three constituents. It extends Theorem 18.6 of [1] (zero-point energy 0 on $\mathbb{R}^{3,3}$ at $b = 0$) to every bounded connection: the cancellation, forced by the perfect-square form (9), survives the splitting line by line, and so does the absence of a ground state. On the coordinate subfields of that theorem (zero-point energy $\pm\frac{3}{2}\hbar\kappa$ or $\pm\frac{1}{2}\hbar\kappa$), G does not preserve the subfield (the zero-momentum flow stays in it only on its intersection with its preimage under G, of dimension 0 or 2), and the theorem does not apply.

Corollary 5.15 (No Fock vacuum with Jordan blocks)**.** *Suppose M is not semisimple at a frequency $\omega \neq 0$, as for every nonzero $b \in \mathfrak{p}_G$. Then no linear symplectic change of variables writes the Hamiltonian on that block as a sum of signed oscillators, and there is no Fock vacuum annihilated by normal-mode lowering operators.*

Proof. A sum of signed oscillators generates a semisimple flow, and symplectic changes of variables act by similarity. □

This is an obstruction. A model block with the same Jordan structure (checked) is $H = \omega(q_1p_2 - q_2p_1) + \frac{1}{2}(q_1^2 + q_2^2)$, whose terms Poisson-commute [38]; that its quantisation has continuous spectrum is inherited and not re-derived.

5.6 Three constituents: the part of $\mathfrak{ps}_W$ that commutes with G_3

We use the generators $E_{6(k-1)+a} = \omega_6^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}^{\otimes(3-k)}$, $\omega_6 = e_1e_2\cdots e_6$, of equation (2) of [1] on $\mathbb{R}^{512}$, with the adjoint action of bivectors on the 18 vectors computed exactly, and set $G_3 = G \oplus G \oplus G$. We follow the numbering of Section 17 of [1]: $W = \{7, 13, 14, 15\}$, the colour directions are $\{10, 11, 12, 16, 17, 18\}$, $E = \mathrm{span}\{7, 8, 9, 13, 14, 15\}$, and $\mathrm{rot}(i, j) = e_{i'}e_{j'} - e_ie_j$ acts by $\frac{1}{2}\,\mathrm{ad}$.

Proposition 5.16 (The binding-compatible algebra)**.** *The E_A satisfy the relations of* $\mathrm{Cl}(9, 9)$*, G_3 is an anti-isometry with $G_3^2 = -\mathbf{1}$, and* $\dim(\mathrm{Cent}(G_3) \cap \mathfrak{so}(9, 9)) = 72$*. For four-dimensional $U \subset \mathrm{span}\{2, 3, 7, 8, 9, 13, 14, 15\}$, neither $\mathfrak{so}(U)$ nor $\mathfrak{so}(6)_C$ meets* $\mathrm{Cent}(G_3)$*, and their sum $\mathfrak{ps}_U$ meets it in dimension $\binom{d}{2}$, $d = \dim(U \cap E)$; for $U = W$ in $\mathfrak{so}(4)_\Delta = \mathrm{span}\{\mathrm{rot}(i, j) : i < j \in W\}$. The elements $T_1^{\pm} = \frac{1}{2}(\mathrm{rot}(13, 14) \pm \mathrm{rot}(7, 15))$, $T_2^{\pm} = \frac{1}{2}(\mathrm{rot}(7, 14) \mp \mathrm{rot}(13, 15))$ and $T_3^{\pm} = \frac{1}{2}(\mathrm{rot}(7, 13) \pm \mathrm{rot}(14, 15))$ satisfy $[T_i^+, T_j^-] = 0$ and $[T_1^{\pm}, T_2^{\pm}] = -T_3^{\pm}$ cyclically. The trace form is negative definite, so $\mathfrak{so}(4)_\Delta = \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$ is compact. Each $T_3^{\pm}$ has eigenvalues $\pm\frac{i}{2}$, four times each, and 0, ten times.*

This restates and re-verifies Theorem 17.21 of [1] (fixed there), here for $U = W$, $\{3, 7, 8, 9\}$, $\{2, 3, 7, 8\}$ (dimensions $6, 3, 1$) and in `x4_check.py` for all 70 coordinate U. Each $\mathfrak{su}(2)$ factor locks a weak rotation of W to the colour rotation of the partners $\{10, 16, 17, 18\}$; reading $\mathfrak{so}(4)_\Delta$ as "Yang and Mills' isospin, doubled" is interpretive, as after Theorem 17.22 of [1].

Proposition 5.17 (Spectra on the eighteen-dimensional frame)**.** *Let $\kappa = 5$ and $m = 1$.*

(i) For $b = \beta T_3^+$ with $\beta = 2$, $\chi_S = (\lambda^2 + 25)^5(\lambda^2 + 16)^2(\lambda^2 + 36)^2$: frequency κ on the five complex singlet directions and $\kappa \mp \beta/2$ on the two $\mathfrak{su}(2)_+$ doublets, of weights $\pm\frac{1}{2}$.

(ii) *For* $b = \beta_+ T_3^+ + \beta_- T_3^-$ *with* $(\beta_+, \beta_-) = (2, 4)$*, the lines are* $\kappa \pm \frac{1}{2}(\beta_+ \pm \beta_-) = 8, 6, 4, 2$*, together with* κ*.*

(iii) *In both cases* $\Xi = -G_3/(2\kappa m)$ *solves the Sylvester equation,* M *is semisimple, the Krein signatures are* (k, k)*, and the zero-point energy vanishes.*

(iv) *The weak-only rotation* $R_{7,13}$ *and the colour-only rotation* $R_{10,16}$ *give a non-semisimple* M*, hence secular growth.*

(v) $b = R_{7,13} \pm R_{10,11} \in \mathfrak{ps}_W$ *has* $[b, G_3] \neq 0$*, and every solution is bounded, for every* $\kappa > 0$*. At* $\kappa = 5$ *the frequencies are* κ *(six pairs) and* $|\omega_j| \approx 4.319,\ 5.050,\ 5.731$*, with* ω_j *the roots of* $\omega^3 + 5\omega^2 - 26\omega - 125$*.*

Proof. (i)–(iv): exact characteristic polynomials and semisimplicity tests on 36×36 integer matrices; Krein signatures numerical. (v) G_3 maps the six-plane $\langle e_7, e_8, e_{13}; e_{10}, e_{11}, e_{16}\rangle$ as G maps $\langle e_1, e_2, e_3; e_4, e_5, e_6\rangle$, and b preserves it and vanishes on its η-orthogonal complement, where the motion is the $b = 0$ binding. On the six-plane b is admissible, with unit rotation vectors $\mathbf{e}_2$ and $\pm\mathbf{e}_3$ for the two triples: equal rates, perpendicular axes, so $\mathbf{a} = \frac{1}{2}(\mathbf{e}_2 \pm \mathbf{e}_3) \perp \mathbf{c} = \frac{1}{2}(\mathbf{e}_2 \mp \mathbf{e}_3)$ and $|\mathbf{a}|^2 = |\mathbf{c}|^2 = \frac{1}{2} \neq \kappa^2 + \frac{1}{2}$. Theorem 5.10(ii) applies with f as stated; M is also exactly semisimple at $\kappa = 5$. □

This is fixed. Every constant $\mathfrak{so}(4)_\Delta$-valued connection is compact and commutes with G_3, so by Theorem 5.6(i) it keeps the interior bounded; the splitting follows the $\mathfrak{su}(2)$ weights, β/κ is free, and by Proposition 5.7 the pattern can be removed along one worldline. The single-factor rotations tear the binding, but boundedness does not single out the locked diagonal: part (v), found by the adversarial check (x4-N2) and explained by Theorem 5.10, is bounded with $[b, G_3] \neq 0$ (the anti-locked $R_{7,13} + R_{10,16}$, $\mathbf{a} = 0$ on its six-plane, is not). What singles out $\mathfrak{so}(4)_\Delta$ is commutation with G_3 (Theorem 17.21 of [1]), that is, removability along a worldline (Proposition 5.9).

5.7 Quantum Yang–Mills facts, and the channels of the paper's vacuum

The only place left for b to generate a scale is the quantum theory. We re-verify the facts used by the earlier session.

Proposition 5.18 (Inherited facts)**.** *(i) For* $\mathfrak{su}(4)$ *with* $\mathrm{tr}(T^A T^B) = \frac{1}{2}\delta^{AB}$*,* $\sum_A T^A Z T^A = \frac{1}{2}(\mathrm{tr}(Z)\mathbf{1} - \frac{1}{4}Z)$ *for every* 4×4 *matrix* Z*:* $\frac{15}{8} = C_F$ *on* $\mathbf{1}$*,* $-\frac{1}{8}$ *on traceless* Z*.*

(ii) *The channel factors* $T_1 \cdot T_2 = \frac{1}{2}[C_2(\mathit{pair}) - C_2(R_1) - C_2(R_2)]$ *are* $+\frac{3}{8}, -\frac{5}{8}$ *on* $\mathbf{10}, \mathbf{6} \subset \mathbf{4} \otimes \mathbf{4}$*;* $-\frac{15}{8}, +\frac{1}{8}$ *on* $\mathbf{1}, \mathbf{15} \subset \bar{\mathbf{4}} \otimes \mathbf{4}$*;* $+\frac{1}{4}, -\frac{3}{4}$ *on* $\mathbf{3}, \mathbf{1} \subset \mathbf{2} \otimes \mathbf{2}$ *of* $\mathfrak{su}(2)$ *(*$C_2 = \frac{15}{8}, \frac{9}{2}, \frac{5}{2}, 4$ *on* $\mathbf{4}, \mathbf{10}, \mathbf{6}, \mathbf{15}$*). Hence the Dirac bilinear* $\overline{(\mathbf{4}, \mathbf{2}, \mathbf{1})} \otimes (\mathbf{4}, \mathbf{1}, \mathbf{2}) = (\mathbf{1}, \mathbf{2}, \mathbf{2}) \oplus (\mathbf{15}, \mathbf{2}, \mathbf{2})$ *has* $\mathfrak{su}(4)$ *factors* $-\frac{15}{8}, +\frac{1}{8}$*, and a Lorentz-scalar Majorana bilinear of* $(\mathbf{4}, \mathbf{1}, \mathbf{2})$*, symmetric in its internal indices, lies in* $\mathrm{Sym}^2(\mathbb{C}^4 \otimes \mathbb{C}^2) = (\mathbf{10}, \mathbf{1}, \mathbf{3}) \oplus (\mathbf{6}, \mathbf{1}, \mathbf{1})$ *(*$30 + 6$*), with factors* $(+\frac{3}{8}, +\frac{1}{4})$ *and* $(-\frac{5}{8}, -\frac{3}{4})$*.*

(iii) $\int_{k^2 \le k_{\max}^2} \frac{d^4k}{(2\pi)^4} \frac{1}{k^2 + M_{\mathrm{dyn}}^2} = \frac{1}{16\pi^2}\left[k_{\max}^2 - M_{\mathrm{dyn}}^2 \ln(1 + k_{\max}^2/M_{\mathrm{dyn}}^2)\right]$ *(Euclidean);* $f_{\mathrm{gap}}(x) = 1 - x^2 \ln(1 + x^{-2})$ *decreases strictly from* 1 *to* 0 *(*$f'_{\mathrm{gap}} < 0$ *since* $\ln(1 + u) > u/(1 + u)$*), so* $1 = \lambda_{\mathrm{c}} f_{\mathrm{gap}}(x)$ *has a root* $x = M_{\mathrm{dyn}}/k_{\max} > 0$ *if and only if* $\lambda_{\mathrm{c}} > 1$*; at* $\lambda_{\mathrm{c}} = \frac{7}{4}$ *it is* $x = 0.5330234964\ldots$*, confirming the earlier* 0.533*.*

(iv) *The ladder critical coupling* $\pi/(3C_F)$ *[9, 21] is* $8\pi/45 \approx 0.5585$ *at* $C_F = \frac{15}{8}$*. For* $SU(2)$ *with* N_f *Dirac doublets* $\beta_0 = (22 - 2N_f)/3$*,* 6 *at* $N_f = 2$ *[12, 27], and* $\Lambda = \mu e^{-8\pi^2/(\beta_0 g^2)}$ *is independent of* μ*.* $SU(4) \to Sp(4)$ *(*$N_f = 2$*) leaves* $15 - 10 = 5$ *Goldstone directions [26].*

All of this is inherited [22, 32], and was computed exactly (generalised Gell-Mann basis, joint Casimir eigenspaces on $\mathrm{Sym}^2(\mathbb{C}^8)$), the root of (iii) with mpmath to 25 digits. The reading of (ii) is interpretive. Under the most-attractive-channel heuristic of one-boson exchange [28] (negative factor: attraction), with a common coupling for $\mathfrak{su}(4)$ and $\mathfrak{su}(2)_R$, which the paper itself adopts only as an assumption (Assumption 19.4 and the remark after Proposition 19.5 of [1]), the exchange favours $(\mathbf{1},\mathbf{2},\mathbf{2})$ and the colour-breaking $(\mathbf{6},\mathbf{1},\mathbf{1})$, and disfavours $(\mathbf{15},\mathbf{2},\mathbf{2})$, which the Georgi–Jarlskog factor -3 needs (Proposition 21.2 of [1]; [11]), and $(\mathbf{10},\mathbf{1},\mathbf{3})$, which carries the paper's Standard-Model vacuum (Section 20 of [1]). That vacuum is a minimum of a scalar potential, not a gauge-exchange condensate, so the heuristic addresses a different, dynamical scenario, in which a gauge-generated vacuum would point away from the paper's.

5.8 Synthesis: where the b-field enters

(a) *The binding frequency comes from the self-adjoint part, never from a connection* (fixed; obstructed for b alone): κ, with $(\kappa G)^2 = -c^2/\ell^2$, sits in the tensor part; b alone gives free drift (Theorem 5.2 and Proposition 5.12).

(b) *A commuting b splits the binding frequency without supplying a scale* (fixed; Zeeman reading interpretive): lines κ, $\kappa \pm \sqrt{w \cdot \overline{w}}$, bounded for admissible b, removable along any worldline, β/κ free (Theorem 5.6 and Proposition 5.7); at three constituents $\mathfrak{so}(4)_\Delta$, the part of the Pati–Salam algebra commuting with the binding root, gives $\mathfrak{su}(2)$-weight lines and is not the only bounded part (Proposition 5.17); the zero-point cancellation survives line by line (Theorem 5.14).

(c) *A purely anticommuting b tears the binding* (obstructed; spectra fixed), with rates $\sqrt{\kappa^2 + |\mathbf{c}|^2}$ (admissible) or $\sqrt{\kappa^2 - \nu^2}$ (pairing boost) carried by Jordan blocks (Theorem 5.8 and Corollary 5.15). A mixed one may or may not: an admissible b keeps every orbit bounded exactly at equal rates of the triples about non-opposite axes, outside $|\mathbf{a}|^2 = \kappa^2 + |\mathbf{c}|^2$ (Theorem 5.10); a boost can be screened gyroscopically (Proposition 5.11); neither gives a scale.

(d) *The only role left for b in generating a scale is quantum* (open). The arena leaves ℓ free (Proposition 5.12) and classical four-dimensional Yang–Mills theory has no dimensionful parameter; a scale arises only by dimensional transmutation, $\Lambda_b = \mu\, e^{-8\pi^2/(\beta_0 g^2)}$ [3, 12].

Conjecture 5.19 (A transmuted binding length)**.** *$\ell = \hbar c/\Lambda_b$ for the scale Λ_b of a Yang–Mills b-field, that is, $\hbar\kappa = \Lambda_b$.*

Reading the rest energy mc^2 as $\hbar\kappa$ is a separate assumption, not part of the conjecture: the spectrum of M does not involve m, and the rescaling of Proposition 5.12 keeps $m\kappa$, not m/κ, fixed. The conjecture is open: nothing in the arena derives it. A mechanism would need κ to be dynamical, as the rotor rate of Section 17 of [1] is, and coupled to a quantum gauge sector with a fixed β_0 and matching coupling; even then, by Proposition 5.18(ii), the most attractive channels are not those of the paper's vacuum.

Corrected so that it is true, the user's sentence reads as follows. In $E = mc^2$, read through the added identification $mc^2 = \hbar\kappa$, the symmetric part is the self-adjoint binding shift κG, with square $-c^2/\ell^2$. The Yang–Mills b-field is the antisymmetric, transport part of the same shift, and it is compact. Classically it generates neither m nor κ: when it commutes with G it splits the interior rate κ into charge-labelled lines; a purely anticommuting b tears the binding, and a mixed one may or may not (Theorem 5.10 and Proposition 5.11). Whether its quantum scale is the ℓ that the arena leaves free is open.

statement	class	checks
Proposition 5.1: decomposition; roots on opposite sides	inherited; fixed (restated: Prop. 3.3, Thm 17.3)	d4-0, d4-1
Theorem 5.2: (i) covariant factorisation; (ii) b connection, κG tensor	fixed; inherited	d4-1, d4-2
Proposition 5.3, (11): $\mathfrak{l}_G \oplus \mathfrak{p}_G$; admissible split; Lemma 5.4	fixed; inherited (Lemma 5.4)	d4-3, d4-4
Theorem 5.5: Roth–virial criterion	fixed	d4-11
Theorem 5.6: Zeeman lines; bounded iff $e^{-b\tau}$ bounded; boost spirals	fixed	d4-5 to d4-7
Proposition 5.7: moving frame; Zeeman reading	inherited; interpretive	d4-13
Theorem 5.8: (i) never semisimple; (ii), (iv) spectra; (iii) pairing boost	obstructed; fixed; fixed (restated: Prop. 16.7)	d4-8 to d4-10
Proposition 5.9: removable part $\mathfrak{l}_G$, invariant $D_\tau G$	fixed	d4-13
Theorem 5.10: admissible b bounded iff $\mathbf{a} \perp \mathbf{c}$, $\mathbf{a} \neq 0$, $\lvert\mathbf{a}\rvert^2 \neq \kappa^2 + \lvert\mathbf{c}\rvert^2$	fixed	d4-26
Proposition 5.11: screening by a boost; dimension 10	fixed (dimension numerical, regular points)	d4-12
Proposition 5.12: (i) free drift; (ii) no scale; (iii) m drops out	inherited; inherited (restated); fixed	d4-25
Proposition 5.13; Theorem 5.14; Corollary 5.15	fixed (restated: Thm 16.1); fixed; obstructed	d4-14 to d4-16
Proposition 5.16: $\mathfrak{so}(4)_\Delta = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$	fixed (restated: Thm 17.21)	d4-17, d4-18
Proposition 5.17: (i)–(iv) spectra, single-factor tearing; (v) unlocked bounded b ("only the locked diagonal" withdrawn)	fixed	d4-19, d4-27
Proposition 5.18: Yang–Mills facts; channel, isospin and magnetic-type readings	inherited; interpretive	d4-20 to d4-24
Conjecture 5.19: $\hbar\kappa = \Lambda_b$ ($mc^2 = \hbar\kappa$ a separate assumption)	open	—

Table 6: Claims of Section 5, their classes, and the checks printed by `d4_bfield.py`; each is re-checked independently by `x4_check.py`, whose findings x4-N1 to x4-N4 are incorporated in Theorems 5.6 and 5.10 and Propositions 5.11 and 5.17. "Restated": fixed in [1], restated and re-verified here.

6 Synthesis: the hypothesis, sentence by sentence

We now return to the hypothesis of Section 1 and test each of its clauses against what Sections 2 to 5 establish. The verdicts are summarised in Table 7. Throughout, the relation $E = mc^2$ itself is inherited; what is at stake is only how the arena realises its three ingredients.

6.1 "The energy is one undivided entity, and it is six-dimensional"

This clause survives, in the field-strength reading of $\mathbb{R}^{3,3}$. Among quadratic forms on the six frame components, exactly one is invariant under the diagonal rotations and under duality, and it is positive: $E_6 = \frac{1}{2}|z|^2 = \frac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2)$ (Section 2.1; this restates Theorem 17.11(iii) of [1]). Its symmetry group is $U(3)$, so the energy is "unitary" in a literal sense, but only as a pointwise quadratic form: $U(3)$ is not a symmetry of Maxwell's equations on the frame, nor of the split-form interior. In the closure of the force law the undivided energy of the *momentum* field, $E_6(\pi) = \frac{1}{2}|p|^2$, is a Casimir, conserved by every Hamiltonian, together with its rest energy $E_0(\pi)$ and its Poynting vector $\mathbf{S}(\pi)$ (Section 4.1). The energy of the *frame* field is not conserved. The other reading of "six-dimensional", three temporal energies forming a vector in V^+, fails: splitting that vector lowers the interior mass, $m_4^2c^4 = \mu^2c^4 - |\mathcal{E}_\perp|^2$ (Section 2.7).

6.2 "Mass emerges through an internal differentiation of that unity"

This clause survives as an identity, not as a process. With no interaction and no second field, the undivided energy of a field point splits as

$$E_6^2 = E_0^2 + |\mathbf{S}|^2, \qquad E_0 = \tfrac{1}{2}|q| = \tfrac{1}{2}|z \cdot z|,$$

the complex Lagrange identity (Section 2.3). In SI units it reads $u^2 = c^2|\mathbf{g}|^2 + u_0^2$, the field-point form of $E^2 = (pc)^2 + (mc^2)^2$, with rest-mass density u_0/c^2. The rest part E_0 is the complete invariant of Lorentz transformations together with duality, the Lorentz scalar of the stress tensor whose 00 component is E_6 (Rainich), and the minimum of E_6 over boosts, attained exactly in the rest frame (Section 2.4). All of this is inherited classical electrodynamics written in the arena's variables. The arena adds which forms play which role: E_6 belongs to the duality class of Theorem 17.11 of [1], E_0 to the Lorentz class, and $E = mc^2$ is the locus, the rest frame, where the two coincide.

The differentiation does not happen in time. In the closure the split $E_6^2 = E_0^2 + |\mathbf{S}|^2$ of the momentum field is fixed by the initial data, because all three terms are Casimirs. What moves is the duality angle and the rotor rate, and the conserved total $H = \frac{1}{2}I_0\kappa^2 + U(\theta)$ exchanges energy between a rate part and a Lorentz-invariant frame part (Section 4.3). The mass of a whole is not produced by interference either: two disjoint counter-propagating null pulses have $q \equiv 0$ everywhere and $M = U > 0$ (Section 2.6).

6.3 "c^2 is the symmetric part"

This clause survives in one precise form. The constant c is the unique isotropic conversion between the temporal and the spatial triple, forced by Schur's lemma (Theorem 3.2 of [1]). It is a conversion, not a symmetry, and its weight in E_6 is a unit convention (Section 2.5). The form in which the clause is exact is the decomposition of a shift operator into its two summands (Theorem 17.3 of [1]):

$$\mathrm{End}(V) = \mathfrak{so}(3,3) \oplus \{X : X^\dagger = X\}, \qquad \text{shift} = b + \kappa G, \qquad (\kappa G)^2 = -\frac{c^2}{\ell^2}\,\mathbf{1}.$$

The η-self-adjoint summand is the symmetric part: it is the summand in which factorisation holds, the one that binds in neutral signature, and the one that carries c, through $\kappa = c/\ell$. The antisymmetric summand is a connection. Reading the first as "c^2" and the second as "b" is interpretive, but the decomposition, and the fact that only the first summand binds, are fixed by the paper.

6.4 "m is what breaks the symmetry"

This clause is true for the right symmetries and false for the ones it is usually read about (Section 3).

1. A nonzero rest energy breaks duality. The duality $U(1)$ generated by G acts on the Lorentz invariant by $q \mapsto e^{2i\theta}q$, so a field with $E_0 \neq 0$ keeps only $\{\pm\mathbf{1}\}$, and $\arg q$ labels the broken directions. At a null field point, duality joins the little group, which becomes $\mathfrak{e}(2)$ with helicity as its compact generator. This is inherited, pulled back to the frame.

2. For Dirac matter the mass breaks the chiral and Pauli–Gürsey symmetries, $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2) \to \mathfrak{u}(1)_J$, with three Goldstone directions, one of them the pseudoscalar mass angle (Theorem 18.5 of [1], re-verified).

3. The mass breaks no spacetime symmetry, not even the six-dimensional one. The Dirac mass form $T = e_1e_2e_3$ is invariant under all of $\mathrm{Spin}(3,3)$ and is the only such kernel

(Corollary 18.4 of [1]). Within $\mathfrak{so}(3,3)$ it is the kinetic term, where c enters, that cuts the symmetry down to the interior Lorentz algebra and J. In this sense the hypothesis is inverted.

4. In the closure the rest energy of the momentum field is exactly the amplitude of the potential for the duality angle, $R_0 = 2|\nu_1 + i\nu_2|\, E_0(\pi)$, with (ν_1, ν_2) the Lorentz-class inertia coefficients of Section 2. A null (massless) momentum field leaves the angle flat and the rate constant; a massive one makes it a pendulum whose squared small-oscillation frequency is $4|\nu_1 + i\nu_2|E_0(\pi)/(mI_0)$ (Section 4.2). This restates Theorem 17.13 and Corollary 17.14 of [1]. Reading the angle as a pseudo-Goldstone boson of duality is interpretive: the breaking is explicit, carried by the Lorentz-class inertia, and every pendulum has this shape.

6.5 "m is generated from the b-particle of Yang–Mills"

Classically, this clause fails, and the failure is sharp (Section 5).

1. The binding rate is never a gauge potential. For every connection b the field equation factorises covariantly, $(D - \kappa G)(D + \kappa G)X = 0$ with $D = \partial_\tau + b$; κG transforms as a tensor, b as a connection. Without κ, a constant b is pure gauge along the worldline: no rest frequency, no mass scale.

2. An admissible Yang–Mills field is compact, since no gauge boson is a boost (Corollary 2.4 of [1]). Its part commuting with G splits the rest frequency into charge-labelled lines κ, $\kappa \pm \beta$, removable along a single worldline by a rotating frame (Larmor's theorem). This is a spectrum without a scale, since β/κ is free. A purely anticommuting part produces Jordan blocks and secular growth. A mixed connection keeps every orbit bounded exactly when it rotates the temporal and the spatial triple at equal rates about axes that are not opposite, outside one exceptional ratio of the rates to κ (Theorem 5.10).

3. At three constituents the part of the Pati–Salam algebra commuting with the binding root is $\mathfrak{so}(4)_\Delta \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ (Theorem 17.21 of [1]): two copies of Yang and Mills' isospin algebra. It splits the rest frequency by $\mathfrak{su}(2)$ weights and supplies no scale.

4. In the selected vacuum the order is the reverse of the hypothesis. The Pati–Salam vector bosons acquire mass by eating nine Goldstone directions (Theorem 22.2 of [1]), rather than generating the mass of anything.

5. At the quantum level, one-gauge-boson exchange favours the Dirac singlet channel $(\mathbf{1}, \mathbf{2}, \mathbf{2})$ and the colour-breaking $(\mathbf{6}, \mathbf{1}, \mathbf{1})$, and disfavours both the $(\mathbf{15}, \mathbf{2}, \mathbf{2})$ needed for the Georgi–Jarlskog factor and the $(\mathbf{10}, \mathbf{1}, \mathbf{3})$ that the paper's vacuum uses. This is inherited group theory applied to the arena's representations.

The one role left open for the b-field is the scale. The algebra fixes c and leaves ℓ, hence κ, free; classical Yang–Mills theory in four dimensions has no scale either, but its quantum theory generates one by dimensional transmutation, $\Lambda_b = \mu \exp[-8\pi^2/(\beta_0 g^2(\mu))]$. The identification $\hbar\kappa = \Lambda_b$, that is $\ell = \hbar c/\Lambda_b$, is a conjecture, not derived. Even granted, reading the rate as a mass, $mc^2 = \hbar\kappa$, is a further assumption: in the interior dynamics m and κ are independent inputs, the spectrum does not depend on m, and the classical rescaling invariant is $m\kappa$.

6.6 Particles, antiparticles, the vacuum and the Goldstone bosons

The original questions of this investigation have the following answers in the arena (Sections 3.4 to 3.6). For Dirac matter at rest the flow is a rotation generated by the six-dimensional mass form itself, $\Psi(\lambda) = e^{-m\lambda T}\Psi_0$, with rate equal to the mass. Particle and antiparticle are its two senses, and charge conjugation, the surplus plane, exchanges them. The binding rotation of the

clause	what the arena gives	verdict
energy undivided, six-dimensional	$E_6 = \frac{1}{2}(\|\mathbf{B}\|^2 + \|\mathbf{E}\|^2)$, unique, positive, symmetry $U(3)$ pointwise; Casimir of the momentum field	holds (inherited / restates [1])
mass from internal differentiation	$E_6^2 = E_0^2 + \|\mathbf{S}\|^2$, $E_0 = \min_{\text{boosts}} E_6$; split fixed by initial data in the closure	holds as an identity, not as a process
c^2 the symmetric part	c a Schur conversion; κG the η-self-adjoint summand of the shift, the only one that binds	holds in the shift decomposition (interpretive reading of a fixed result)
m breaks the symmetry	breaks duality and the chiral algebra; breaks neither Lorentz nor $\mathrm{Spin}(3,3)$	true for internal symmetries, false for spacetime ones
m generated from the b-field	b is the connection summand: splits or tears the binding, never creates it; vector bosons are massive by eating Goldstones	fails classically; scale by transmutation open

Table 7: The hypothesis against the arena.

frame also has two senses, but parity exchanges them, not charge conjugation, so its rate $\kappa = c/\ell$ is a conversion rate and its two senses are not a particle–antiparticle pair. The Fock vacuum of one momentum is unique and neutral, with energy $-2E$ ($-2mc^2$ at rest). The frame has no ground state, its spectrum being $\hbar\kappa\,\mathbb{Z}$, but its zero-point energy cancels in neutral signature, and this cancellation survives, frequency by frequency, every connection whose motion stays bounded (Section 5). Goldstone directions: three at one constituent in the Dirac phase; nine at three constituents, all eaten, with no Majoron because $[J, \Phi_0] = [J\Gamma, \Phi_0] = 2[t_3^R, \Phi_0]$; and one axion-like direction of charge $1 - \Gamma$ at the electroweak stage, which a realistic potential must lift explicitly.

6.7 The hypothesis, restated so that the algebra supports it

Collecting the verdicts, the strongest statement of the hypothesis that the arena supports is the following. *The undivided energy is the duality-invariant, unitary energy* $\frac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2)$ *of the six field components. The constant c is the unique isotropic conversion between times and lengths, and it enters the dynamics through the* η*-self-adjoint part of the shift, the binding root. The mass is the Lorentz-and-duality invariant* $\frac{1}{2}|q|$*, the minimum of the energy over boosts, and* $E = mc^2$ *is the rest frame, where the two coincide. What the mass breaks is duality and, for matter, chirality, never Lorentz invariance or* $\mathrm{Spin}(3,3)$*. The Yang–Mills b-field is the antisymmetric part of the same shift: it can split or destabilise the binding, but it cannot create it, and the only role left for it in generating mass is to supply, by dimensional transmutation, the one scale* ℓ *that the algebra leaves free.*

6.8 Open problems

1. A mechanism fixing ℓ, and with it κ, from the arena or from the quantum dynamics of a b-field, so that $\hbar\kappa = \Lambda_b$ becomes a derivation; and a reason, absent so far, to identify the interior rate with a particle mass.

2. A dynamical process in which rest energy is produced rather than carried: in the closure the rest energy of the momentum field is a Casimir, while that of the frame field drifts without bound.

3. A common Lorentz action for the two readings of $\mathbb{R}^{3,3}$: the field-strength reading, where E_6 and E_0 live, and the interior reading, where the Dirac mass form lives, exclude each

other on the same six directions (Proposition 12.6 of [1]).

4. The classification of compact connections that keep the binding bounded, beyond the commuting ones and the equal-rate family found here.

7 How the statements were verified

Every finite-dimensional statement of Sections 2 to 5 was checked by computation, in two independent passes. The scripts are in `investigations/rest_energy/verify/` and run with `python3` from any directory, using `numpy`, `sympy`, `scipy` and `mpmath`.

1. *Derivation scripts*, one per section (`d1_unitary_energy.py`, `d2_symmetry_breaking.py`, `d3_closure_dynamics.py` and `d4_bfield.py`), were written together with the sections. They rebuild the realisation of [1], Appendix A, from its definition and print one `PASS` line per claim of their section, with the claim's label.

2. *Adversarial scripts* `x1_check.py` to `x4_check.py` were written afterwards by separate authors whose task was to refute the claims. They import nothing from the derivation scripts, rebuild every convention from the paper's sources, and wherever possible use a different method: symbolic where the first pass was numerical, random tests where it used special cases, a different integrator for the dynamics. Their findings, including one refuted claim and several reclassifications, were applied before the note was assembled.

3. `run_all.py` runs all eight scripts and writes `RESULTS.md`; it exits with status zero only if every script does.

Algebraic statements were checked exactly, in integer or rational arithmetic, since every blade of the realisation is a signed permutation matrix. Statements about spectra and Jordan structure were checked with rational data; floating-point inputs were not used for semisimplicity tests, where they produce false negatives. Dynamical statements were checked by numerical integration with relative tolerances of 10^{-11} to 10^{-12}, and elliptic and Mathieu quantities with `mpmath` and `scipy.special`. Statements of [1] that are cited without re-verification are marked as such in the text. The corrections made to [1] itself in the course of this work are listed, with their own checks, in `investigations/errata_2026-10-17/`.

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